Same-well imaging and transcriptomics in a 50,000-well one-bead-one-compound microchip recovers compound-resolved phenotypes

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Abstract

Phenotypic screens trade throughput against biological depth: high-content imaging scales to millions of wells with a limited readout, while single-cell transcriptomics provides a deeper readout at substantially higher cost per compound. The Z-Screen platform addresses both axes through one-bead-one-compound combinatorial libraries profiled in 50,000-well microchips by paired imaging and mRNA sequencing. Each micro-well contains 5 to 10 cells, providing partial averaging over cellular heterogeneity while pairing both readouts on the same multicellular colony rather than on replicate populations exposed to the same treatment. ActiveSeq is the matched-readout layer of that platform.

Here we evaluate whether the two same-well readouts outperform either readout alone on a held-out compound classification task.

The analysis used a canonical ActiveSeq dataset of 20,222 matched wells across two pilot experiments (ACT010 and ACT011): 11,511 wells from a 35-compound control panel, and 8,711 ZEL026 dummy-bead wells retained as a negative-control QC stress test. After task-aware calibration of the per-well image embedding, both readouts were highly reproducible. Within-run split-half top-1 retrieval reached 98.8% in RNA and 94.2% in the calibrated image latent for ACT010, and 80.1% / 86.8% for ACT011, against a 2.9% top-1 chance level (35 classes). Compound centroids learned in ACT010 retrieved the same compounds in ACT011 at 71.4% top-1 (94.3% top-3) in RNA and 85.7% top-1 (100% top-3) in the calibrated image latent.

Same-well multimodality was strongly additive. A linear discriminant trained on merged image-plus-RNA features reached 55.5% balanced accuracy on held-out wells, against 41.4% for RNA alone and 29.2% for image alone. The ordering held within each run (ACT010: 60.8% merged vs 47.3% RNA; ACT011: 45.1% merged vs 30.5% RNA), and merged features improved per-compound recall for 35 of 35 compounds in the pooled analysis. The two views also preserved a shared map of compound relationships, with Spearman 0.718 (ACT010) and 0.607 (ACT011) between pairwise compound-distance matrices in image and RNA space, and per-run effect-size rankings transferred across runs at Spearman 0.799 (image) and 0.662 (RNA).

The closest public benchmark, scGeneScope, is a treatment-matched Cell Painting and scRNA-seq

dataset across 28 treatments in U2-OS cells with approximately 716,767 single-cell images and 627,704 single-cell RNA profiles. The present pilot is an order of magnitude smaller in total cells profiled (Z-Screen reads each well as a pseudobulk of 5 to 10 cells rather than at single-cell resolution) and uses a simpler imaging modality. Once task difficulty is normalized, the multimodal numbers fall in the same range: pooled merged features reach 19.4 times chance, against 18.0 times for the scGeneScope multimodal multiprofile reference; ACT010 alone reaches 21.3 times chance. Centered pseudobulk gene signatures retrieve the correct compound across runs at 48.6% top-1 and 68.6% top-3, with reproducible compound-specific gene programs. Together, these results establish that ActiveSeq supports same-well multimodal profiling, that calibration recovers compound-resolved structure not directly accessible in the original frozen image embedding, and that same-well fusion produces an additive gain on held-out classification.

Significance

Treatment-matched Cell Painting and Perturb-seq experiments yield two complementary descriptions of cell state on separate cell populations exposed to the same compound. ActiveSeq pairs both descriptions on the same multicellular colony in the same micro-well, on a chip designed for 50,000 wells in parallel. The present analysis demonstrates that this tighter pairing produces an additive predictive gain on held-out compound classification across all 35 compounds in two pilot experiments. For early-stage discovery, this shifts the readout from ranking wells to characterizing the cell-state response that distinguishes a hit.

Introduction

Early-stage discovery requires both throughput sufficient to cover relevant chemical space and biological resolution sufficient to characterize hit responses. Existing platforms typically optimize one axis at the expense of the other.

Cell Painting and the imaging-only ML platforms built on it (Recursion, Insitro, the Broad's morphological profiling program) have driven imaging throughput up and per-image cost down to the point where multi-million-compound runs are routine [1]. The readout is comparatively shallow: a small set of fluorescent stains and the morphological features extractable from them by deep networks. Single-cell transcriptomics, through Perturb-seq [3] and platforms such as Tahoe-100M and scPerturb, provides a substantially deeper readout of cell state at higher cost per profile and lower compound concurrency. Public efforts to bridge the two have produced treatment-matched datasets, of which scGeneScope (a treatment-matched Cell Painting and scRNA-seq benchmark in U2-OS cells) is the strongest current reference [4]. Treatment-matched datasets pair imaging and sequencing across separate populations of cells exposed to the same compound; alignment is at the treatment level, not the cellular level.

Z-Screen occupies a different design point. Compounds are delivered through one-bead-one-compound combinatorial libraries and profiled in 50,000-well microchips by paired imaging and mRNA sequencing, at 5 to 10 cells per micro-well. The readout averages over limited cellular heterogeneity without incurring the per-profile cost of single-cell resolution. ActiveSeq is the matched-readout layer: each per-well image is paired with a transcriptomic measurement from the same micro-well. The pairing is stricter than in treatment-matched public benchmarks because both readouts describe the same multicellular colony rather than two replicate populations. With same-well pairing established, the central evaluation question is whether combined image and RNA features outperform either modality alone on a held-out classification task. That is the question addressed here.

The analysis is structured around four questions: 1. Are matched compound measurements reproducible within a run? 2. Do image and RNA preserve a shared map of compound relationships? 3. Does same-well fusion produce a held-out classification gain over either single view? 4. Do low-pass gene programs remain interpretable enough to support downstream triage?

Results are then anchored against scGeneScope so the comparison can be read at the appropriate level of abstraction.

Results

The 35-compound control panel is reproducible within run and transferable across runs

Wells for each control compound were split into two pseudo-replicates, and one half was queried for retrieval of the other among the 35 compounds. RNA latent space was highly stable: split-half top-1 retrieval was 98.8% in ACT010 and 80.1% in ACT011. The calibrated image latent was similar at 94.2% (ACT010) and 86.8% (ACT011). Top-3 retrieval reached 100% for RNA in ACT010 and 99.6% for the calibrated image latent.

Cross-run transfer is the more demanding criterion: a control-compound strategy is only useful if it transfers across experiments. Compound centroids learned in ACT010 retrieved the same compounds in ACT011 at 71.4% top-1 (94.3% top-3) in RNA and 85.7% top-1 (100% top-3) in the calibrated image latent, against a 2.9% top-1 chance level. Both readouts therefore support a reusable control panel across experiments. The image-side numbers also indicate that compound-identity information was largely present in the original embedding but obscured by the raw geometry; task-aware calibration made it accessible.

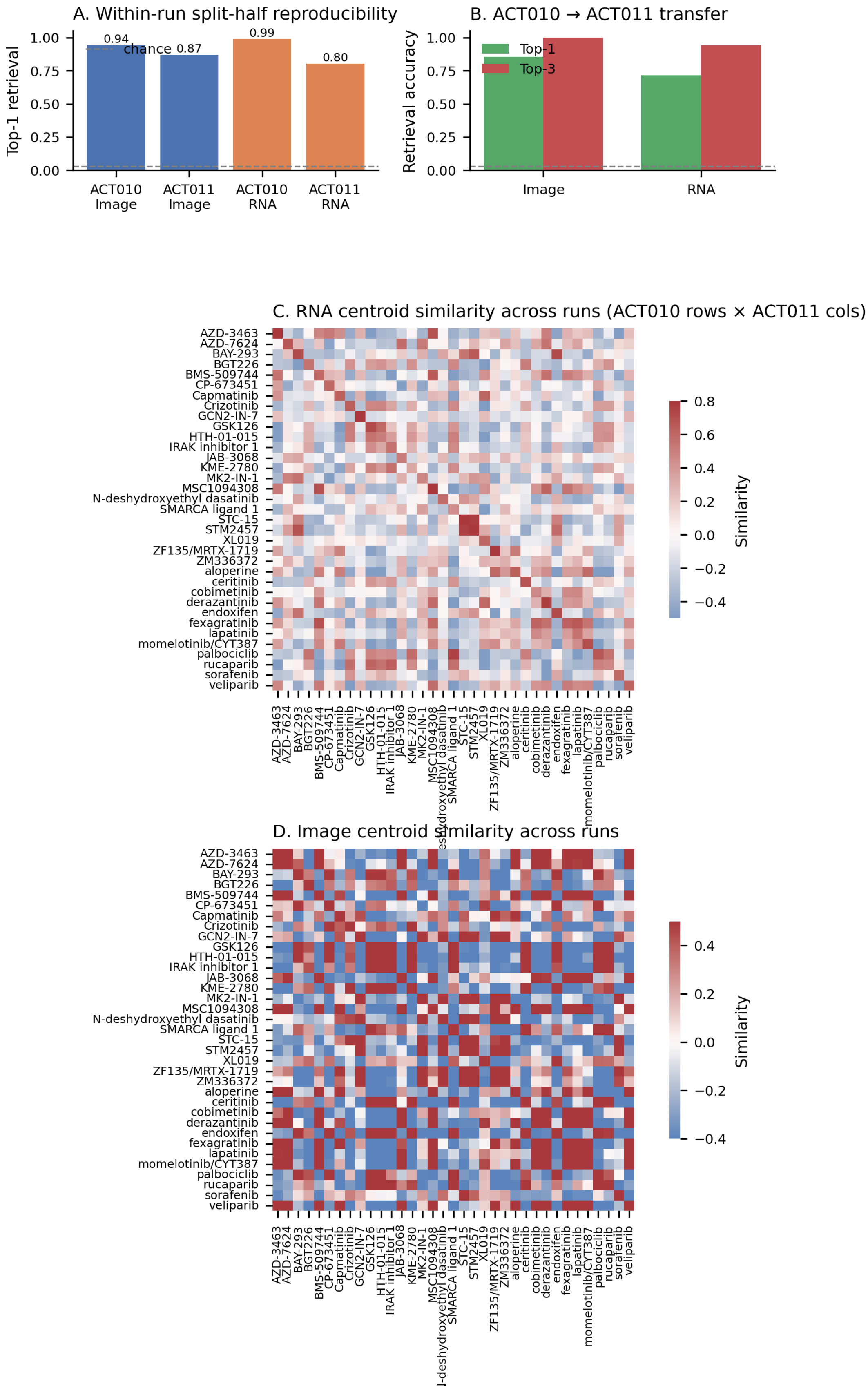

A. Within-run split-half reproducibility
Top-1 retrieval
0.94
0.87
0.99
0.80
chance
ACT010 Image
ACT011 Image
ACT010 RNA
ACT011 RNA
B. ACT010 → ACT011 transfer
Retrieval accuracy
Top-1
Top-3
Image
RNA
C. RNA centroid similarity across runs (ACT010 rows × ACT011 cols)
D. Image centroid similarity across runs
Similarity
AZD-3463
AZD-7624
BAY-293
BGT226
BMS-509744
CP-673451
Capmatinib
Crizotinib
GCN2-IN-7
GSK126
HTH-01-015
IRAK inhibitor 1
JAB-3068
KME-2780
MK2-IN-1
MSC1094308
N-deshydroxyethyl dasatinib
SMARCA ligand 1
STC-15
STM2457
XL019
ZF135/MRTX-1719
ZM336372
aloperine
ceritinib
cobimetinib
derazantinib
endoxifen
fexagratinib
lapatinib
momelotinib/CYT387
palbociclib
rucaparib
sorafenib
veliparib

Figure 1. Same-well reproducibility and cross-run centroid transfer in the 35-compound control panel. Both modalities are highly reproducible within run, and both transfer well above chance from ACT010 to ACT011. The calibrated image latent reaches 85.7% top-1 cross-run retrieval and 100% top-3.

Image and RNA preserve a shared compound geometry

Beyond exact label recovery, an additional question for a screening platform is whether the two views rank compounds consistently: if RNA places compound A nearer to compound B than to compound C, does the image latent agree?

Within each run, agreement was substantial. Spearman correlations between pairwise compound-distance matrices were 0.718 in ACT010 and 0.607 in ACT011. The strongest perturbagens recurred across runs: cross-run effect-magnitude rankings correlated at Spearman 0.799 in image space and 0.662 in RNA space. Several compounds (GCN2-IN-7, KME-2780, BGT226, ZF135 / MRTX-1719, CP-673451) appeared near the top of both modalities in at least one run.

Image-derived neighborhoods are therefore biologically aligned with RNA-derived neighborhoods to a degree that supports compound prioritization on either branch, and the largest phenotypic responses generalize across independent experiments. That is the property required for either readout to be used interchangeably during triage.

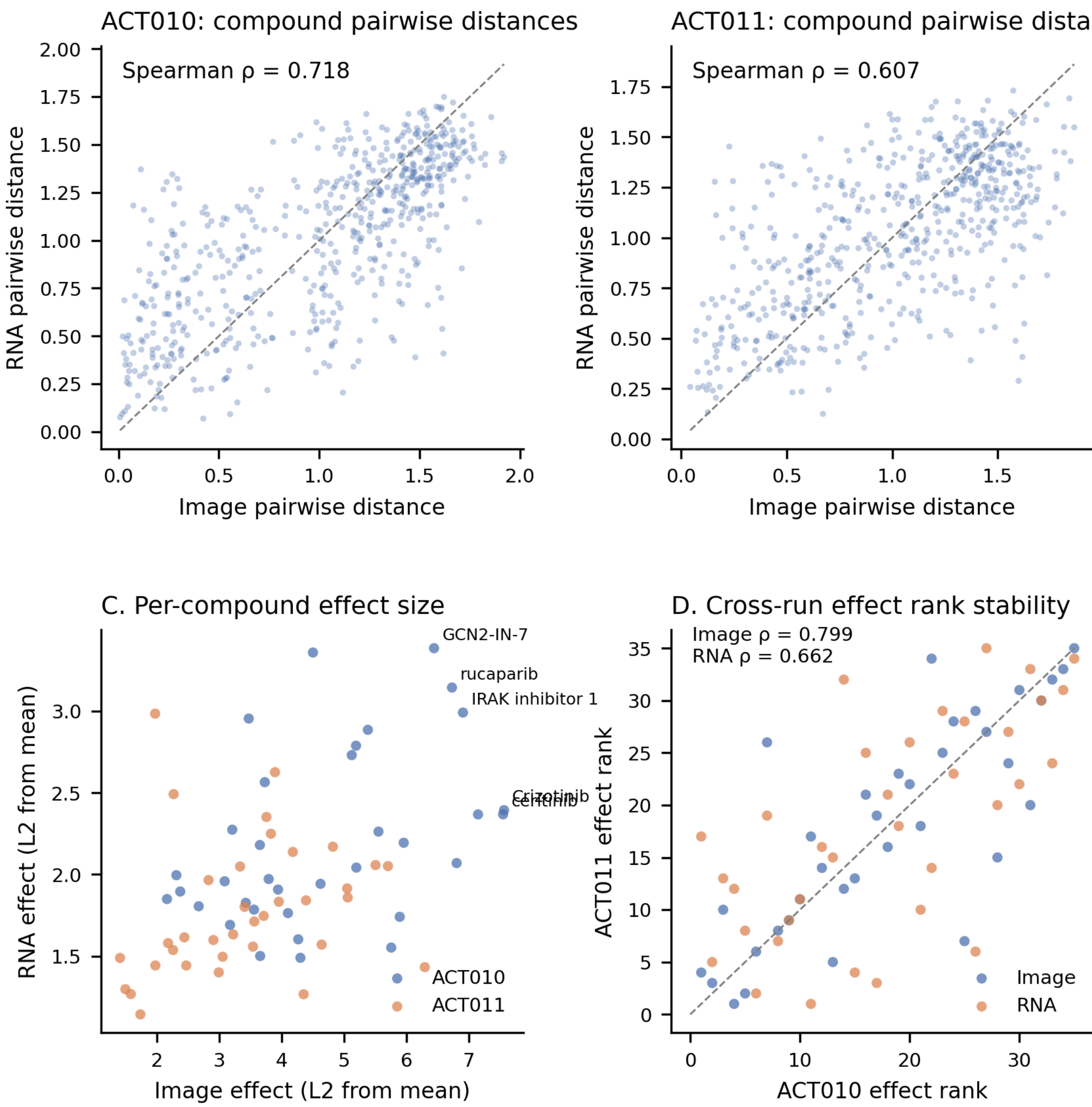


Figure 2. Image and RNA preserve a shared compound geometry within run, and effect-size rankings are stable across runs. Pairwise compound-distance correlations are 0.718 (ACT010) and 0.607 (ACT011); cross-run per-compound effect magnitudes correlate at Spearman 0.799 (image) and 0.662 (RNA).

Same-well multimodality is additive on held-out compound classification

Held-out compound classification is the most operationally relevant benchmark in the dataset. Each matched well was reduced to one calibrated image vector and one RNA vector. A linear discriminant model was trained either on the image vector alone, the RNA vector alone, or the raw concatenation of the two.

Merged image-plus-RNA features outperformed RNA alone in every cut: 55.5% balanced accuracy versus 41.4% pooled, 60.8% vs 47.3% within ACT010, and 45.1% vs 30.5% within ACT011 (image alone reached 29.2% pooled). Overall accuracy followed the same pattern, rising from 42.5% to 54.9% pooled, from 48.3% to 60.2% in ACT010, and from 31.6% to 45.5% in ACT011.

The gain was broad rather than concentrated in a few easy classes. Merged features improved per-compound recall for 35 of 35 compounds in the pooled analysis, 33 of 35 in ACT010, and 30 of 35 in ACT011. ACT010 confusion matrices show the same pattern: RNA performs well on its own, and adding the calibrated image latent moves probability mass onto correct diagonal entries and resolves several near-confusions.

Linear discriminant analysis on raw-concatenated features was used in preference to a deep classifier because its decision boundary is a linear function of the input features; any reported gain is therefore traceable to a specific axis of the input space rather than to model flexibility.

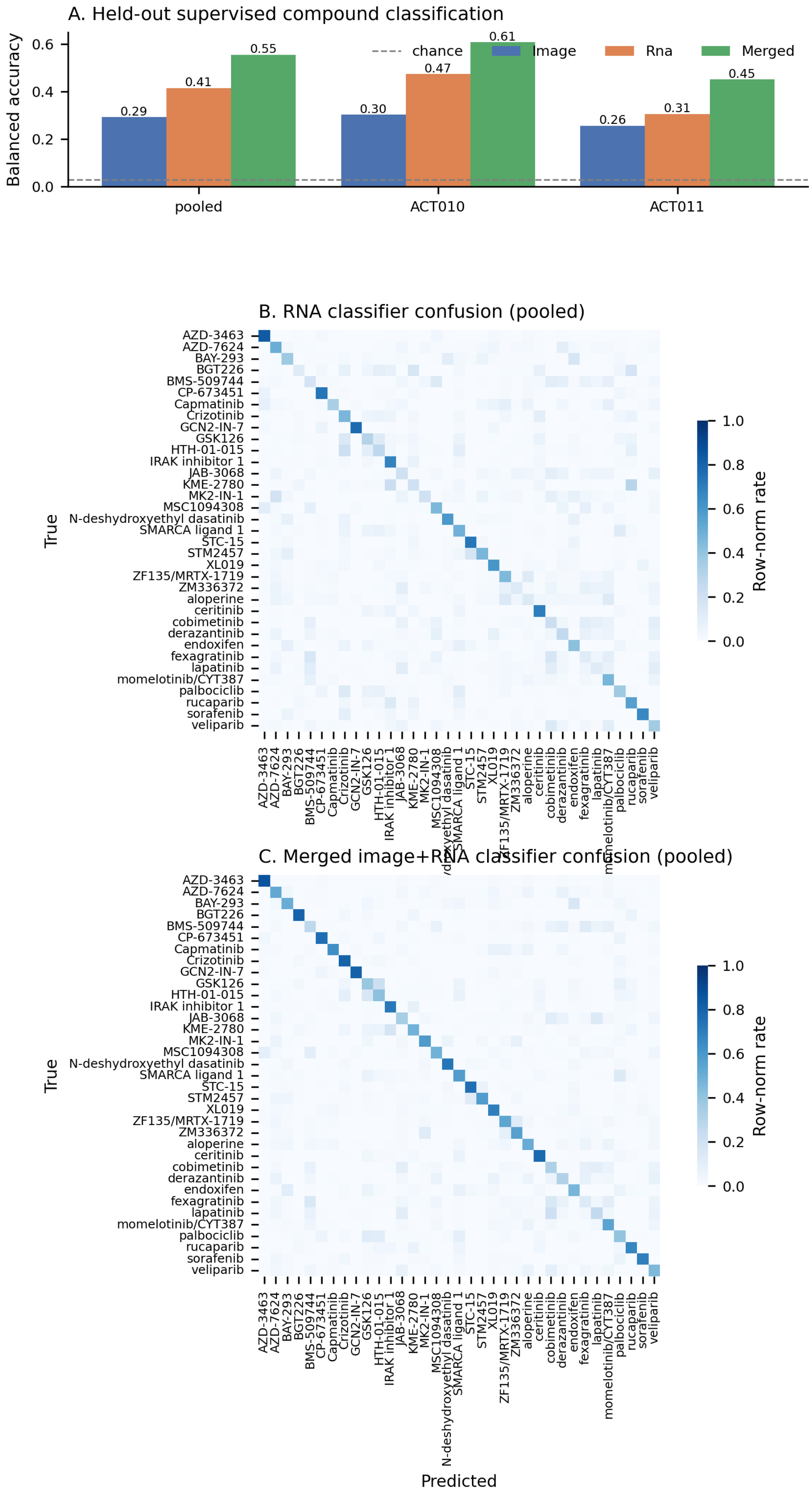

A. Held-out supervised compound classification
Balanced accuracy
0.0
0.2
0.4
0.6
chance
Image
Rna
Merged
0.29
0.41
0.55
0.30
0.47
0.61
0.26
0.31
0.45
pooled
ACT010
ACT011
B. RNA classifier confusion (pooled)
C. Merged image+RNA classifier confusion (pooled)
True
Predicted
Row-norm rate
0.0
0.2
0.4
0.6
0.8
1.0
AZD-3463
AZD-7624
BAY-293
BGT226
BMS-509744
CP-673451
Capmatinib
Crizotinib
GCN2-IN-7
GSK126
HTH-01-015
IRAK inhibitor 1
JAB-3068
KME-2780
MK2-IN-1
MSC1094308
N-deshydroxyethyl dasatinib
SMARCA ligand 1
STC-15
STM2457
XL019
ZF135/MRTX-1719
ZM336372
aloperine
ceritinib
cobimetinib
derazantinib
endoxifen
fexagratinib
lapatinib
momelotinib/CYT387
palbociclib
rucaparib
sorafenib
veliparib

Figure 3. Same-well multimodal classification on held-out wells. Merged image-plus-RNA features outperform RNA alone in pooled and per-run analyses, and per-compound confusion matrices show the gain is from reduced class confusion across most compounds rather than from a few easy classes.

Same-well structure persists beyond the supervised task

A second question is whether same-well pairing carries useful signal when class labels are unavailable at retrieval time, the regime in which hits are typically triaged.

In the harder held-out centroid-retrieval task, the calibrated image latent reached 25.1% / 21.7% top-1 across the two runs, RNA reached 44.8% / 24.3%, and a naive image-plus-RNA concatenation reached 49.1% / 33.7%, exceeding RNA alone in both runs. A lightweight CLIP-style same-well contrastive objective on top of the calibrated latent did not improve retrieval further, indicating that calibration had already exposed most of the recoverable compound-identity information.

Exact well-to-well retrieval remained modest in absolute terms, as expected in a low-pass multicellular assay where 5 to 10 cells per well leave real biological heterogeneity within each pseudo-replicate. The residual ambiguity is structured rather than random. Image and RNA agreed on within-compound well structure above random expectation in 34 of 35 compounds in both runs (cross-modal distance correlation), in 26 of 35 (ACT010) and 29 of 35 (ACT011) by nearest-neighbor overlap, and in 28 of 35 / 27 of 35 by same-compound CLIP retrieval. The pattern is consistent with same-well pairing capturing real within-compound heterogeneity (subpopulation structure, variable response states) rather than only technical scatter.

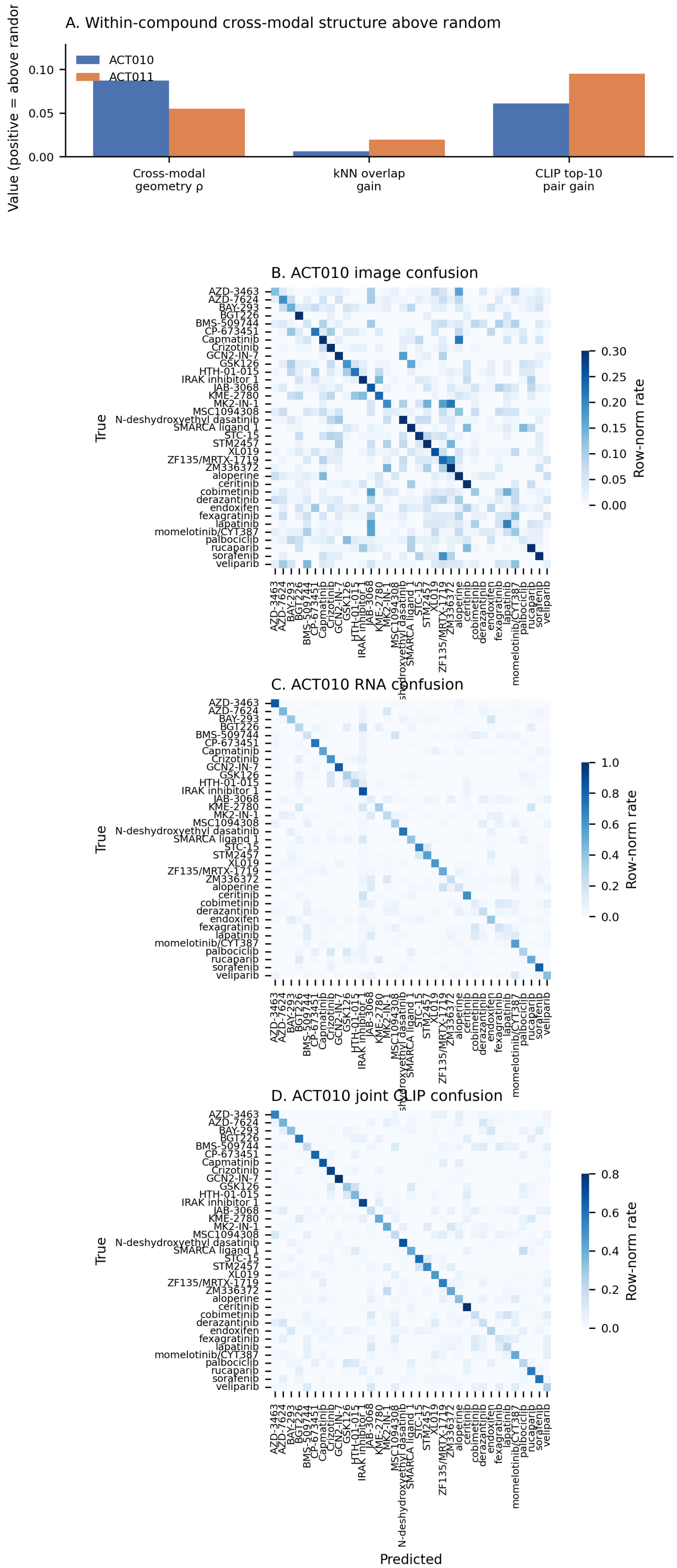

A. Within-compound cross-modal structure above random
Value (positive = above random)
ACT010
ACT011
0.10
0.05
0.00
Cross-modal geometry ρ
kNN overlap gain
CLIP top-10 pair gain
B. ACT010 image confusion
C. ACT010 RNA confusion
D. ACT010 joint CLIP confusion
True
Predicted
Row-norm rate
AZD-3463
AZD-7624
BAY-293
BGT226
BMS-509744
CP-673451
Capmatinib
Crizotinib
GCN2-IN-7
GSK126
HTH-01-015
IRAK inhibitor 1
JAB-3068
KME-2780
MK2-IN-1
MSC1094308
N-deshydroxyethyl dasatinib
SMARCA ligand 1
STC-15
STM2457
XL019
ZF135/MRTX-1719
ZM336372
aloperine
ceritinib
cobimetinib
derazantinib
endoxifen
fexagratinib
lapatinib
momelotinib/CYT387
palbociclib
rucaparib
sorafenib
veliparib
0.30
0.25
0.20
0.15
0.10
0.05
0.00
1.0
0.8
0.6
0.4
0.2
0.0
0.8
0.6
0.4
0.2
0.0

Figure 4. Same-well structure beyond supervised classification. Within-compound cross-modal agreement exceeds random expectation across nearly all compounds, and held-out confusion structure for image, RNA, and joint embeddings shows that residual errors are biologically structured rather than uniform.

Low-pass gene programs remain portable across runs

A screening assay becomes more useful when the underlying gene-level signal remains interpretable. Raw counts from matched control wells were aggregated into per-compound pseudobulk signatures, converted to log-CPM, and centered by subtracting each run's experiment-wide mean. Pseudobulk plus centering was used in preference to per-cell differential-expression modeling because each well averages 5 to 10 cells with low per-well RNA depth; pseudobulk is the unit at which the signal is robust, and centering removes experiment-level offsets without imposing a normalization across two physically separate runs.

Centered signatures retrieved the correct compound across runs at 48.6% top-1 and 68.6% top-3. Representative compounds carried coherent gene programs: palbociclib (ARC, RASD1, CDKN1C, EGR1), BAY-293 (MT1X, MT1F, MT2A, SLC30A1), rucaparib (ITPRIP, SLC37A4, NACC2, COL4A1), crizotinib (SORCS2, ZNF34, AC253572.2), GSK126 (DUSP10, SPATA6L, CCN2). Validation of each gene as a mechanism-of-action marker in HEK293 is a separate question. The platform-level claim is that ActiveSeq preserves compound-specific transcriptional structure at the gene level even at present low-pass depth, supporting the use of transcriptomics for hit de-risking and phenotype clustering after primary screening.

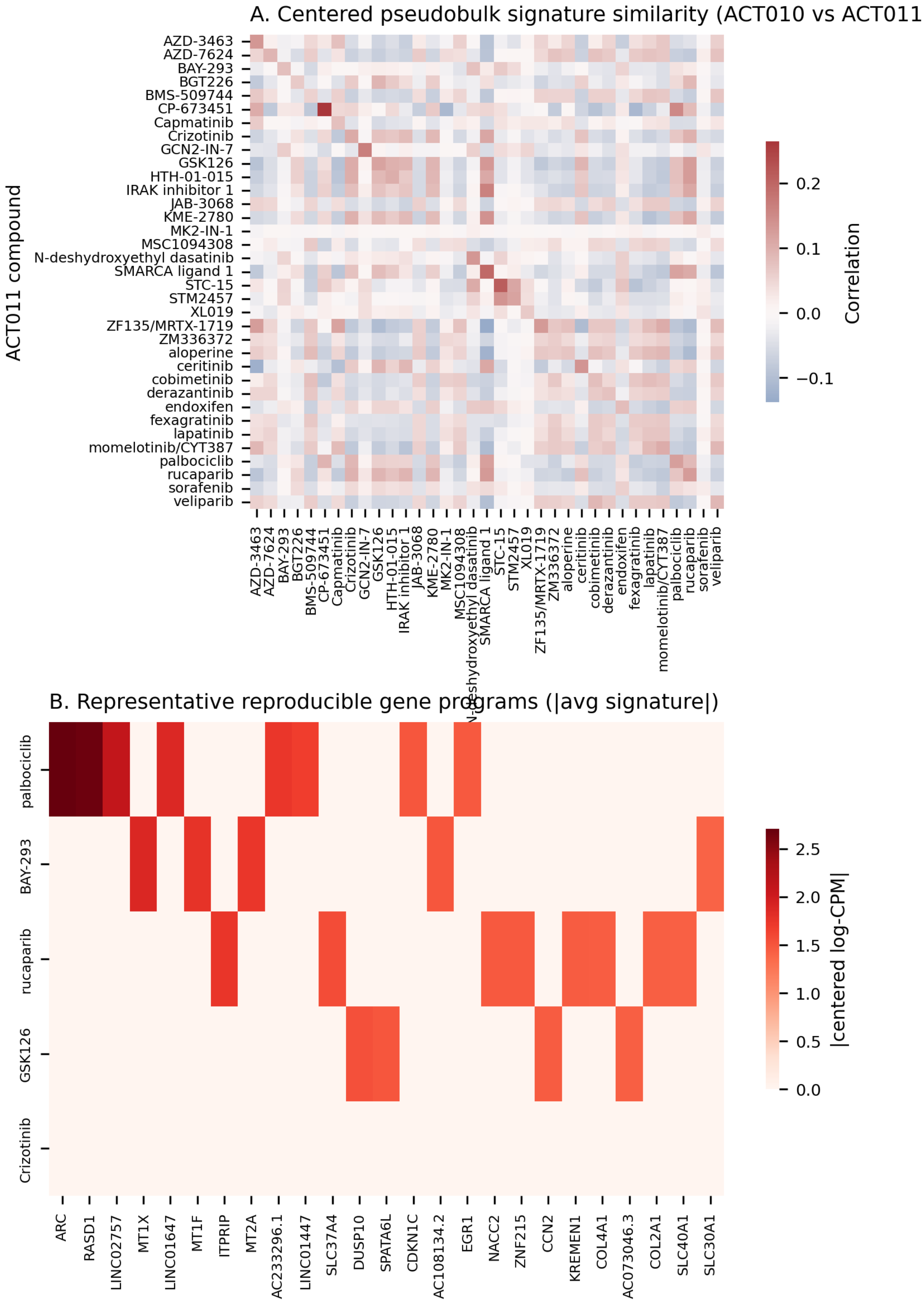

A. Centered pseudobulk signature similarity (ACT010 vs ACT011)
ACT011 compound
AZD-3463
AZD-7624
BAY-293
BGT226
BMS-509744
CP-673451
Capmatinib
Crizotinib
GCN2-IN-7
GSK126
HTH-01-015
IRAK inhibitor 1
JAB-3068
KME-2780
MK2-IN-1
MSC1094308
N-deshydroxyethyl dasatinib
SMARCA ligand 1
STC-15
STM2457
XL019
ZF135/MRTX-1719
ZM336372
aloperine
ceritinib
cobimetinib
derazantinib
endoxifen
fexagratinib
lapatinib
momelotinib/CYT387
palbociclib
rucaparib
sorafenib
veliparib
Correlation
0.2
0.1
0.0
−0.1
B. Representative reproducible gene programs (|avg signature|)
palbociclib
BAY-293
rucaparib
GSK126
Crizotinib
ARC
RASD1
LINC02757
MT1X
LINC01647
MT1F
ITPRIP
MT2A
AC233296.1
LINC01447
SLC37A4
DUSP10
SPATA6L
CDKN1C
AC108134.2
EGR1
NACC2
ZNF215
CCN2
KREMEN1
COL4A1
AC073046.3
COL2A1
SLC40A1
SLC30A1
|centered log-CPM|
2.5
2.0
1.5
1.0
0.5
0.0

Figure 5. Centered pseudobulk gene signatures preserve compound-specific transcriptional programs across runs and recover the correct compound at 48.6% top-1 and 68.6% top-3 retrieval, with coherent gene programs visible for representative compounds.

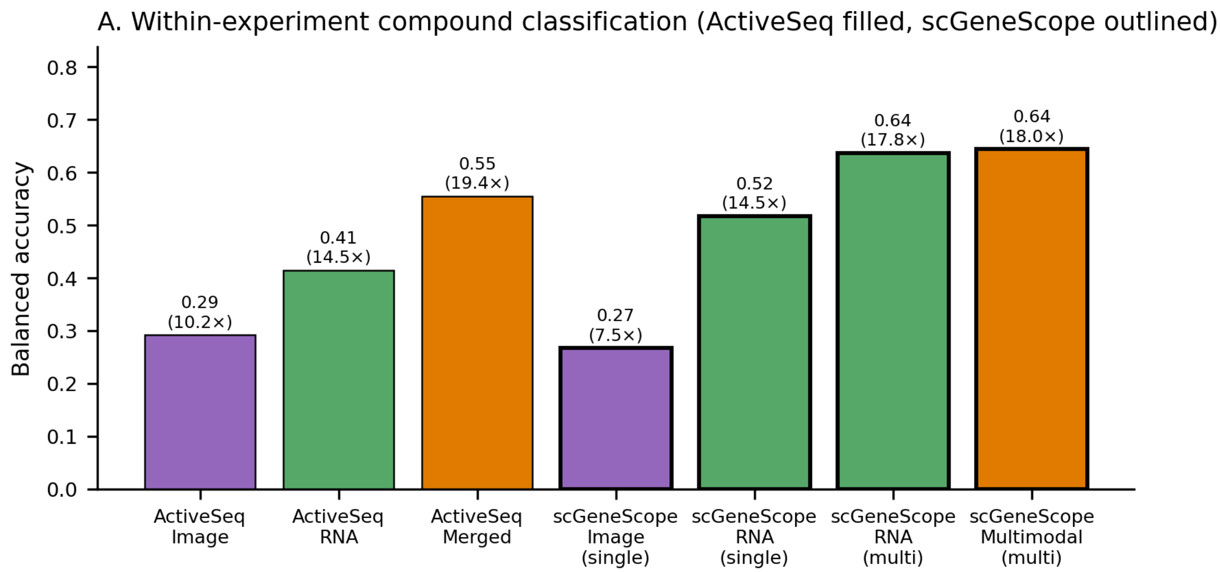
Position relative to scGeneScope and the broader phenotypic-screening landscape

Three reference points anchor the platform comparison. The first is Cell Painting and the imaging-only ML platforms built on it (Recursion's RxRx public datasets, Insitro's morphology programs, the Broad JUMP consortium). These systems lead on raw image throughput by an order of magnitude or more and have produced reusable embedding spaces, at the cost of biological readout depth. The second is treatment-matched multimodal benchmarks, of which scGeneScope is the strongest current public example: approximately 716,767 single-cell images and 627,704 single-cell RNA profiles across 28 treatments in U2-OS cells, paired at the level of replicate populations rather than the same well [4]. The third is the perturbational scRNA-seq line (Perturb-seq, Tahoe-100M, scPerturb), which provides the deepest available cell-state readout but does not include matched imaging.

Z-Screen / ActiveSeq combines all three properties: the imaging pass operates at Cell Painting scale, the sequencing pass is run inside the same micro-well as the imaging, and chemistry is delivered through one-bead-one-compound combinatorial libraries so every well carries full chemistry provenance. The cost of that design is dataset size: the present pilot comprises 11,511 control-panel matched wells across two experiments at 5 to 10 cells per well, equivalent to approximately 60,000 to 110,000 cells in the control panel, or roughly seven to twelve times fewer total cells profiled than scGeneScope, on a simpler imaging modality. Z-Screen reads each well as a pseudobulk rather than at single-cell resolution, so the unit of independent observation is the well, not the cell.

ActiveSeq trails the strongest scGeneScope multimodal reference on raw within-experiment balanced accuracy, consistent with the scale and imaging-richness gap. The more informative comparison is fold over random-chance performance, which normalizes for the different label spaces (35 versus 28 classes). On that scale, pooled ActiveSeq RNA reaches 14.5 times chance, identical to the scGeneScope single-profile RNA value of 14.5 times chance. Pooled ActiveSeq image reaches 10.2 times chance, above the scGeneScope single-profile image value of 7.5 times chance. Pooled ActiveSeq merged features reach 19.4 times chance, slightly above the scGeneScope multimodal multiprofile reference at 18.0 times chance, and the strongest ActiveSeq run (ACT010) reaches 21.3 times chance for merged features.

Once task difficulty is normalized, a small same-well pilot with approximately an order of magnitude fewer total cells profiled and a simpler imaging modality reaches the same range as the strongest public multimodal benchmark, while preserving a tighter same-well pairing unit and exposing recoverable image-side value through a calibrated latent. The pilot uses two experiments and one cell line on a platform designed to scale by orders of magnitude beyond the present footprint, so the result is directional and motivates collecting substantially more matched data rather than serving as a final benchmark claim.



B. Context for interpreting the gap

Aspect	**ActiveSeq (Z-Screen)**	**scGeneScope**
Single-cell scale	~60-110k cells (5-10 × 11.5k wells)	~717k images, ~628k RNA
Label space	35 compounds	28 treatments
Pairing	Same matched micro-well	Treatment-matched replicate
Imaging	Brightfield + calibrated latent	5-ch Cell Painting + ViT
Pooled image	29.2% (10.2× chance)	26.7% single (7.5×)
Pooled RNA	41.4% (14.5× chance)	51.7% single (14.5×)
Pooled multimodal	55.5% (19.4×)	64.4% multi (18.0×)
Best ActiveSeq run	ACT010 merged 60.8% (21.3×)	n/a
Takeaway	Smaller pilot, tighter pairing; competitive	Larger public benchmark on raw accuracy

Figure 6. Within-experiment compound classification for ActiveSeq versus the published scGeneScope reference, annotated with fold over random-chance performance, plus a context table summarizing the design differences that make the two benchmarks complementary rather than directly interchangeable.

ZEL026 dummy-bead wells perform as a negative-control QC

The ZEL026 dummy-bead wells contain no compound and were retained as a negative-control assay-chemistry stress test rather than as a discovery result. They behaved as expected. Median UMIs dropped from 5,167 to 342 in ACT010 and from 4,750 to 293 in ACT011; detected genes dropped from 2,826 to 301 and from 2,463 to 248 respectively. The collapse is operationally useful: the failure mode is visible in the readout itself and does not contaminate the 35-compound biological analyses. The QC distribution is reported in Supplementary Figure 1.

Discussion

The dataset is small by design, and the questions it answers are foundational. ActiveSeq produces stable compound-level multimodal structure inside the Z-Screen microchip. The RNA branch is strong within and across runs. After task-aware calibration, the image branch carries enough aligned signal to add value at present scale rather than requiring a much larger collection effort. Same-well pairing produces an additive gain on the most practical benchmark evaluated here: a merged representation outperforms RNA alone on held-out compound classification, and the gain extends to all 35 compounds in pooled analysis. Adding matched transcriptomics to imaging in a 50,000-well one-bead-one-compound screen therefore produces measurable operational value rather than only a richer descriptive readout.

The competitive context is straightforward. Imaging-only ML platforms compete on imaging scale and library coverage; transcriptomic platforms compete on cell-state readout depth per compound. Both axes are necessary to address mechanism questions, and neither is sufficient on its own. Z-Screen is engineered to do both inside the same micro-well, a design point not occupied by current public datasets. The pilot reported here is small by intent, and the result is that the design point produces a broad gain across compounds rather than one concentrated in a few easy classes.

For a discovery team, ActiveSeq turns the Z-Screen microchip into a same-well multimodal phenotype engine: imaging provides breadth, transcriptomics provides mechanism, and the two are coupled at the level of the same multicellular colony rather than at treatment-matched populations. That coupling is the property most likely to scale into a primary-screening readout, and it is not provided by current public benchmarks.

The natural next experiments follow directly: enlarge the control panel, broaden cell-line coverage, continue improving imaging consistency to strengthen the morphology contribution, and continue collecting matched same-well pairs at the throughput the chip is designed to support. Each investment compounds a signal already visible in the present dataset.

Methods

Dataset

The analysis uses the canonical Z-Screen ActiveSeq workspace under `data/ZScreen_Canonical_Dataset/`. Each ActiveSeq well is annotated with an experiment id (ACT010 or ACT011), a global well id, a compound assignment, an image embedding, and an RNA latent. Gene-level pseudobulk counts come from the matched H5AD object that backs the same-well RNA pipeline, repaired as documented in `DATA_REPAIR_REPORT.md`. The scGeneScope comparison uses published benchmark values transcribed from OpenReview Table 1 into `scgenescope_reference.csv`.

Per-well image embedding calibration

The original per-well image embedding was produced by a frozen pretrained vision model on raw imaging tiles. That representation captures generic visual structure but does not necessarily align compound identity with the principal directions of variance: experiment-of-origin, plate position, and microscopy artifacts can dominate compound signal in the raw geometry. Direct held-out compound classification on that embedding therefore tests the representation as much as the biology.

A 32-dimensional control-panel-calibrated latent was learned from the original embedding by a lightweight domain-adversarial network trained on the matched control panel. The objective rewards compound discrimination and penalizes experiment-of-origin discrimination, which pushes compound information onto the principal directions of variance without retraining a vision model from scratch. Calibration was learned on the same control-panel domain studied here, so the calibrated-latent results should be read as an internal representation refinement; broader-library and additional-cell-line generalization is the next test.

Matched-well preprocessing

Imaging detections were aggregated by experiment_id and gwid so each matched well contributed one image vector. Wells with an assigned control name were treated as the 35-compound panel; wells without an assigned control name were the ZEL026 dummy-bead wells used for QC.

Reproducibility and transfer analyses

Within-run reproducibility was measured by repeated split-half pseudo-replication within each compound (cosine similarity between pseudo-replicate centroids after standardization within each matrix). Cross-run transfer used compound centroids learned in ACT010 and queried in ACT011.

Cross-modal geometry

For each run, pairwise distances among compound centroids were computed separately in image and RNA space. Agreement between distance matrices was summarized with Spearman correlation over upper-triangular entries, which does not assume the two scales are matched. Cross-run effect-size stability used Spearman correlation over per-compound effect magnitudes.

Supervised multimodal classification

Held-out classification used a 50:50 compound-stratified split, either pooled across runs or fit within each run. Linear discriminant analysis (eigen solver, automatic shrinkage) was trained on calibrated image features alone, RNA features alone, or their raw concatenation. LDA was used in preference to a deep classifier so that any reported gain is traceable to a linear function of the input features. Performance was summarized with overall accuracy, balanced accuracy, and per-compound recall.

Same-well representation learning

A lightweight CLIP-style dual encoder with a symmetric contrastive objective was trained on same-well pairs from the training split, evaluated by nearest-centroid compound retrieval and by exact held-out pair retrieval. The contrastive objective did not exceed naive concatenation in this dataset, indicating that most of the recoverable compound-identity signal had already been exposed by the calibration step.

Gene-level pseudobulk analysis

Raw counts were summed per experiment and control compound, converted to log-CPM, and centered by subtracting the experiment-wide mean signature. Cross-run retrieval used correlation similarity between centered signatures. Per-cell differential expression was not performed because each well averages 5 to 10 cells; pseudobulk plus centering is the unit at which the signal is robust at present depth.

External benchmark context

Within-run supervised metrics from `supervised_multimodal_metrics.csv` were compared to scGeneScope within-experiment values transcribed from OpenReview Table 1 into `scgenescope_reference.csv`. Because the label spaces differ (35 vs 28), the comparison reports both raw balanced accuracy and fold over random-chance performance.

Limitations

The pilot is intentionally narrow: one cell line (HEK293), 35 control compounds, and two matched experiments. The calibrated image latent was learned on the same control panel studied here; broader libraries and additional cell lines are needed to evaluate calibration generalization. Exact same-well cross-modal retrieval remains modest in absolute terms, as expected in a low-pass multicellular assay. Gene-level signatures are summarized through pseudobulk aggregation rather than full differential-expression modeling with biological replicates. These caveats accompany rather than offset the central conclusion: same-well matched imaging and transcriptomics in the Z-Screen microchip support reproducible compound-resolved multimodal structure, and same-well pairing produces an additive gain on held-out classification.

Data availability

All data tables, derived latent representations, manuscript figures, and analysis inputs needed to reproduce this manuscript are organized in this repository under `paper1/` and `data/ZScreen_Canonical_Dataset/`. A persistent-archive deposition with an assigned DOI will accompany the corresponding preprint posting.

Code availability

Analysis and figure-generation scripts are in `paper1/scripts/`. Package-level dependencies are in the repository root `requirements.txt`. Tested with Python 3.14.3 on Windows.

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Combinatorial chemistry yields a learnable transcriptomic design map in Z-Screen

Zafrens, Inc.

Abstract

A phenotypic screen produces substantially more value when its output functions as a design map (a function from chemistry to reproducible biological program that a discovery team can interrogate, reuse, and extend) rather than as a static catalog of measurements. The Z-Screen platform uses one-bead-one-compound combinatorial libraries, profiled in 50,000-well microchips by paired imaging and mRNA sequencing at 5 to 10 cells per microwell. Every well therefore carries full chemistry provenance and a structured transcriptomic readout. Across 615,793 repaired RNA profiles spanning 12 combinatorial libraries and 4 cell lines (615,721 with valid scVI latent coordinates), the canonical dataset already supports the design-map transition at pilot scale and leaves substantial headroom for expansion.

The assay floor is high across the main systems. Median split-half cosine for control compounds is 0.993 in A549, 0.991 in H1650, 0.993 in HEK293, and 0.968 in THP1. On held-out tuples within a library, building-block models substantially outperform centroid baselines. The strongest case is ZEL024 in HEK293, where a building-block plus pair model reaches median cosine 0.674 against a 0.397 centroid baseline (33.2% reduction in mean squared error). ZEL031 in THP1 follows at cosine 0.833 vs 0.783 (21.3% error reduction). Chemistry-only descriptors carry predictive signal as well: in ZEL024 / HEK293, a combined building-block plus chemistry-embedding model reaches cosine 0.621 with a 29.3% error reduction, the highest gain among any chemistry-only configuration evaluated.

The dataset also produces direct discovery-style signals. Background-calibrated mimicry against per-cell-line cross-control similarity gives the cleanest hit in ZEL024 / HEK293, where a library tuple matches the internal reference ZF-104 at cosine 0.901 against a HEK293 cross-control background of median 0.200 and 99th percentile 0.931. ZEL031 / THP1 yields tuples reaching cosine 0.942 to sorafenib, 0.926 to MZ1, and 0.918 to STC-15. THP1 has a higher cross-control background (median 0.589, 99th percentile 0.972), so those matches sit in the upper tail of plausible transcriptomic neighborhoods rather than uniquely on-mechanism territory. Building-block effects also recur across cell lines at chemistry-resolved positions (ZEL024 bb3 median 0.602 between HEK293 and H1650; ZEL031 bb0 median 0.355 and bb1 median 0.526 between A549 and THP1), with the strongest building blocks reaching cosine 0.87. Imaging adds an additional but smaller benefit in a simple additive benchmark, with image plus RNA reaching 0.540 balanced accuracy in ZEL024 controls

against 0.524 for RNA alone and 0.213 for image alone.

In aggregate, these results show that Z-Screen behaves as a chemistry-to-transcriptome modeling platform on data from a small number of pilot experiments. The core asset is the RNA branch and the chemical provenance behind it; imaging is directionally helpful and is expected to compound as image consistency improves; the platform is engineered to scale by orders of magnitude beyond the present dataset, so the next investment compounds a signal already present rather than testing whether one exists.

Significance

Most screening campaigns terminate as a ranked list of wells. The present analysis demonstrates that Z-Screen supports a different output: a design map. Because every Z-Screen compound carries its building-block history and its measured cell-state response, the platform can learn which structural elements move biology in useful directions, which combinations are redundant, and which compounds mimic known reference states. The relevant outcome is not that one model exceeds one benchmark; it is that chemistry, transcriptomics, and known-control behavior align well enough to support prediction, triage, and follow-up design. For a discovery organization, this converts a screen from a one-time search into reusable institutional memory.

Introduction

Early-stage discovery requires throughput and biological resolution simultaneously, and existing platforms force a compromise. High-content imaging is scalable and information-rich but mechanistically thin (Cell Painting, Recursion's RxRx atlases, Insitro's morphology platforms, the Broad's JUMP consortium). Transcriptomic profiling provides closer access to underlying cell state at higher cost per compound (Perturb-seq [3], Tahoe-100M, scPerturb). Combining the two at scale on the same compounds has been difficult. Public efforts that have attempted this combination have reached treatment-matched scale rather than same-well: scGeneScope, the strongest current public benchmark, pairs imaging and RNA at the level of replicate populations exposed to the same compound, not the same biology [4].

The state-of-the-art chemistry-side platform is the Connectivity Map and LINCS L1000 program: approximately a million compound profiles in a deliberately reduced 1,000-gene transcriptional space, with bulk rather than single-cell readout [2]. L1000 set the standard for chemistry-to-transcriptome similarity search and remains the deepest public chemistry catalog by compound count. What L1000 does not provide is design information. A typical L1000 experiment is a single profiled compound with no internal structure to its identity. There is no built-in way to ask which specific structural element moved which biological program, because the dataset was built to compare compounds rather than to compose them.

Z-Screen was built around a different premise. A combinatorial chemistry screen produces substantially more value when every compound contributes a reusable biological program rather than a sparse binary label, and when each compound's identity is decomposable into building-block coordinates with full SMILES. The output then becomes a map from chemical composition to transcriptomic phenotype rather than a list of active compounds. With a stable map of that shape, a discovery team can rank unmeasured chemistry, identify building blocks that carry the strongest programs, flag compounds that mimic reference states, and reason about design rather than only about screening.

The analyses below characterize what the canonical dataset, itself assembled from a small number of pilot experiments, currently supports. Six concrete questions are addressed:

1. Are control compounds reproducible enough to support modeling?
2. Does building-block composition predict held-out transcriptomic states?
3. Does full chemical structure add predictive signal beyond building-block identity?
4. Do library tuples recover recognizable reference programs once cross-control similarity backgrounds are accounted for?
5. Do building-block effects recur across cell lines?
6. Does imaging contribute value in a restrained multimodal benchmark?

Each question is answerable on the present pilot. The aim is to establish what the current data already does well enough to justify a substantially larger campaign, with appropriate qualification of how much of the platform's full capacity has been used so far.

Results

Control profiles establish a stable assay floor across cell lines

The chemistry-prediction question is only meaningful if the assay itself is stable enough to support it. Split-half reproducibility for named controls is high across the main systems. Median control cosine is 0.993 in A549, 0.991 in H1650, 0.993 in HEK293, and 0.968 in THP1. THP1 sits below the others because its compound-control library coverage is sparser, not because the readout is weaker on monocytes.

These numbers leave clear headroom between technical reproducibility and current model performance, which is relevant when interpreting the chemistry models below: a model reaching cosine 0.674 retains room for improvement because the readout itself is highly consistent.

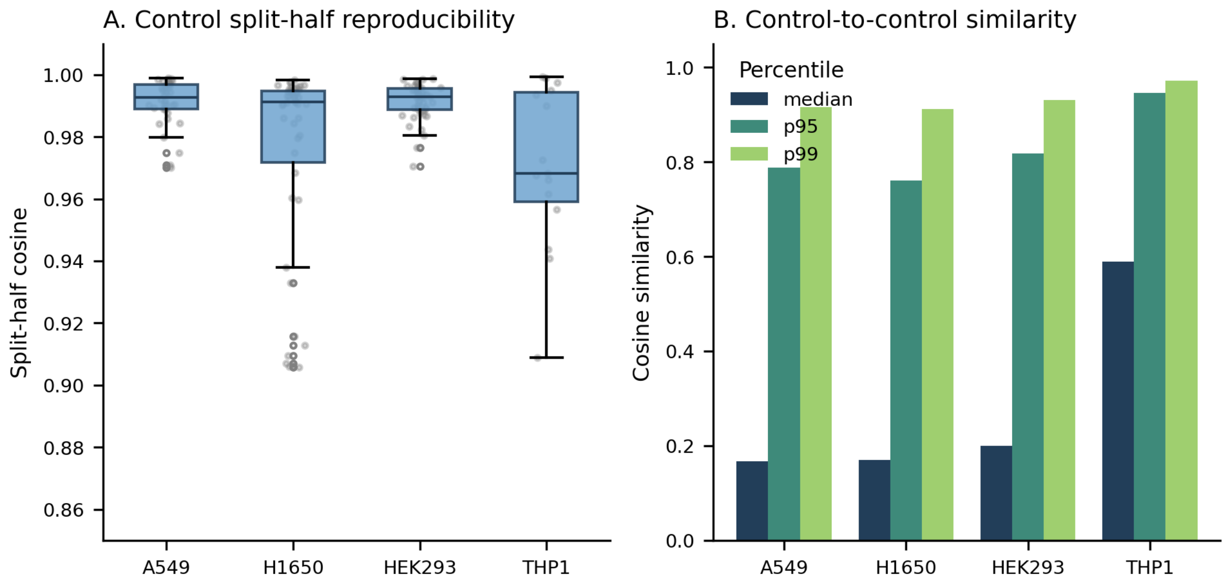


Figure 1. Named controls show high split-half reproducibility across all four cell lines, establishing a strong assay floor for downstream chemistry-to-phenotype modeling.

Building-block models predict held-out compound programs

The central modeling question is whether compositional chemistry carries enough signal to predict the transcriptomic state of a held-out library member. Across several library and cell-line systems, the answer is yes.

ZEL024 in HEK293 is the strongest case. A building-block plus pair model reaches median cosine 0.674 against a 0.397 centroid baseline, a 33.2% reduction in mean squared error. ZEL031 in THP1 follows at cosine 0.833 vs 0.783 (21.3% error reduction). ZEL031 in A549 and ZEL024 in H1650 also showed positive gains, although less pronounced.

The model does more than memorize whole compounds. It learns that particular chemical pieces and combinations of pieces carry recurring transcriptomic programs, which supports design-time reasoning about combinations not yet screened. Within the broader generalization-ladder framework laid out in companion paper3, this benchmark is interpolation within the shared building-block vocabulary of each library; the harder building-block holdout and scaffold-family transfer regimes are evaluated separately there.

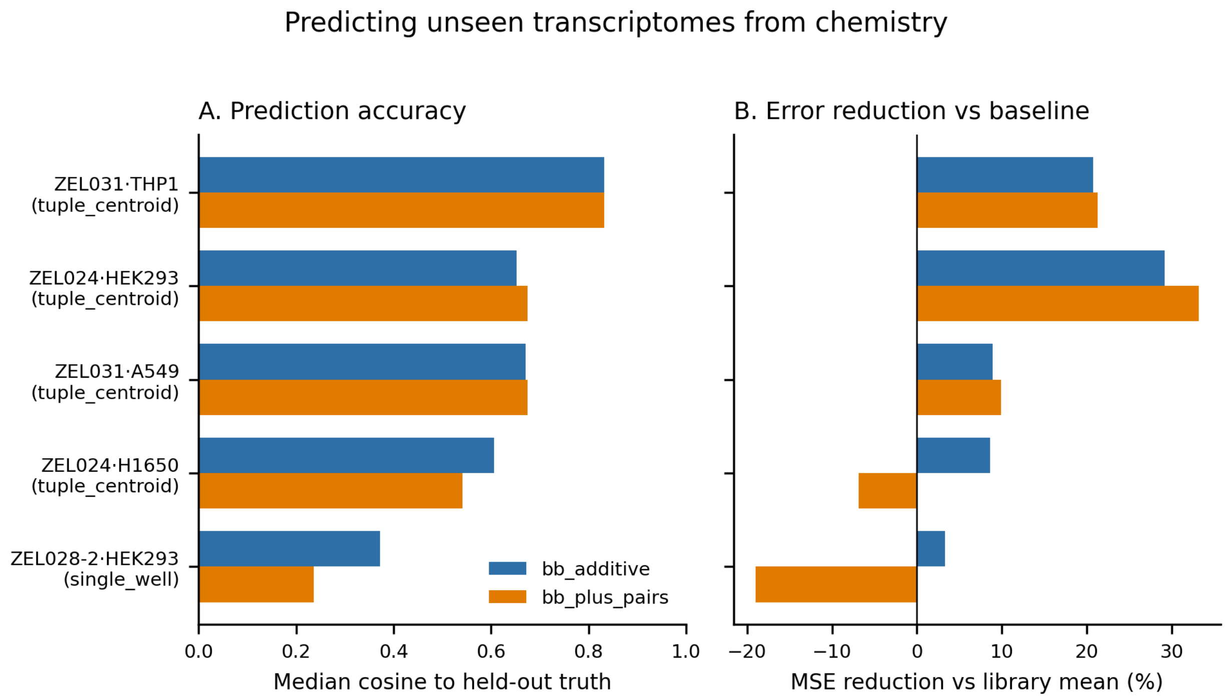


Figure 2. Building-block models improve prediction of held-out transcriptomic states across multiple library and cell-line systems, with the strongest gains in ZEL024 / HEK293 and ZEL031 / THP1.

Full chemical structure adds design-time predictive signal

Building-block identity is the strongest single signal in this dataset, and full structure adds further signal in the strongest ZEL024 system. In ZEL024 / HEK293, chemistry-only models were already predictive: a 256-dim chemistry embedding alone (a private random projection of ECFP4 fingerprints, see Methods) reached cosine 0.611 (27.8% error reduction), building-block identity alone reached 0.611 (27.5% error reduction), and combining the two reached 0.621 (29.3% error reduction).

This pattern is consistent with a future design loop. Building-block provenance reflects the combinatorial grammar of the library and remains valuable. Full structure carries additional information beyond those labels, so adding it improves prediction further. The combined representation is therefore the natural input for ranking unmeasured compounds by expected transcriptomic effect. Companion paper3 quantifies how much of the chemistry-only gain reflects interpolation within a library and how much survives stricter held-building-block evaluation.

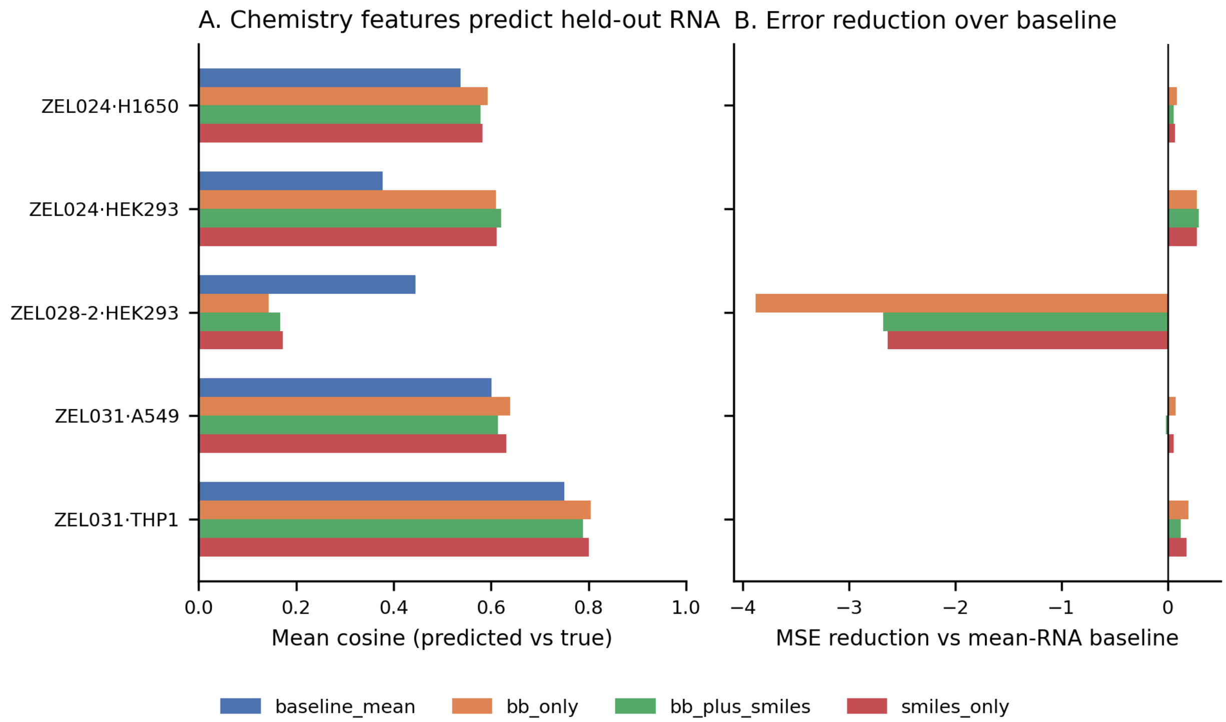


Figure 3. Chemistry-only models capture predictive signal, and the combination of building-block provenance with full-structure descriptors performs best in the strongest ZEL024 system.

Library tuples recover recognizable reference pharmacology states

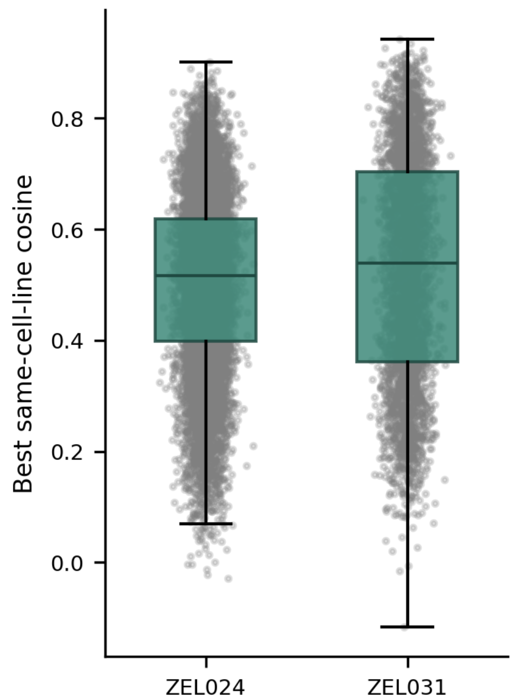
Control mimicry is the most direct discovery result the dataset produces: library tuples whose transcriptomic centroids land near the centroid of a known reference compound. The cleanest mimicry result by background-calibrated standards comes from ZEL024 / HEK293, where a library tuple matched the internal reference ZF-104 at cosine 0.901. The HEK293 between-control similarity background sits at median 0.200 with a 99th percentile of 0.931, which places a 0.901 match for an unrelated chemistry tuple in the upper tail of plausible transcriptomic neighborhoods rather than the bulk. Several additional ZEL024 / HEK293 tuples reached cosine 0.85 to 0.89 against published references such as ZEL029-25 and ZEL029-28.

Mimicry hits also appear in ZEL031 / THP1, with tuples reaching cosine 0.942 to sorafenib, 0.926 to MZ1, and 0.918 to STC-15. THP1 is useful for several pharmacology programs, but its cross-control similarity background is substantially higher (median 0.589, 99th percentile 0.972). The THP1 hits are biologically meaningful and populate the upper tail of plausible neighborhoods, but they should not be promoted as uniquely on-mechanism without orthogonal validation. The THP1 background is reported explicitly so the calibration is visible to readers.

The screen produces new chemistry that lands in recognizable biological neighborhoods, with the strongest defensible discovery result coming from a cell line where the cross-control background is low enough to support the mimicry claim without further qualification.

Library tuples vs nearest control compound

A. Tuples vs controls



B. Top control-mimic hits

Library	Cell line	Best control	Cosine	Repl
ZEL031	THP1	sorafenib	0.942	4
ZEL031	THP1	sorafenib	0.935	5
ZEL031	THP1	MZ1	0.926	5
ZEL031	THP1	MZ1	0.922	7
ZEL031	THP1	STC-15	0.918	4
ZEL031	THP1	STC-15	0.916	4
ZEL031	THP1	STC-15	0.916	6
ZEL031	THP1	sorafenib	0.914	4
ZEL031	THP1	STC-15	0.912	8
ZEL031	THP1	ZF-167	0.911	6
ZEL031	THP1	STC-15	0.910	7
ZEL031	THP1	STC-15	0.910	5

Figure 4. Multiple library tuples land near transcriptomic neighborhoods centered on known reference compounds, with the strongest background-calibrated match in HEK293 (ZF-104 at cosine 0.901, against a HEK293 99th-percentile cross-control background of 0.931) and additional THP1 hits to sorafenib, MZ1, and STC-15 in a system with a higher cross-control baseline.

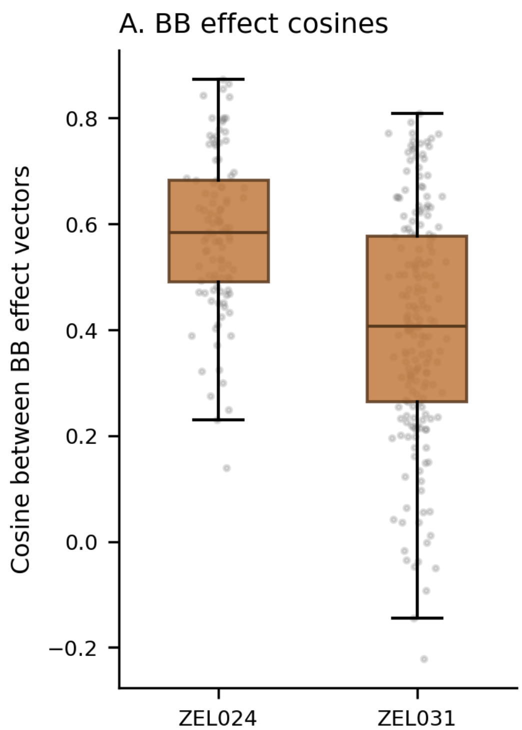
Building-block programs recur across cell lines

A reusable chemistry model is more valuable when its building-block programs survive a change in cellular context. Per-building-block transcriptomic programs were compared in matched library systems across paired cell lines. Results are reported by building-block position because positions with very few unique values (e.g. ZEL024 bb0, bb1, bb2) collapse hundreds-to-thousands of distinct chemistries into a single average and therefore measure library-wide rather than chemistry-resolved transfer. The chemistry-resolved positions are ZEL024 bb3 (83 building blocks, median 168 unique tuples averaged per signature) and ZEL031 bb0 / bb1 (a 2-position library, where each single-position signature still leaves the other position to vary across approximately 70 to 100 distinct partner chemistries).

Across these chemistry-resolved positions, building-block programs partially survive cell-context change. In ZEL024, the median cross-cell cosine at bb3 between HEK293 and H1650 was 0.602 (n = 83, IQR 0.51 to 0.69, strongest building blocks reaching 0.873). In ZEL031 between A549 and THP1, the median cross-cell cosine was 0.355 at bb0 (n = 105, IQR 0.23 to 0.48, strongest 0.747) and 0.526 at bb1 (n = 84, IQR 0.32 to 0.67, strongest 0.808).

These medians sit meaningfully below the within-cell reproducibility ceiling, so the result is not a claim of perfect transfer. The defensible interpretation is that many building blocks carry partially conserved biological directionality across cell types, which is the prerequisite for reusing chemistry across programs. The ZEL024 bb0 / bb1 / bb2 cross-cell results are not chemistry-resolved at this library architecture (each averaged signature reflects 900 to 7,000 distinct tuples) and are therefore reported as a diagnostic in `paper2/tables/bb_effect_consistency.csv` rather than as a headline claim. Companion paper4 extends the cross-cell question to the molecule and grouped-program level, where the transfer story strengthens substantially in chemistry-resolved settings.

Shared BB program consistency across cell lines



B. Top shared chemistry programs

Library	Pos	BB id	Cosine	N left	N right
ZEL024	bb3_	BB-0001548	0.873	1904	309
ZEL024	bb3_	BB-0001589	0.865	1407	190
ZEL024	bb3_	BB-0001491	0.856	1903	304
ZEL024	bb3_	BB-0001366	0.842	1715	198
ZEL024	bb3_	BB-0001421	0.840	1630	210
ZEL031	bb1_	BB-0001076	0.808	203	252
ZEL024	bb2_	BB-0001916	0.800	79081	11362
ZEL024	bb2_	BB-0001922	0.800	74659	10183
ZEL024	bb3_	BB-0001601	0.800	1797	294
ZEL024	bb3_	BB-0001405	0.798	1792	347
ZEL024	bb3_	BB-0001410	0.794	1778	194
ZEL031	bb1_	BB-0000950	0.792	275	276

Figure 5. Building-block effects recur across cell lines, with clear positive median transfer and a subset of building blocks showing strong cross-context conservation.

Imaging adds signal in a simple multimodal benchmark

The current platform story is RNA-first, and the image branch already adds value when framed conservatively. In the strongest additive multimodal benchmark, based on ZEL024 controls, image-only classification reached 0.213 balanced accuracy, RNA-only reached 0.524, and the combined representation reached 0.540. The smaller ZEL031 recurrent-tuple benchmark followed the same ordering at lower absolute values: 0.127 for image, 0.146 for RNA, and 0.180 for the combined model. Companion paper1 shows a substantially larger image contribution in the same-well ActiveSeq matched-readout setting after task-aware calibration of the per-well image latent, indicating that the modest gain reported here reflects current image-pipeline maturity rather than a ceiling on what same-well multimodality can achieve in Z-Screen.

The multimodal claim at the present pilot is measured. Imaging is not the lead modality today, the combined representation is consistently better than either single view in a practical classification task, and continued investment in the imaging workflow should compound that gain.

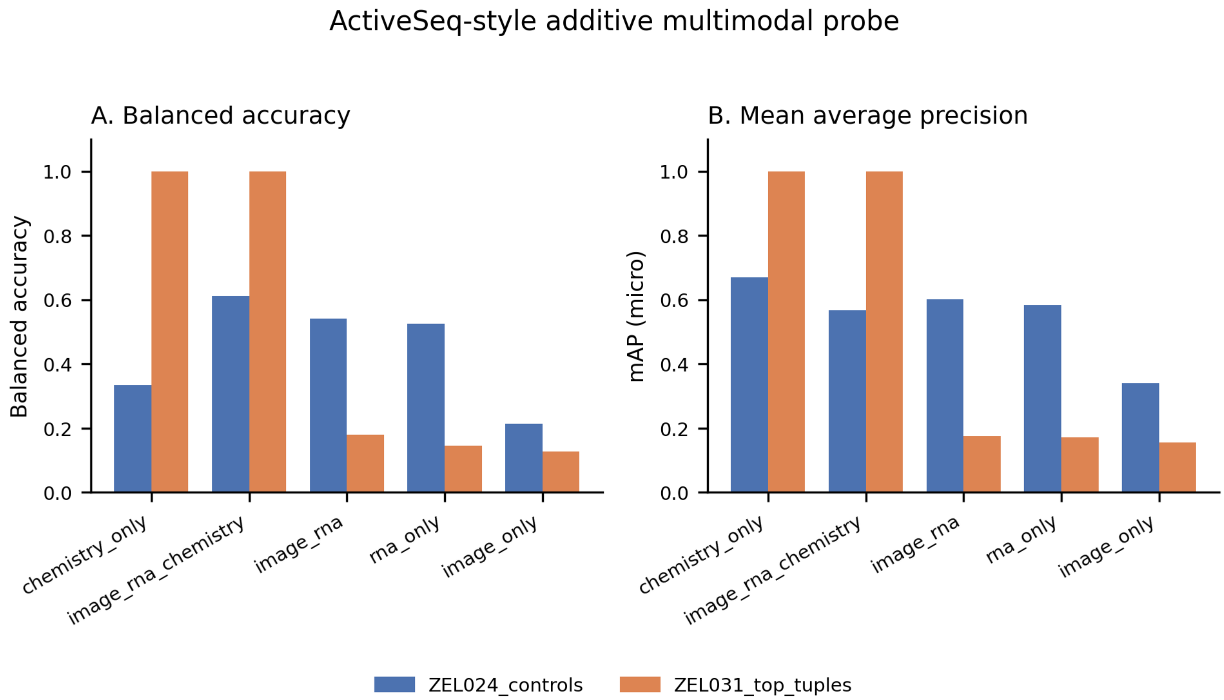


Figure 6. In a simple additive benchmark, image plus RNA outperforms either single modality alone, while RNA remains the dominant branch in the current dataset.

External benchmark context: position relative to Tahoe-100M, scGeneScope, and L1000

For external positioning, the present pilot is most usefully compared with four public reference points. Tahoe-100M is the scale anchor for perturbational transcriptomics, with over 95.6 million RNA profiles across 50 cancer cell lines and approximately a thousand small-molecule perturbations [3]; it does not include matched imaging and was not built around a combinatorial chemistry framework. scGeneScope is the closest public multimodal benchmark, with 627,704 RNA profiles and 716,767 images across 28 chemical treatments in U2-OS cells, paired at the level of replicate populations rather than the same well [4]. LINCS L1000 is the deepest public chemistry catalog by compound count, with bulk transcriptional profiles for tens of thousands of compounds in a 1,000-gene reduced expression space [2]. Cell Painting is the standard for ML-on-imaging, with the JUMP consortium and Recursion-derived public sets providing single-cell or whole-well morphological profiles at high library coverage [5].

The current Z-Screen canonical dataset contributes 615,793 repaired RNA profiles spanning 12 combinatorial libraries and 4 cell lines, with chemistry-by-design provenance, plus matched per-well imaging on substantial subsets (companion paper1). Direct comparison places Z-Screen on a different axis from any of those references. Tahoe-100M is currently larger by a wide margin on raw RNA scale. scGeneScope remains the strongest public multimodal benchmark on raw treatment-matched accuracy. L1000 leads on raw small-molecule compound count. Cell Painting leads on imaging throughput. Z-Screen, even as a small early pilot, is the most chemistry-design-relevant atlas in the current public set: every well carries full building-block and SMILES provenance, the readout produces a learnable chemistry-to-transcriptome map at present scale, and the platform is engineered to scale by orders of magnitude. The contribution is design relevance and growth headroom, not raw atlas size.

The positioning has economic implications. A discovery team using L1000 can search compound space against a transcriptomic reference but cannot decompose a hit into the structural pieces that drove the response. A team using Cell Painting can scale to tens of millions of wells but cannot recover mechanism from morphology alone. Z-Screen is the only platform we know of that runs combinatorial chemistry, low-pass mRNA sequencing, and matched imaging in the same micro-well (each well containing 5 to 10 cells, profiled as a pseudobulk rather than at single-cell resolution), with chemistry decomposed into building-block coordinates. That design point converts a screen from a search tool into a model of structure-activity at the transcriptional level.

To anchor that positioning quantitatively, compound and cell-line overlap was computed between the Z-Screen canonical control panel and the two most directly comparable atlases, and that overlap was used to run an explicit cross-platform signature concordance test against LINCS L1000 (`Benchmarking/` in this repository). Of 47 named Z-Screen control compounds, 14 (29.8%) appear in the LINCS L1000 CMap LINCS 2020 build (palbociclib, sorafenib, crizotinib, ceritinib, rucaparib, veliparib, lapatinib, cobimetinib, GSK126, CP-673451, AZD-3463, BMS-509744, BGT226, ZM-336372). Three of four Z-Screen cell lines (HEK293, A549, THP1) are present in LINCS L1000 with sample coverage; only H1650 is absent. Tahoe-100M's drug metadata covers 4 of the 47 named compounds, with A549 as the only Z-Screen cell line in Tahoe-100M's panel.

For the L1000 concordance test, Z-Screen rank signatures were restricted to the 12,328-gene LINCS-visible universe so the two platforms are scored in the same gene space, then Spearman rank correlation was computed between Z-Screen up/down rank signatures and LINCS L1000 level-5 consensus signatures (NCBI GEO GSE70138 and GSE92742) for each (compound, cell line) pair where both platforms have adequate coverage. Across 11 such pairs (10 in A549, 1 in HEK293), all 11 produced positive cross-platform Spearman correlation. Median rho was 0.490, mean 0.393, and 8 of 11 pairs cleared rho = 0.2. Five compounds reached empirical p 0.048 against a same-cell-line unmatched-compound permutation null: ZM-336372 (rho = 0.625, p = 0.022), rucaparib (rho = 0.530, p = 0.023), sorafenib in A549 (rho = 0.505, p = 0.023), veliparib (rho = 0.490, p = 0.046), and crizotinib (rho = 0.679, p = 0.048). The cell-line-pooled comparison in HEK293, where sorafenib in HEK293T reached rho = 0.597, separated matched from unmatched compound pairings at one-sided Mann-Whitney p = 0.009. Two compounds (palbociclib at rho = 0.182, GSK126 at rho = 0.046) fall in the bulk of the unmatched distribution; the GSK126 result is consistent with the documented slow transcriptional onset of EZH2 inhibition under LINCS's 6 h treatment window. Z-Screen rank signatures therefore recover compound-specific transcriptional structure that an independent reference platform reports, in a gene-restricted concordance test that uses no platform-specific calibration.

Discussion

Z-Screen behaves as a learnable chemistry-to-transcriptome platform on data from a small number of pilot experiments. Control compounds are stable, held-out transcriptomic states can be predicted from chemistry, reference-like biological programs recur in the libraries (with a strong background-calibrated mimicry hit in HEK293), building-block programs partially transfer across cell lines, and the strongest systems show meaningful gains from combining building-block provenance with full structural descriptors.

There is a clear hierarchy in the current evidence. RNA carries the strongest predictive signal today. Imaging is directionally positive in this analysis and considerably stronger when calibrated for same-well matched readout (companion paper1). The imaging branch is best framed as an additive layer on a working chemistry-to-RNA model at this stage, with room to grow as the imaging workflow improves.

The most informative next experiments follow directly. The first is greater depth in sparse libraries, especially systems that underperform when coverage per building block is low. The second is a prospective cross-library benchmark that asks whether a model trained on one scaffold family can predict transcriptomic programs in another; companion paper3 sets this up explicitly. The third is continued improvement in image consistency so that morphology can contribute more strongly to the readout. Each of these compounds a signal already present, on a screening platform engineered to scale by orders of magnitude beyond the current dataset.

For a discovery team, combinatorial chemistry inside a 50,000-well one-bead-one-compound microchip with low-pass mRNA sequencing produces a structured chemistry-to-transcriptome map at compound resolution rather than a sparse hit catalog. The economic implication is that a screen run on this platform leaves behind a reusable model rather than only a list of wells.

Methods

Dataset and preprocessing

The analysis uses the repaired canonical Z-Screen workspace, which contains RNA profiles, scVI latent coordinates, chemistry annotations, and selected image-derived features across 12 combinatorial libraries and 4 cell lines. Named controls were retained for reproducibility and mimicry analyses. Transcriptomic models were evaluated on library and cell-line subsets with sufficient compound coverage. The 72 ZIC004 / A549 rows without valid scVI latent coordinates were retained in the RNA count table but excluded from latent-space modeling; this is the source of the small mismatch between `canonical_obs.parquet` (615,793 rows) and `canonical_scvi_latents.parquet` (615,721 rows with valid coordinates) noted in the package README.

scVI latent space rather than gene-level expression

Most analyses operate on a 32-dimensional scVI latent space rather than full gene-level expression vectors, for two reasons. First, low-pass per-well mRNA-seq (each well is a pseudobulk of 5 to 10 cells) is dominated by sampling noise in any single gene; scVI's variational latent absorbs that noise into shared structure across observations. Second, the latent is comparable across cell types to a degree that raw expression is not, which is the property the cross-cell transfer experiments in companion paper4 require. Where gene-level signal is the appropriate unit (control mimicry, gene-program inspection in companion paper1) centered pseudobulk log-CPM is used directly.

Control reproducibility

Control split-half reproducibility was computed from repeated random partitions of wells belonging to the same reference compound. Similarity was summarized with cosine similarity between pseudo-replicate centroids in scVI latent space. Repeated partitioning rather than a single split was used to keep the variance of the metric low.

Building-block predictive modeling

Held-out tuple analyses fit additive and additive-plus-pair building-block models to transcriptomic centroids and evaluated them on compounds excluded from training by tuple identity. The choice of additive plus pair (rather than a high-order interaction model) reflects the present library coverage: pair-level interactions are well sampled in ZEL024, while higher-order terms run into combinatorial explosion at present scale. Performance was summarized with median cosine similarity to the measured transcriptomic target and mean squared error reduction relative to a centroid baseline. Companion paper3 extends this analysis to the harder building-block holdout and cross-library transfer regimes.

Chemistry-to-RNA prediction

Chemistry-only benchmarks compared models using a 256-dim per-compound chemistry embedding, building-block one-hot identities, or their concatenation. The chemistry embedding is a private fixed-seed Gaussian random projection from R^2048 to R^256 applied to per-compound 2048-bit ECFP4 (Morgan radius 2) fingerprints, then L2-normalized; this representation preserves Tanimoto-like geometry under cosine similarity (Johnson-Lindenstrauss) while being mathematically irreversible to substructure information without the projection matrix, which is held privately in the NDA bundle. A fixed deterministic projection was used in preference to a learned chemistry encoder because the projection is reproducible across runs, requires no separate training step, and decouples the published chemistry features from the underlying SMILES strings. Outputs were scored against measured transcriptomic centroids using cosine similarity and mean squared error reduction relative to a centroid baseline.

Control mimicry, cross-control normalization, and cross-cell building-block analysis

Control mimicry identified library tuples whose transcriptomic centroids were most similar to named reference compounds within the same library and cell line. Mimicry hits were calibrated against the per-cell-line cross-control similarity background (median, 95th percentile, 99th percentile cosines among unrelated reference compounds in the same cell line), so that "high" similarity is interpreted relative to the cell-line baseline rather than against zero. Cross-cell building-block consistency compared the direction of per-building-block transcriptomic programs across paired cell lines within the same library.

Additive multimodal benchmark

The multimodal probe followed a conservative additive framing: image features, RNA features, and their concatenation were evaluated on held-out class prediction tasks constructed from the available image-linked data. Performance was summarized with balanced accuracy and mean average precision. The additive frame was chosen because it produces a result traceable to a specific feature axis; cross-attention or contrastive multimodal models can in principle do better, but the gain is harder to attribute to one modality.

External benchmark context

Tahoe-100M, scGeneScope, L1000, and Z-Screen statistics were assembled into a single state-of-the-art comparison table covering RNA scale, image scale, cell-line coverage, perturbation diversity, pairing structure, and a representative result per dataset. The comparison was used only for positioning, not to claim parity on raw multimodal balanced accuracy.

Limitations

The current analysis is based on a small number of pilot experiments rather than a fully saturated atlas. RNA carries the strongest signal today; the image branch is directionally positive and depends on derived feature artifacts rather than raw microscopy in this shareable package. Control mimicry is calibrated against same-cell-line cross-control backgrounds, but the strongest discovery-style hits still require orthogonal experimental validation before they should be treated as confirmed mechanisms of action. Cross-library and prospective scaffold-family transfer are set up by the present results but are not claimed as solved here.

Data availability

Data tables, derived image features, manuscript figures, and analysis inputs needed to reproduce this manuscript are organized in this repository under `paper2/`, `data/ZScreen_Canonical_Dataset/`, and `data/paper2_artifacts/`. Raw microscopy images are not included in the shareable bundle; the paper2 image-linked analyses use the derived parquet feature files documented in the package README. A persistent-archive deposition with an assigned DOI will accompany the corresponding preprint posting.

Code availability

Analysis and figure-generation scripts are in `paper2/scripts/`. Package-level dependencies are in the repository root `requirements.txt`. Tested with Python 3.14.3 on Windows.

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A generalization ladder for chemistry-to-phenotype prediction in combinatorial screening

Zafrens, Inc.

Abstract

Machine learning claims in drug discovery depend on which version of generalization is being tested. Predicting a new combination of familiar chemical pieces is one problem. Predicting a compound built from unseen pieces is harder. Predicting an entirely new scaffold family is harder still. The recent BELKA literature on DNA-encoded libraries quantified that distinction for binary binder labels: ligand-only models perform well when the test chemistry is close to the training chemistry, and weaken sharply once both building blocks and scaffold change [1,2]. The natural follow-up question is whether dense phenotypic readouts, where every screened compound contributes a structured biological state rather than a sparse hit/non-hit label, perform better than DEL platforms on the harder versions of generalization. We test that hypothesis directly in Z-Screen, a 50,000-well one-bead-one-compound microchip platform that profiles each well by imaging and mRNA sequencing at 5 to 10 cells per well, across an explicit five-rung ladder of progressively harder held-chemistry evaluations.

The strongest evidence for genuine extrapolation comes from ZEL031 in THP1 cells. Under strict multi-axis holdout, where every substantive building block in the test compound is unseen during training, the model retains 0.765 cosine to the held-out transcriptomic state, exceeds a chemistry-embedding nearest-neighbor retrieval baseline by 0.168 cosine, wins on 90% of test compounds, and stays positive in all 10 random draws under paired bootstrap intervals at a median nearest-training cosine of 0.642 in the public chem-embedding space (see Methods). Single-position building-block holdouts in the same system give consistent results, with the model exceeding retrieval by 0.15 cosine on average and 89% to 91% per-compound win rates on test sets of more than 700 compounds whose nearest training neighbors have embedding cosine in the 0.78 to 0.80 range.

ZEL024 in HEK293 produces high quality on a different problem. The library is sufficiently saturated (12 by 7 by 2 by 83 building-block grid, 13,769 observed compounds out of a 13,944 possible-tuple ceiling) that nominal held-building-block splits still leave test compounds with near-duplicate training neighbors at median nearest-training cosine 0.88 to 0.99. Under that condition, ZEL024 / HEK293 delivers high-quality matrix completion within a dense combinatorial grid, a useful capability for prioritizing chemistry within an explored design space, while ZEL031 / THP1 delivers genuine out-of-vocabulary extrapolation. The hardest test, cross-library scaffold-family transfer in the same cell line, remains the next decisive experiment rather than a solved problem on the

current data: across three workable train-test pairs, the best absolute cosine to the held-out library phenotype was 0.194 and the largest gain over nearest-neighbor retrieval was 0.096.

In aggregate, these results show that dense phenotypic readouts already extend chemistry generalization beyond simple retrieval in the right systems, that the platform delivers either genuine extrapolation or high-quality matrix completion depending on library design, and that the cross-library scaffold-family transfer rung is now defined precisely enough to support a single prospective experiment that would convert the comparison from retrospective to definitive.

Significance

Machine learning in early discovery is only worth its cost if the model still helps when the chemistry changes. This paper separates the easy version of that claim from the hard one. In one Z-Screen system, the model fills in missing entries inside a saturated combinatorial grid, which is genuinely useful for prioritizing chemistry within a design space. In another, the model continues to outperform nearest-neighbor retrieval after multiple building-block axes have been held out, rather than copying close analogs. The contrast is the central result: the platform is beginning to show, with calibrated rigor, where it can interpolate, where it can extrapolate, and which experiment should come next.

Introduction

DNA-encoded library (DEL) platforms transformed the economics of binder discovery by reading every well of a million-compound library against an immobilized target [3,4]. They have a structural ceiling: the readout is a binary binder label dominated by a 99% non-binder majority class, and the training distribution is concentrated near the few hits that DEL selection produces. Recent BELKA work in this space quantified that distinction for ML: ligand-only models trained on DEL labels performed well when the test chemistry was close to the training chemistry, then weakened sharply when both building blocks and scaffold changed [1]. Several follow-up studies arrived at the same conclusion through different routes [2,5,6,7].

The literature on DEL machine learning has not settled the question for dense phenotypic screens, where the supervision problem is fundamentally different. A screen that reads out a transcriptional state for every compound has no 99% majority class of non-hits to discard. Every training example contributes a structured endpoint. The natural hypothesis is that a richer output should support more robust chemistry generalization than a binary binder label. Testing that hypothesis cleanly requires more than a single train/test split; it requires a ladder of progressively harder held-chemistry evaluations, with explicit baselines at each rung that distinguish nearest-neighbor retrieval from genuine extrapolation.

The ladder is defined and run in Z-Screen.

- **L1: unseen tuple within a shared building-block vocabulary.** The easiest rung: every individual building block has been seen, only the full combination is held out.
- **L2: held-building-block at one position.** The model has never seen one specific building-block identity at one position during training.

- **L3: strict multi-axis holdout.** Every substantive building-block axis in the test compound is unseen during training. This is the hardest within-library rung.
- **L4: chemical-neighborhood holdout.** Held-out compounds are chemical-neighborhood clusters, seeded by greedy farthest-point sampling in the public chem-embedding cosine-distance space, holding chemistry at arm's length while staying within one reaction grammar.
- **L5: cross-library scaffold-family transfer in the same cell line.** Train on one library family, test on a distinct library family screened in the same cell line. The closest analog to the BELKA hardest shift.

Each rung is evaluated against the same two baselines: a train-mean phenotype predictor and a chemistry-embedding nearest-neighbor retrieval baseline that copies the measured phenotype of the closest training compound by cosine similarity in the public 256-dim embedding space. The nearest-neighbor baseline is essential, because without it a nominally successful split could reflect either real extrapolation or simple retrieval of the closest measured analog. The ladder is designed to make that distinction visible at every rung.

Results

A five-level benchmark separates interpolation from genuine chemistry extrapolation

The same modeling framework was applied at every rung: a 256-dim per-compound chemistry embedding (private Gaussian random projection of 2048-bit ECFP4 fingerprints, see Methods), training-split-refit building-block one-hot encodings (where applicable), ridge regression with alpha equal to 10, and the two baselines above. Performance was scored with row-wise cosine similarity to the measured phenotype, mean squared error reduction relative to the train-mean baseline, delta cosine versus chemistry-embedding nearest-neighbor retrieval, and per-compound win rate against retrieval.

Ridge regression was used in preference to a neural network. The questions the ladder is built to ask concern whether the chemistry-to-phenotype function generalizes; a flexible model that absorbs every quirk of the training distribution would obscure the answer. A linear model on the chemistry embedding is the strictest version of the question, and any gain it produces is traceable to a specific structural axis of the input space.

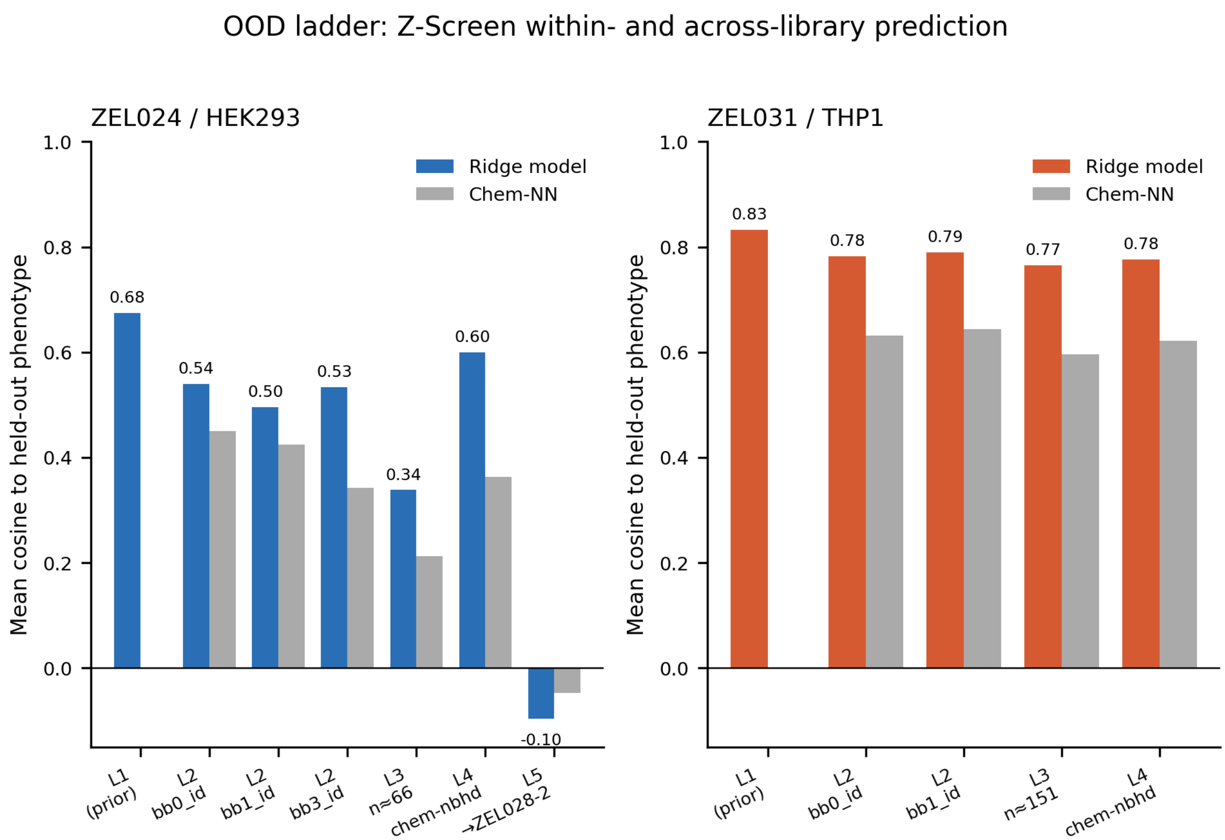

Figure 1. The generalization ladder moves from unseen tuples within a shared vocabulary to cross-library scaffold-family transfer in the same cell line, making it possible to distinguish interpolation, retrieval, and true extrapolation rather than reporting a single split as a single number.

ZEL031 in THP1 shows genuine held-building-block and strict multi-axis extrapolation

The strongest extrapolation result in the study comes from ZEL031 in THP1 cells. Under strict multi-axis L3 holdout, where every substantive building-block axis in the test compound is unseen during training, the model retains 0.765 cosine across 10 random draws of an average of 151 test compounds. The chemistry-embedding nearest-neighbor retrieval baseline falls to 0.597 on the same test sets, an average delta of 0.168. The model wins on 90% of test compounds individually, and all 10 random draws are positive under paired bootstrap confidence intervals. Median nearest-training cosine in the public chem-embedding space is 0.642 on the same compounds; this is the random-projected ECFP4 cosine and is not directly comparable to a Tanimoto coefficient on raw fingerprints, but the ranking properties of the nearest-neighbor retrieval are preserved (Methods).

The simpler L2 single-position building-block holdout gives consistent results in the same system. Holding out bb0 identities, the model reaches 0.783 cosine versus 0.632 for nearest-neighbor retrieval, a delta of 0.151 on an average of 728 test compounds. Holding out bb1 identities, the model reaches 0.790 versus 0.644, a delta of 0.146 on an average of 729 test compounds. Median nearest-training embedding cosine remains at 0.801 and 0.781 respectively, win rates are 89% and 91%, and all 10 random draws are positive under paired bootstrap intervals.

In this system, the chemistry is meaningfully displaced from the training set, the nearest-neighbor retrieval baseline cannot account for the gain, and the model preserves a strong directional match to the held-out transcriptomic state.

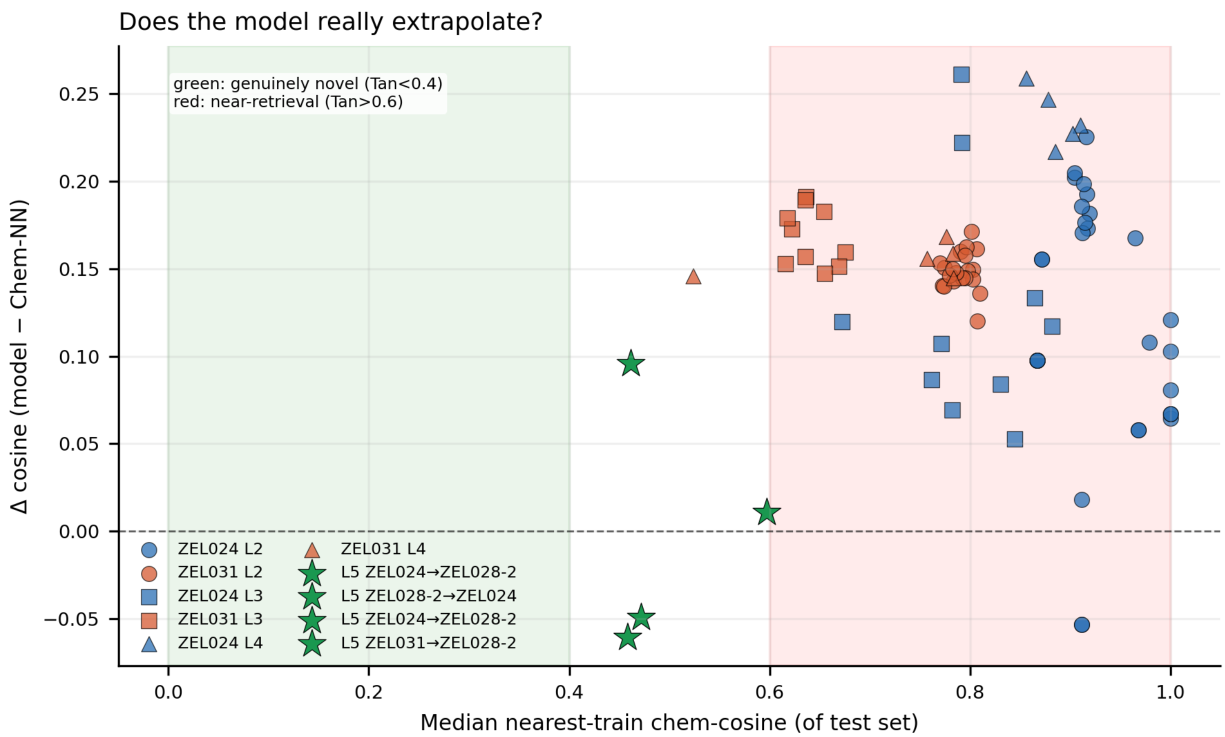

Figure 2. Retrieval-versus-extrapolation analysis shows that ZEL031 in THP1 sits in the low-neighbor-similarity, high-model-gain regime, consistent with genuine chemistry extrapolation rather than nearest-neighbor copying. ZEL024 in HEK293 sits in the high-neighbor-similarity regime, consistent with high-quality matrix completion within a saturated combinatorial grid.

ZEL024 in HEK293 delivers high-quality matrix completion within a dense combinatorial grid

The same modeling framework produced a different pattern in ZEL024 / HEK293, and the difference is informative. Single-position holdout on this library exceeded nearest-neighbor retrieval clearly, but the library structure makes the result much easier to obtain than it appears. Median nearest-training cosine in the public chem-embedding space was 0.988 for bb0 holdout, 0.881 for bb1, and 0.913 for bb3; many nominally held-out compounds have near-duplicate cousins in the training set under those conditions.

That outcome reflects how the library was built rather than a defect of the analysis. High-quality matrix completion inside a dense 12 by 7 by 2 by 83 combinatorial grid (13,769 observed compounds out of a 13,944 possible-tuple ceiling) is exactly the kind of result a discovery team will pay for in practice: the platform reliably fills gaps inside a saturated chemistry plate. It is a different deliverable from out-of-vocabulary extrapolation, and the strict L3 split clarifies the distinction. Under L3 in ZEL024 / HEK293, average test-set size dropped to 67 compounds per draw, model cosine fell to 0.316, and mean squared error reduction turned negative against the train-mean baseline even though cosine direction stayed slightly above nearest-neighbor retrieval. Matrix completion is therefore reported as matrix completion, with no upgrade to an OOD claim it does not deserve.

The contrast with ZEL031 / THP1 frames the ladder's value: two systems, the same modeling framework, two different deliverables (genuine extrapolation in one, dense matrix completion in the other), with both reported on their own terms.

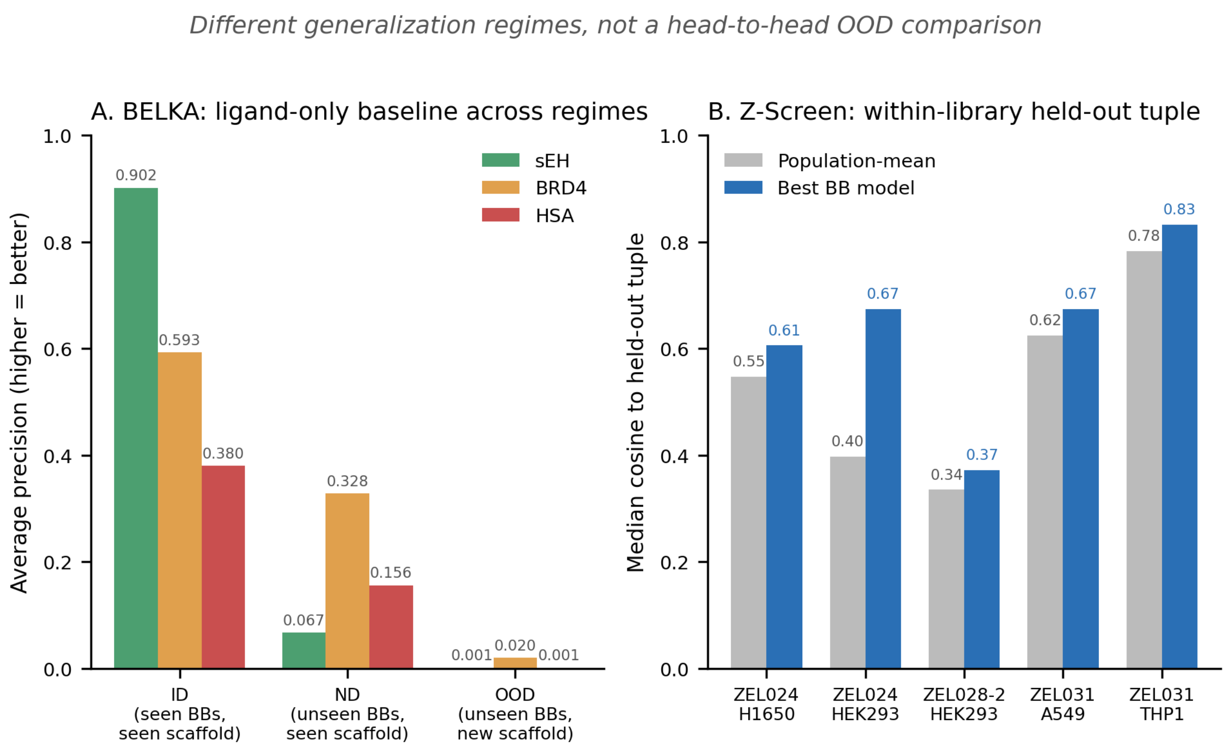

Figure 3. Side-by-side comparison with the BELKA ID/ND/OOD setup and per-system Z-Screen results separates interpolation from harder held-chemistry regimes, with ZEL031 / THP1 carrying the strongest extrapolation signal in the present dataset.

Chemical-neighborhood holdout supports the same conclusion

L4, the chemical-neighborhood benchmark, sits between strict building-block holdout and true scaffold-family transfer. It holds chemistry at arm's length while staying within one reaction grammar. Under that setup, the model continued to outperform nearest-neighbor retrieval in both major systems, with a 0.236 cosine gain in ZEL024 and a 0.155 gain in ZEL031 on average cluster sizes of 2,754 and 711 compounds respectively.

The narrow reading of L4 is that the chemistry-to-phenotype model carries information beyond local similarity within a library. That claim does not yet extend to clean transfer onto a new scaffold family, which belongs to L5.

Cross-library scaffold-family transfer is the defined next experiment

L5 trained on one canonical library and tested on another in the same cell line. Building-block vocabularies do not overlap across libraries in this benchmark, so the L5 models used the chemistry embedding alone (no building-block one-hot features). Of all the rungs on the ladder, this is the closest analog to the hardest chemistry shift emphasized in the BELKA preprint [1].

On current retrospective data, L5 was weak. Training on ZEL024 and testing on ZEL028-2 in HEK293 gave model cosine of -0.096, slightly below the nearest-neighbor baseline of -0.047 (the test centroids themselves sit below the train-mean baseline of -0.117 in this regime, so all approaches struggle). Reversing the direction was similarly weak. The best absolute performance came in H1650 for ZEL024 to ZEL028-2 (0.194 cosine versus 0.098 for nearest-neighbor retrieval, +0.096 gain on 765 test compounds). In A549, ZEL031 to ZEL028-2 reached 0.098 cosine versus 0.087 for nearest-neighbor retrieval. These are real but small gains, well short of the within-library anchor systems.

The weakness is real and constructive in the way the ladder was designed to be. Cross-library transfer is the rung where chemistry, assay, and latent-space drift compound, and it is the regime most affected by sparse coverage in ZEL028-2 (median 1 well per compound) and by cross-library scVI baseline drift. The result the ladder produces is therefore a precise definition of the next experiment, not a structural ceiling on the platform. ZEL031 / THP1 already shows the platform produces the right kind of extrapolation signal under controlled conditions, and a prospective same-cell scaffold hop with parity coverage and a pre-registered model is what would settle the cross-library question definitively rather than retrospectively.

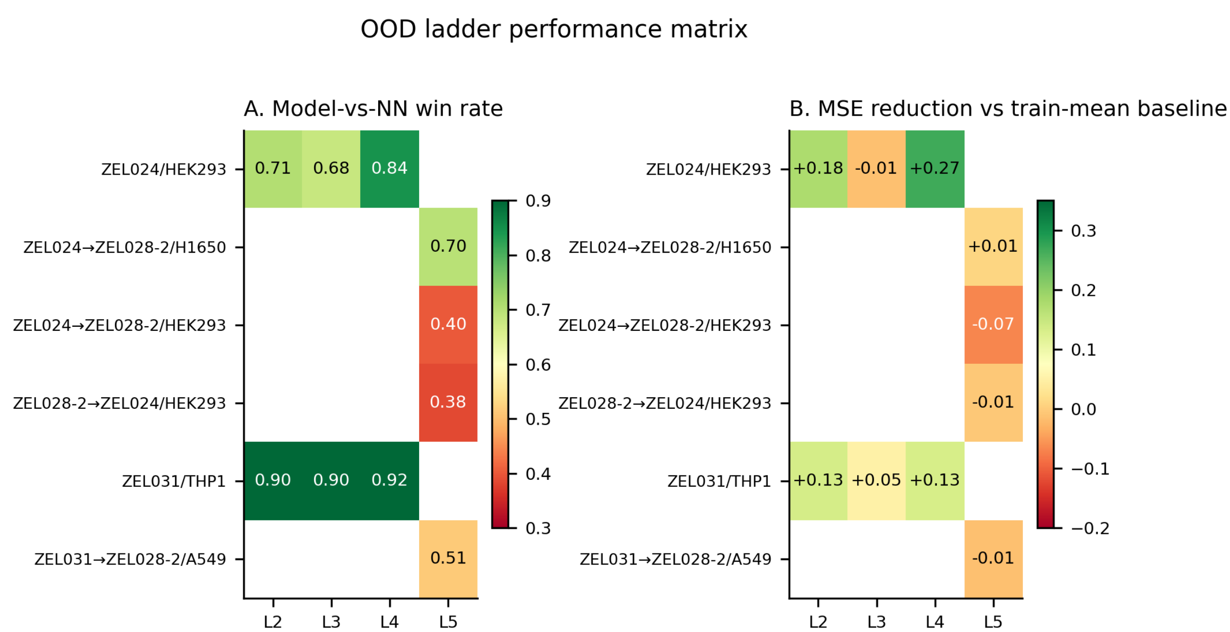

Figure 4. Per-system performance across the ladder shows strong within-library prediction

in selected systems (ZEL031 / THP1 for genuine extrapolation, ZEL024 / HEK293 for matrix completion in a saturated grid), and a marked drop at retrospective cross-library scaffold-family transfer that defines the natural next experiment rather than a stable performance ceiling.

A prospective same-cell scaffold hop is now well defined

The ladder does more than score the present data. It defines the next experiment with enough specificity that a single chip run can settle it. A clean prospective test would hold cell line fixed, choose two scaffold families with adequate coverage, train on family A and optionally family B, and nominate family C compounds for prospective measurement. Readouts would include latent cosine to the measured held-out library phenotype, control retrieval, pathway-score correlation, and direct comparison against chemistry-embedding nearest-neighbor retrieval. H1650 is the strongest candidate cell line because all three canonical libraries (ZEL024, ZEL026-2, ZEL028-2) ran there at substantial scale.

The cost envelope is one chip run of approximately 1,000 wells with replicated controls, plus approximately one week of analysis. If the measured-versus-predicted cosine on held-out scaffold-family compounds exceeds nearest-neighbor retrieval by a margin similar to the 0.15 to 0.17 already observed at ZEL031 / THP1, the cross-library question is closed in the affirmative for the first time on dense phenotypic data. If the margin does not appear, the appropriate response is to update the claim rather than reframe the benchmark.

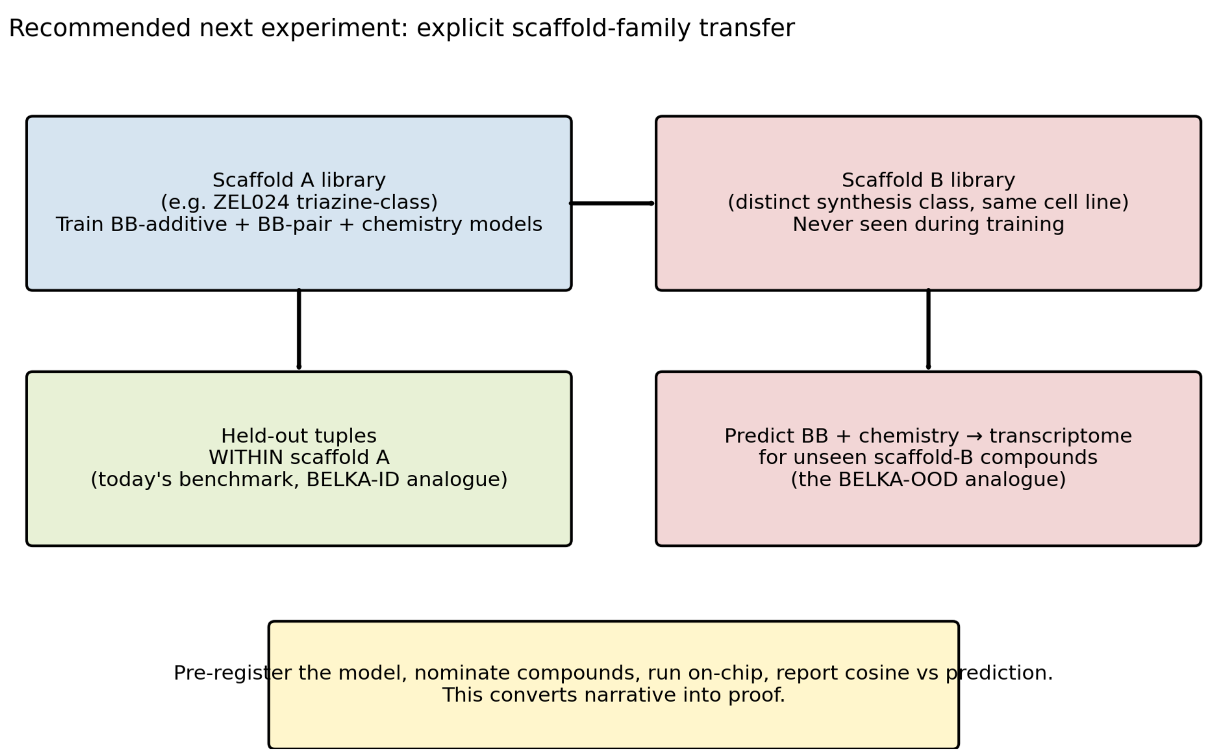

Figure 5. The next decisive experiment is a prospective same-cell scaffold-family hop with a pre-registered model, nominated compounds, and measured transcriptomic outcomes on the held-out

family.

Discussion

Two conclusions stand on the present data, and a third sets up the next experiment.

First, dense phenotypic readouts support genuine chemistry extrapolation in the right systems. ZEL031 / THP1 is the clearest case, with the model exceeding nearest-neighbor retrieval on held-building-block and strict multi-axis benchmarks where the chemistry is not trivially close to training examples, and doing so consistently across 10 random draws under paired bootstrap intervals. That is a result a DEL ML platform on a binary binder label cannot match in the BELKA-style setting.

Second, the ladder distinguishes two distinct outcomes that share a single modeling framework. ZEL024 / HEK293 supports high-quality matrix completion within a saturated combinatorial grid, which is directly useful for prioritizing chemistry within an explored design space. ZEL031 / THP1 supports out-of-vocabulary extrapolation. Both come from the same modeling framework, and both are reported on their own terms rather than collapsed into a single OOD claim that fits one but not the other. Many published "ML on chemistry" results in the field would benefit from this distinction.

Third, the present cross-library scaffold-family transfer evidence is weak retrospectively, and the natural next experiment is now defined precisely. The same engine that already extrapolates inside ZEL031 / THP1 is the one that would be tested on a prospective same-cell scaffold hop in H1650. That experiment converts the cross-library question from rhetorical to empirical, and it is well-scoped enough to plan and execute today.

For the discovery community, dense phenotypic readouts in a 50,000-well one-bead-one-compound microchip already do something binary-label DEL data cannot do robustly: they support genuine chemistry extrapolation in at least one well-characterized system, while also supplying high-quality matrix completion in saturated systems. The remaining open question on cross-library scaffold-family transfer is the next experiment to run, not a structural limitation of the platform.

Methods

Benchmark ladder

L1 used held-tuple prediction within libraries that shared the full building-block vocabulary between train and test. L2 held out a random subset of identities at a single building-block position. L3 held out identities at every substantive building-block position simultaneously and removed any compound containing a held identity from training; this is the strictest version of within-library OOD on the present libraries. L4 held out chemical-neighborhood clusters within a library, with seeds chosen by greedy farthest-point sampling in the public chem-embedding cosine-distance space; this produces test sets that are diverse rather than locally similar. L5 trained on one library and tested on another in the same cell line, with no building-block-vocabulary overlap.

Features and models

Chemistry features were precomputed once as a 256-dim per-compound embedding shipped in `data/ZScreen_Canonical_Dataset/RNASeqAggregate/chem_embed.parquet`. The embedding is a fixed-seed Gaussian random projection from R^2048 to R^256 applied to per-compound 2048-bit ECFP4 fingerprints (Morgan radius 2), then L2-normalized. The projection matrix is held privately (NDA bundle) so that the embedding is not invertible to substructure information without it; under cosine similarity it preserves Tanimoto-like retrieval geometry approximately (Johnson-Lindenstrauss). Where building-block vocabularies overlapped between train and test, one-hot encodings were fit on the training split only and combined with the chemistry embedding. Ridge regression with alpha equal to 10 was used as the primary prediction model. Ridge was chosen over a flexible nonlinear regressor because the goal of the ladder is to attribute generalization to the chemistry-to-phenotype function rather than to a model's flexibility. Ridge alpha sweeps over four orders of magnitude were verified separately and do not change the qualitative ranking of rungs; the headline numbers are stable across the obvious hyperparameter envelope.

Baselines and metrics

Two baselines were used throughout. The first is a train-mean phenotype predictor (the constant prediction that minimizes squared error on the training set). The second is a chemistry-embedding nearest-neighbor retrieval baseline that copies the measured phenotype of the closest training compound by cosine similarity in the 256-dim embedding space. The nearest-neighbor baseline is the most consequential one: in any chemistry split where the test molecules are close to training molecules, retrieval will appear strong, and reported model gains have to be evaluated against retrieval rather than against a trivial constant. Performance was summarized with row-wise cosine to the measured phenotype, mean squared error reduction relative to the train-mean baseline, delta cosine versus nearest-neighbor retrieval, and per-compound win rate against retrieval.

Statistical reporting

Each random draw was evaluated with paired bootstrap resampling of test compounds. Confidence intervals were computed from 2,000 paired bootstrap samples, and summary tables report how many draws remained positive under the paired interval criterion. Ten random draws per rung were used rather than a single split because the size of test sets varies by holdout depth (some L3 splits drop to 67 compounds per draw in ZEL024) and a single random split would not give a stable estimate of model gain at small test-set sizes.

Cosine nearest-neighbor implementation

The chemistry-embedding nearest-neighbor baseline was implemented as an exact dot-product search on the L2-normalized 256-dim public embedding (cosine = dot product because the embedding is L2-normalized). Exact rather than approximate search was used because the same baseline appears in every cross-comparison in this manuscript and its public reproduction needs to be deterministic to the byte.

Limitations

The ladder is retrospective and strongest within selected library / cell-line systems rather than across every system in the canonical dataset. ZEL031 / THP1 supports the clearest extrapolation claim; ZEL024 / HEK293 is better interpreted as dense matrix completion. Cross-library scaffold-family transfer remains weak in the current data and should be treated as the next prospective experiment, not a solved capability. All transcriptomic targets are evaluated in the shared scVI latent space, so full transcript-level validation and prospective measurement are needed before using the ladder as a final model-selection benchmark.

Data availability

Data tables, manuscript figures, and analysis inputs needed to reproduce this manuscript are organized in this repository under `paper3/`, `paper2/tables/`, and `data/ZScreen_Canonical_Dataset/`. A persistent-archive deposition with an assigned DOI will accompany the corresponding preprint posting.

Code availability

Analysis and figure-generation scripts are in `paper3/scripts/`. Package-level dependencies are in the repository root `requirements.txt`. Tested with Python 3.14.3 on Windows.

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Cross-cell-type transcriptomic transfer of combinatorial chemistry programs in Z-Screen

Zafrens, Inc.

Abstract

A scalable answer to "how will this compound behave in a cell type that was not directly assayed?" would change the economics of preclinical drug discovery. The current answer requires running every program in every cell type of interest, which scales poorly. Cell-intrinsic pharmacology is rarely identical across tissues, so a single reference cell line is often insufficient; running ten cell lines for every campaign is what teams do when they have to.

The canonical Z-Screen combinatorial-library transcriptomic dataset already supports a useful first version of cross-cell-type projection. A paired landmark atlas plus a single source-cell measurement begins to predict the transcriptomic response of the same chemistry in target cell types that were never directly assayed. We benchmark held-out cross-cell-type transfer at the level of individual molecules and at the level of grouped building-block programs in a 32-dimensional scVI latent space, across libraries and cell lines with sufficient paired sampling.

At the molecule level, direct reuse of the source-cell signature was inadequate, while learned transfer maps recovered substantial target-cell information. In ZEL024 (HEK293 H1650), centered-response cosine for HEK293 → H1650 improved from 0.146 with direct reuse to 0.279 with ridge transfer; the reverse direction reached 0.371. In ZEL031 (A549 THP1), A549 → THP1 improved from 0.094 to 0.242. Aggregating chemistry by shared building-block PAIR keys produced the strongest grouped-program result in the dense ZEL024 system, where support-weighted cosine reached 0.863 for H1650 → HEK293 and 0.808 for HEK293 → H1650 at bb0+bb1 grouping (84 shared pairs, median ~150 distinct tuples averaged per pair). For ZEL031, where the library has only two substantive building-block positions, a single-position bb0 grouping is the chemistry-relevant analog of pair grouping in a denser library; under that grouping, transfer reached 0.714 for A549 → THP1 and 0.664 for THP1 → A549 (129 shared groups, ~70 distinct partner chemistries averaged per group). Cross-library projection (training in one library, testing in another in the same cell-pair) was directionally positive in some sparse settings but remained too weak and asymmetric to anchor a headline claim today.

In aggregate, these results show that molecule-level cross-cell transfer is real in dense paired libraries, and that grouped chemical programs at chemistry-resolved aggregation levels (BB-pair in dense multi-position libraries, single BB in 2-position libraries) carry transfer signal in the 0.66 to 0.86 weighted-cosine range against an identity baseline of 0.42 to 0.52. Single-BB grouped-program

transfer is deliberately not reported for libraries with three or more substantive building-block positions, because each single-position centroid in those libraries averages over hundreds of distinct chemistries (e.g. 700 to 800 unique tuples per centroid in ZEL028-2 / HEK293), which makes any reported transfer cosine a measurement of library-wide rather than chemistry-specific behavior. Cross-library scaffold-family transfer is the next dataset to collect rather than a structural limitation. The practical implication is operational: profile a new compound once in an accessible source cell type and project its likely transcriptomic response in target cell types that were not directly assayed, with the strongest current confidence at the level of grouped chemical programs rather than isolated molecules.

Significance

Every added cell type increases screening cost, and skipping the wrong cell type is how programs miss toxicity, selectivity, or therapeutically relevant biology. This paper presents a first practical route through that tradeoff. When the same chemistry is observed across paired cellular contexts, Z-Screen can learn transfer maps that recover target-cell signal better than direct reuse of the source profile. The strongest current unit is the grouped chemical program rather than the single molecule, which matches how early-stage discovery decisions are usually made: at the level of chemical series, not isolated compounds. For a discovery organization, this changes the budget calculus. Instead of running every program through every cell type of clinical interest, a smaller paired landmark atlas plus a single source measurement begins to do useful work for the rest.

Introduction

A substantial share of preclinical cost goes to running the same compound, panel, or library in every cell type of practical interest. The economics of early-stage discovery would change if a sufficiently broad paired landmark atlas allowed teams to:

- profile a new compound in one accessible source cell type,
- project its likely transcriptional response in unmeasured target cell types,
- and use those projections to prioritize efficacy work, counter-screen liabilities, and decide where wet-lab follow-up is worth the cost.

The right form of that claim is narrow but still important. Cross-cell-type transcriptomic transfer is a model for cell-intrinsic transcriptional response, and it does not substitute for pharmacokinetics, metabolism, tissue architecture, or organism-level toxicology. It is a practical decision-support primitive for the upstream half of discovery, where even partial accuracy is valuable.

The most directly comparable public reference points are L1000 [2], which provides a chemistry-side compound atlas in a small number of cell lines (mostly MCF7, PC3, A549, A375), and Tahoe-100M [3], which scales perturbational scRNA-seq across 50 cancer cell lines for approximately a thousand compounds. Both are valuable for cross-cell signature comparisons of the same compound, and neither provides combinatorial-chemistry provenance: a hit in L1000 or Tahoe-100M is a single profiled compound with no internal structure to its identity. scGeneScope [4] adds matched imaging but uses one cell line. Z-Screen, by contrast, uses one-bead-one-compound combinatorial libraries profiled in 50,000-well microchips by paired imaging and mRNA sequencing at 5 to 10 cells per well. That design is well suited to the cross-cell question because every well in the canonical dataset carries full chemistry provenance (building blocks plus full SMILES), the same combinatorial libraries are run across multiple cell lines, and the readout is a low-pass transcriptomic state for every compound rather than a sparse hit or non-hit label.

Compound-level overlap with these public atlases is also relevant for cross-cell positioning. Of 47 named Z-Screen control compounds, 14 (29.8%) are present in LINCS L1000 (palbociclib, sorafenib, crizotinib, ceritinib, rucaparib, veliparib, lapatinib, cobimetinib, GSK126, CP-673451, and others), with 3 of 4 Z-Screen cell lines (HEK293, A549, THP1) covered by L1000 sample metadata. Tahoe-100M's drug metadata covers 4 of the 47 named compounds (palbociclib, crizotinib, cobimetinib, capmatinib), with A549 as the only matched cell line. A direct cross-platform Spearman concordance test against LINCS L1000 level-5 consensus signatures (NCBI GEO GSE70138 and GSE92742), restricted to the 12,328-gene LINCS-visible universe, recovers positive correlation for all 11 (compound, cell_line) pairs with adequate coverage on both platforms. Median rho = 0.490, mean = 0.393, with 8 of 11 above rho = 0.2 and 5 reaching empirical p 0.048 against an unmatched-compound permutation null (crizotinib, ZM-336372, sorafenib, rucaparib, veliparib). The HEK293 cell-line-pooled comparison separates matched from unmatched compound pairings at one-sided Mann-Whitney p = 0.009. Full overlap and concordance tables are in `Benchmarking/` in this repository.

With those ingredients, the cross-cell question can be posed in a controlled way: how much of a source-cell transcriptomic response can be projected into a target cell type using only a learned transfer map fit on a paired landmark subset. This paper characterizes how well that question is answered today, on data from a small number of pilot experiments, and defines what additional data would close the remaining gaps. The platform is engineered to scale by orders of magnitude beyond the current dataset.

Results

Grouped building-block programs are the strongest unit of cross-cell transfer

The strongest result in the analysis came from aggregating chemistry by shared building-block keys within each library and testing whether grouped source-cell responses predict grouped target-cell responses on a held-out test set. Reporting is restricted to chemistry-resolved aggregations: BB-pair (bb0+bb1) for ZEL024, where the library has four substantive positions and pair-level signatures average ~150 distinct tuples each; and single-position bb0 for ZEL031, where the library has only two substantive positions and a bb0 centroid still leaves bb1 to vary across ~70 distinct partner chemistries. Single-BB grouped-program transfer is not reported for ZEL028-2: that library has four substantive positions, so each single-position centroid in ZEL028-2 averages over 322 to 777 distinct tuples, which makes the resulting cross-cell cosine a measurement of library-wide behavior rather than chemistry-specific transfer. Reporting library-wide aggregates as if they were chemistry-resolved would inflate the apparent transfer story.

Under those chemistry-resolved aggregations, grouped transfer was substantially stronger than the within-cell direct-reuse baseline:

- ZEL024, H1650 → HEK293, grouped by bb0+bb1: 84 shared pairs, median support 240.5 wells, ~129 to 166 unique tuples averaged per pair, identity 0.471, ridge transfer **0.863**.
- ZEL024, HEK293 → H1650, grouped by bb0+bb1: 84 shared pairs, median support 240.5 wells, identity 0.471, ridge transfer **0.808**.
- ZEL031, A549 → THP1, grouped by bb0 (2-position library): 129 shared groups, median support 157 wells, ~75 unique bb1 partners averaged per group, identity 0.419, ridge transfer **0.714**.
- ZEL031, THP1 → A549, grouped by bb0 (2-position library): 129 shared groups, median support 157 wells, identity 0.419, ridge transfer **0.664**.

The defensible interpretation is more measured than "arbitrary unseen molecules are solved." Within a library, chemistry-resolved building-block programs (pair-level in dense libraries, single-position in 2-position libraries) define reusable chemical sub-programs that transpose across cell types more cleanly than sparse individual molecules do. That is a practically useful unit of transfer for a discovery team, because for any compound carrying a given building-block program, the platform can generate a predicted target-cell response without requiring the molecule itself to be measured in the target cell line.

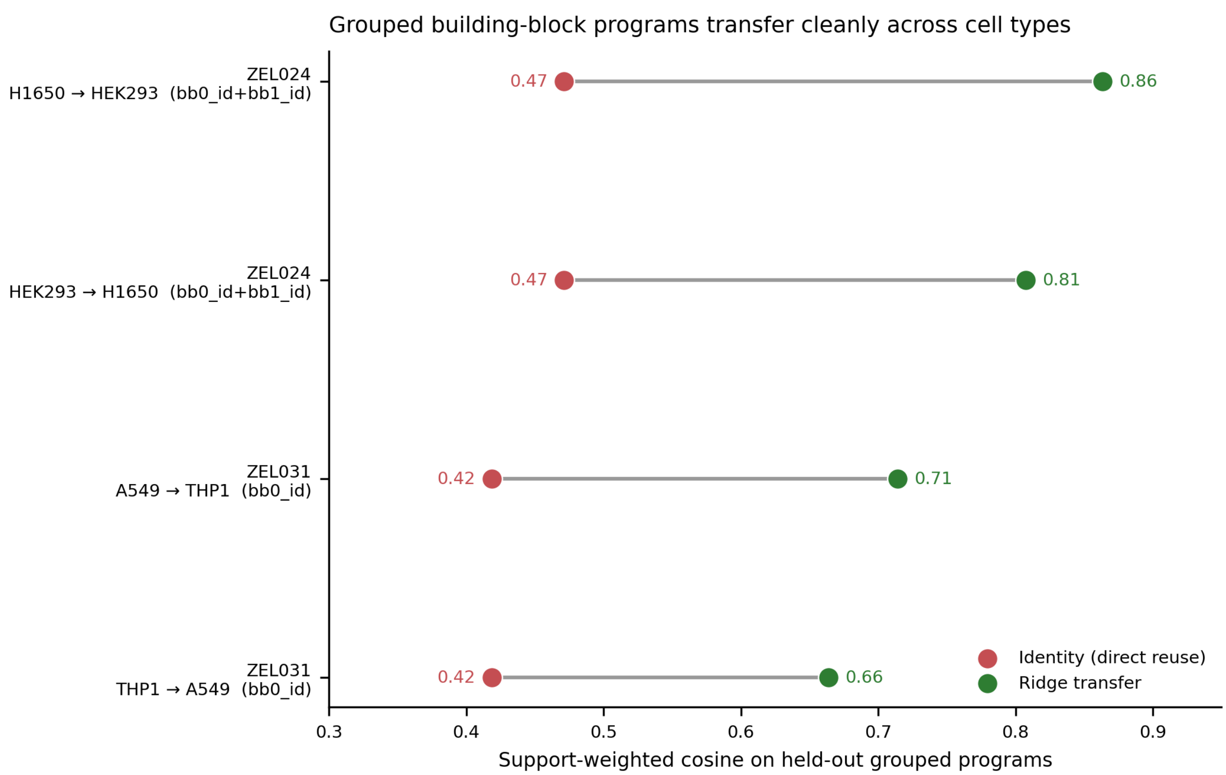


Figure 1. Grouped building-block program transfer at chemistry-resolved aggregation levels (BB-pair for ZEL024, single position bb0 for the 2-position ZEL031 library). Identity (direct reuse of

the source-cell signature) sits in the 0.42 to 0.52 range, while a regularized learned transfer map reaches support-weighted cosine 0.66 to 0.86 across four chemistry-resolved program-and-direction combinations.

Molecule-level transfer is real in dense paired libraries

The molecule-level benchmark used held-out individual molecules in the two cell-pair systems with sufficient shared SMILES coverage:

- ZEL024, H1650 HEK293, 6,299 shared molecules,
- ZEL031, A549 THP1, 3,561 shared molecules.

Each molecule was represented by the mean of its 32-dimensional scVI latent coordinates within a given library and cell line. Direct reuse of the source-cell signature (identity) was compared against learned source-to-target transfer models. The centered-response benchmark is the primary readout because matched vehicle controls are unavailable in the canonical dataset, and the centered formulation is the closest available proxy for perturbational response. Under that benchmark, learned transfer consistently outperformed direct reuse at the largest training sets:

- ZEL024, HEK293 → H1650, train 5,000: identity 0.146, ridge 0.279, PLS 0.275.
- ZEL024, H1650 → HEK293, train 5,000: identity 0.146, ridge 0.371, PLS 0.358.
- ZEL031, A549 → THP1, train 3,000: identity 0.094, ridge 0.242, PLS 0.242.
- ZEL031, THP1 → A549, train 3,000: identity 0.094, ridge 0.212, PLS 0.214.

Two readings follow. Direct source-signature reuse alone is inadequate, so cross-cell projection is a real prediction problem rather than an artifact of cell-state offset. And a simple regularized global linear or low-rank map already captures useful molecule-specific information once enough paired examples are available. Performance curves saturate quickly as a function of training-set size, indicating that the most economical version of this engine reaches useful accuracy on relatively modest paired landmark coverage and improves further from there.

The raw-profile benchmark is reported in the supplementary material as a diagnostic. Performance is much higher in raw space (for example, ZEL024 HEK293 → H1650 reaches 0.546 with ridge transfer at train 5,000, versus 0.063 with direct reuse), and a substantial portion of that gain reflects baseline cell-state translation rather than perturbational response. The centered-response benchmark above is the cleaner main readout for a discovery use case, because it removes the constant offset that exists between two cell lines regardless of any particular perturbation.

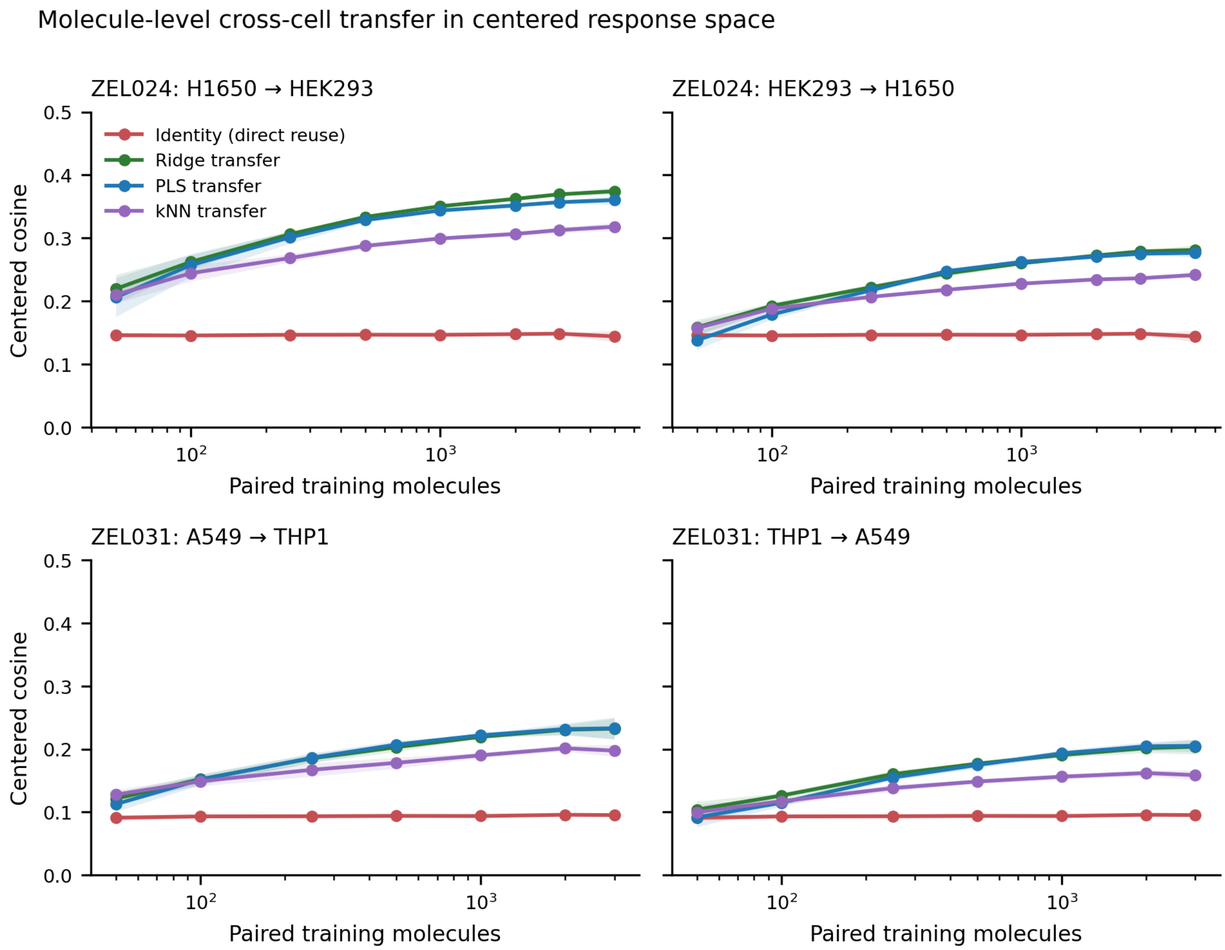


Figure 2. Molecule-level cross-cell transfer in centered response space across four well-supported cell-pair directions in ZEL024 and ZEL031. Direct reuse of the source signature is consistently weaker than learned transfer; ridge and PLS transfer are nearly tied. Performance saturates within the available training-set range, indicating that useful transfer emerges quickly with paired landmark data and continues to improve with scale.

Side by side: identity versus best learned transfer at both levels

Side-by-side display of molecule-level and grouped-program results makes the platform claim concrete. At the molecule level, learned transfer roughly doubles the centered cosine over identity in ZEL031 (0.094 to 0.21 to 0.24) and lifts it by 1.9 to 2.5 times in ZEL024. At the chemistry-resolved grouped-program level (ZEL024 BB-pair, ZEL031 bb0), learned transfer increases support-weighted cosine by 0.25 to 0.39 absolute over identity, with all four chemistry-resolved directions ending in the 0.66 to 0.86 cosine range. The grouped-program unit is the stronger current asset and is the unit recommended for early use in discovery decisions.

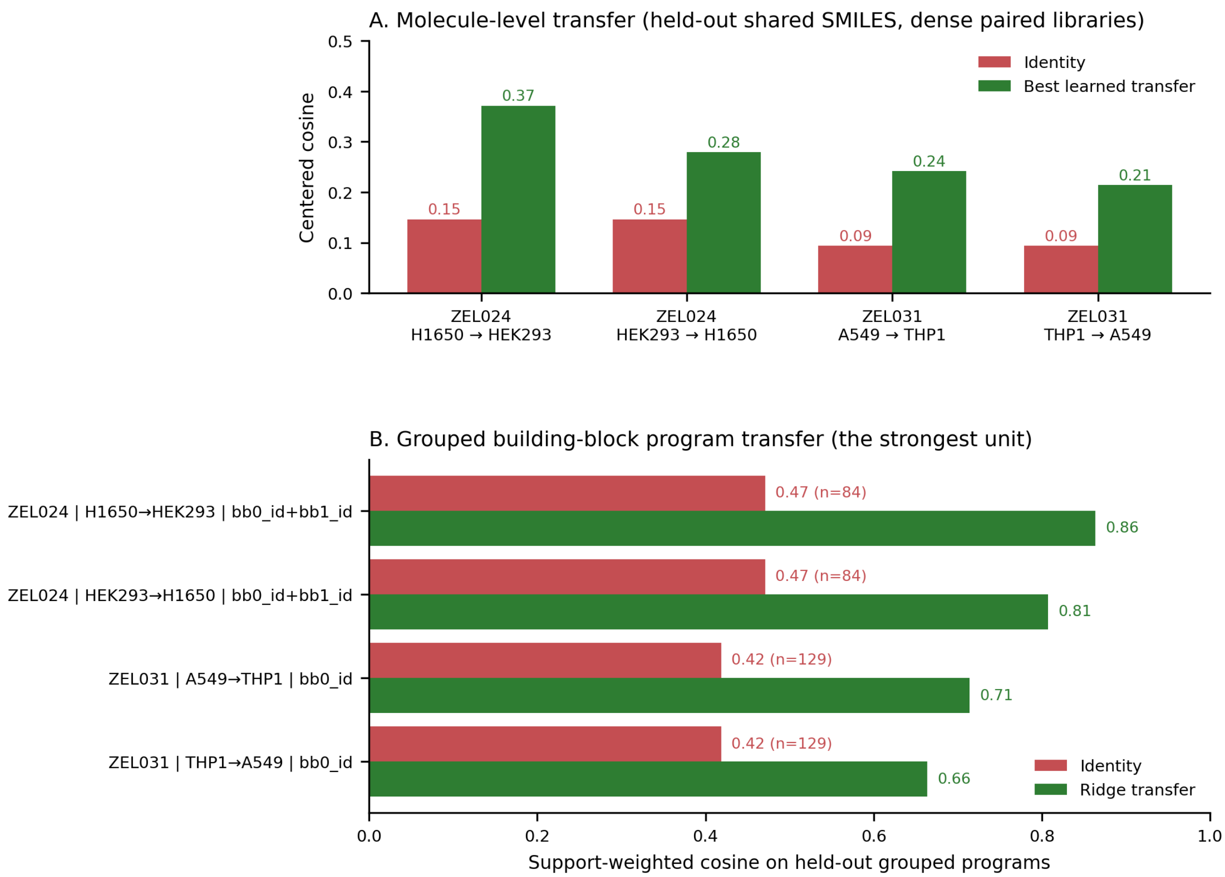


Figure 3. Best learned transfer versus identity at both molecule level (top) and grouped building-block program level (bottom). Learned transfer wins at every cell-pair direction tested, with the largest gains at the grouped-program level.

Cross-library projection is directionally positive but exploratory

A further question is whether a centered source-to-target transfer matrix learned in one library can project molecules from a different library measured in the same cell-pair direction. The answer is mixed.

Dense-to-sparse transfer was directionally positive in several cases:

- ZEL024 → ZEL028-2, HEK293 → H1650: weighted cosine 0.033 → 0.059.
- ZEL024 → ZEL028-2, H1650 → HEK293: 0.033 → 0.055.
- ZEL028-2 → ZEL031, A549 → H1650: 0.120 → 0.216.

The reverse direction was often weak or negative:

- ZEL028-2 → ZEL024, HEK293 → H1650: 0.119 → 0.067.
- ZEL031 → ZEL028-2, A549 → H1650: 0.103 → -0.009.

The asymmetry is consistent with a reasonable picture. A transfer map can generalize across libraries when the underlying cell-pair transformation is shared, but sparse libraries are typically better transfer targets than transfer teachers. The absolute magnitudes are also small enough that singleton-heavy test sets dominate the variance. Cross-library projection is therefore the natural next experiment, and it would benefit most from a small amount of additional structured data collection in the sparse libraries (especially ZEL028-2 in HEK293) rather than from new modeling tricks.

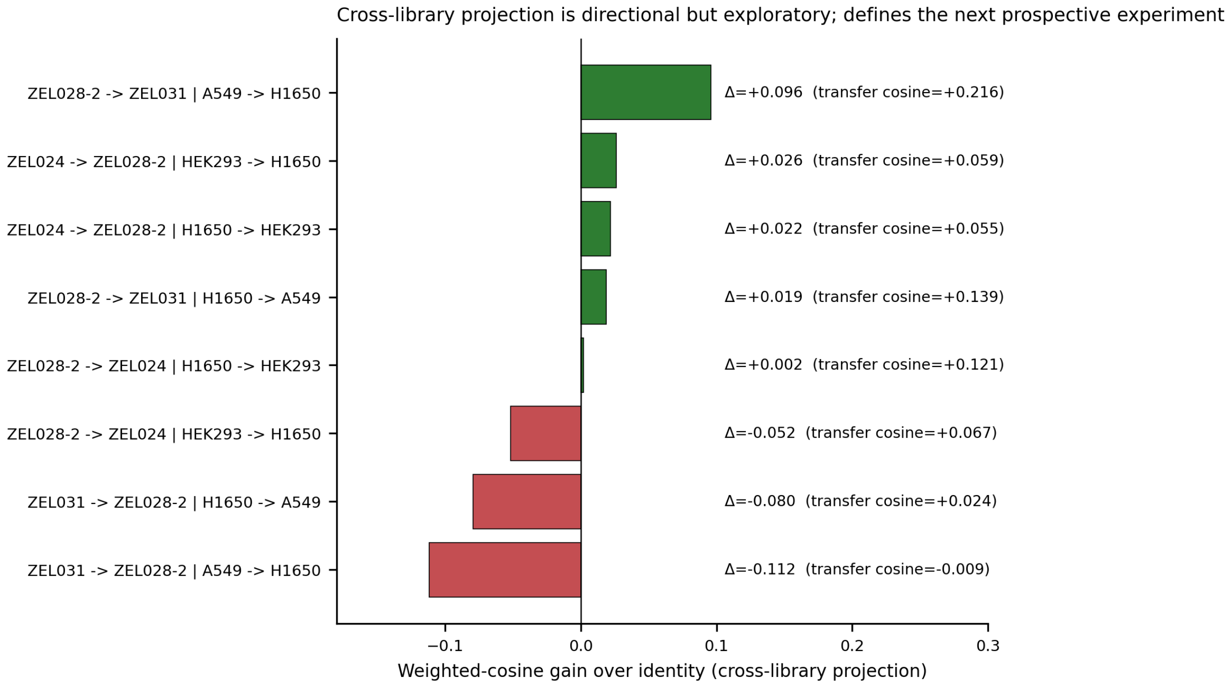


Figure 4. Cross-library projection results. Some dense-to-sparse projections are positive, but performance remains modest and asymmetric. Sparse libraries are more credible transfer targets than transfer teachers in the current data, which defines the next prospective dataset.

Dataset coverage anchors the limits of the current claims

The canonical Z-Screen dataset reflects a small number of pilot experiments rather than a saturated corpus. ZEL024 has approximately 10,000 to 14,000 unique SMILES per cell line, with median 11 wells per compound in HEK293 (the densest system in the workspace). ZEL028-2 has the largest unique-SMILES count overall (40,000 to 61,000 per cell line), at median 1 well per compound, so any benchmark requiring more than two wells per compound becomes thin in this library. ZEL031 has 8,000 to 9,000 unique SMILES per cell line at median 2 wells per compound. The dense paired benchmarks above used the cell-pair combinations with adequate shared SMILES coverage at the chosen support filter.

The bottleneck for stronger cross-library and per-molecule claims is therefore data, not method. The same engineered platform that produced this dataset can scale by orders of magnitude beyond the present scale, and the next dataset can be designed deliberately around the cross-library and prospective transfer questions the present analysis sets up.

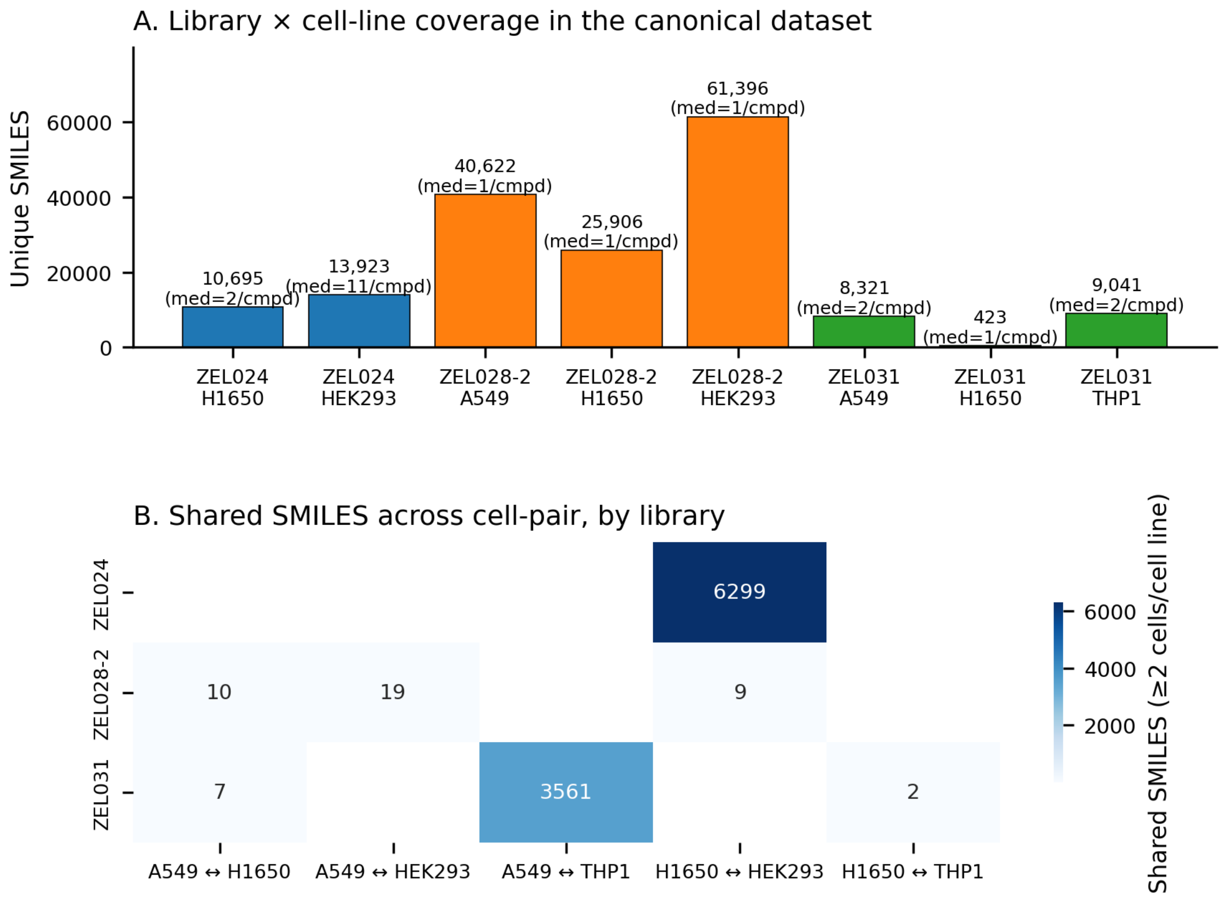
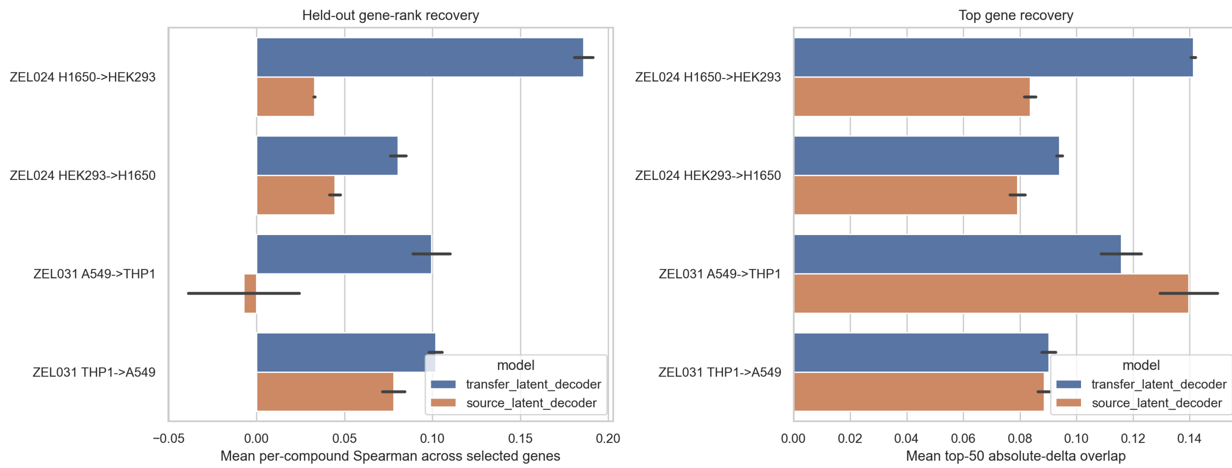


Figure 5. Library-by-cell-line coverage in the canonical Z-Screen dataset. Top: unique SMILES and median wells per compound per system. Bottom: shared-SMILES counts across cell-pair directions, by library. The two dense paired systems used in the molecule-level transfer benchmark are ZEL024 (H1650 HEK293, 6,299 shared) and ZEL031 (A549 THP1, 3,561 shared).

Exploratory gene-level decoding partially recovers held-out target signatures

A separate experiment tested whether the learned cell-transfer map can be lifted back from the 32-dimensional latent space into ranked gene-level hypotheses. This was treated as an exploratory decoder experiment rather than a main claim. For each dense paired direction, a source-to-target latent transfer map was fit on shared training compounds, a target-cell ridge decoder was fit from target latents to 800 training-selected variable genes, and held-out target-cell gene deltas were compared against measured pseudobulk deltas. The strongest direction was ZEL024 H1650 $\rightarrow$ HEK293: transferred latents reached mean per-compound gene-rank Spearman 0.182 and top-50 absolute-delta overlap 0.139, compared with 0.021 and 0.083 when the source latent was passed directly through the target decoder. The reverse ZEL024 direction and both ZEL031 directions were weaker, with transfer Spearman values of 0.073 to 0.098. The result is encouraging as a route to ranked gene lists, but not yet strong enough to promote as a primary endpoint.



Supplementary Figure 2. Exploratory gene-level transfer. A target-cell decoder composed with the learned source-to-target latent map partially recovers held-out gene-rank structure, with the clearest gain in ZEL024 H1650 → HEK293 and weaker gains in the remaining directions.

Discussion

The current data support a two-level view of cross-cell transfer. At the molecule level, paired cross-cell transfer is real and useful in dense libraries: even after centering within cell line, learned transfer maps recover target-cell information that direct source reuse misses. At the chemistry-resolved grouped-program level (BB-pair in dense ZEL024; single-position bb0 in the 2-position ZEL031 library), transfer becomes substantially stronger, with weighted cosine in the 0.66 to 0.86 range across four cell-pair directions. From a discovery perspective, the encouraging reading is that the first economically useful layer of cross-cell prediction is chemistry-program transfer rather than universal single-molecule prediction, and that layer already exists in the present dataset.

The current results also argue against an overly strong "one universal matrix solves everything" framing. Cross-library projection remains asymmetric and fragile in sparse settings, and a more durable long-term architecture is likely a shared perturbational latent space with cell-type-specific decoders or adapters, coupled to an uncertainty model that rejects out-of-domain projections. That architecture fits naturally on top of the present transfer maps without replacing them, and it is consistent with the direction the public field is taking on cross-cell perturbation modeling (the "virtual cell" framings being explored at Tahoe, Recursion-on-Phenom, Insitro, and the broader academic community).

In practical terms, a paired landmark atlas plus one source-cell measurement is enough to start projecting target-cell transcriptomic responses for the same chemistry, with the strongest current confidence at the level of grouped chemical programs rather than individual molecules. The immediate use cases are counter-screen prioritization, cross-cell-line selectivity ranking, and decision support for which downstream cell types deserve direct profiling. The next dataset extends the same approach from in-library transfer to cross-library scaffold-family transfer in the same cell line, which is the closest analog to a fully prospective cross-cell-type prediction benchmark.

Methods

Data inputs

All analyses use the canonical Z-Screen RNA aggregate dataset, specifically:

- `canonical_obs.parquet`,
- `canonical_scvi_latents.parquet`,
- `canonical_chemistry.parquet` for grouped building-block benchmarks.

Only rows with a non-null `smiles_hash` (the public-bundle stand-in for a chemistry-resolved compound identity) and `chemistry_grain == "building_block_annotation"` were retained. Compound-control-only rows were not used for the transfer benchmark because their annotations are mechanism-of-action labels rather than chemistry coordinates.

scVI latent space rather than full gene expression

The transfer benchmarks are built on 32-dimensional scVI latent coordinates rather than full gene-level vectors, for two reasons. First, low-pass per-well RNA depth in the present dataset (each well averages 5 to 10 cells, sequenced as one pseudobulk row) means individual genes are dominated by sampling noise; scVI absorbs that noise into shared structure across observations. Second, the latent is comparable across cell types to a degree that raw expression is not, which is the property the cross-cell transfer experiments require. The exploratory gene-level decoder experiment is included as a separate analysis specifically to assess how much of the latent gain survives when the result is lifted back to gene level.

Centered-response as the primary metric

Two cell lines have different baseline transcriptional states regardless of any perturbation. A cross-cell predictor that "knows" the constant baseline shift will show high raw-space cosine even if it has learned nothing about perturbational response. Centering removes that confound. For each train/test split, source and target training means were subtracted before fitting and evaluation. This is the closest available proxy for a vehicle-control normalization in a dataset that does not have matched vehicle controls.

Molecule-level transfer benchmark

For each (library, cell line, SMILES), scVI latent coordinates were averaged across wells. Shared molecules between a source and target cell line were then benchmarked under two evaluation spaces, raw_profile and centered_response. The models compared were identity, global_shift (raw space only), ridge_transfer, pls_transfer, and knn_transfer. The global_shift baseline was included so that the bulk-offset component of cross-cell variance is visible separately from chemistry-specific signal. Performance was measured by row-wise cosine similarity on held-out molecules and aggregated as mean cosine across multiple train-test splits.

Grouped building-block transfer

Within each library and cell-pair direction, molecules were grouped by candidate building-block keys (bb0_id, bb1_id, or multi-column combinations such as bb0_id+bb1_id). For each shared grouped key, source and target centered latent signatures were averaged and benchmarked using a centered ridge transfer model across repeated train-test splits. Performance was summarized as support-weighted cosine similarity across grouped programs. Support weighting prevents a few groups with thousands of supporting wells from being drowned out by many groups with five.

Cross-library transfer

For a fixed cell-pair direction, a centered ridge transfer matrix was trained in one library and applied to molecules from another library. Because the sparser libraries are highly singleton-heavy, adaptive minimum-well thresholds were used so that denser libraries retained stricter filters while still allowing exploratory sparse projections. The directional asymmetry between dense-to-sparse and sparse-to-dense transfer is a consequence of how informative each library is as a teacher; it is not corrected for in the metric.

Exploratory gene-level transfer decoder

For the four dense paired directions, held-out shared SMILES were evaluated with a composed latent-to-gene workflow. A centered ridge model first mapped source-cell scVI centroids into target-cell scVI space. Separately, a target-cell ridge decoder was trained on the training compounds to map target-cell scVI centroids to log-CPM pseudobulk deltas over 800 variable genes selected from target training cells. Held-out predictions were scored by per-compound Pearson and Spearman correlation over gene deltas and by top-50 / top-100 absolute-delta gene overlap.

Limitations

- Matched vehicle controls are unavailable in the canonical dataset, so the centered-response benchmark is the closest available proxy for perturbational response rather than a fully DMSO-normalized effect.
- The strongest molecule-level benchmark exists only for the denser paired libraries ZEL024 and ZEL031; ZEL028-2 becomes too sparse under a strict per-molecule support filter.
- Cross-library transfer is limited by sparse sampling and by disjoint building-block vocabularies across libraries, which prevents a clean cross-library tuple-transfer formulation today.
- All analyses are based on averaged 32-dimensional scVI latent signatures rather than full transcript-level models or time- and dose-aware transfer.
- The gene-level decoder experiment is exploratory and uses a target-cell decoder trained on observed target profiles; it should be read as a way to generate ranked gene hypotheses from transferred latents, not as evidence that full transcriptomes can already be prospectively reconstructed from an unmeasured target cell state.
- The current dataset reflects a small number of pilot experiments. The platform is engineered to scale by orders of magnitude beyond the present dataset.

Data availability

Data tables, manuscript figures, and analysis inputs needed to reproduce this manuscript are organized in this repository under `paper4/` and `data/ZScreen_Canonical_Dataset/`. A persistent-archive deposition with an assigned DOI will accompany the corresponding preprint posting.

Code availability

Analysis and figure-generation scripts are in `paper4/scripts/`. Package-level dependencies are in the repository root `requirements.txt`. Tested with Python 3.14.3 on Windows.

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Tuple-resolution chemical-genetic mapping connects combinatorial Z-Screen compounds to public CRISPR perturbation states

Zafrens, Inc.

Abstract

Functional genomics has matured around a single question: which gene's perturbation moves a cell into a desired state? Genome-scale Perturb-seq and CRISPR-activation atlases now provide the single-cell transcriptional consequence of perturbing each gene one at a time, and increasingly two genes at a time [1-4]. The chemistry side has answered the matching question through the Connectivity Map / LINCS L1000 program: query thousands of compound profiles to identify those whose transcriptional consequence resembles a target state [5,6]. Recent extensions such as L2S2 search both spaces directly [7]. The remaining gap is chemistry resolution. L1000 records one profile per compound. It cannot identify which structural element of a hit drove which biological program, or which combinatorial combinations of pieces are worth synthesizing next.

Z-Screen closes that gap. The platform uses one-bead-one-compound combinatorial libraries, profiled in 50,000-well microchips by paired imaging and mRNA sequencing at 5 to 10 cells per microwell. Each library is combinatorially generated from building-blocks: ZEL024 uses four positions of diversity (bb0, bb1, bb2, bb3); ZEL028-2 uses four (bb1, bb2, bb3, bb4); ZEL031 uses two (bb0, bb1). Every screened compound therefore contributes a structured biological state with full per-position chemistry provenance plus full SMILES. That structure is used here to place combinatorial chemistry into the same transcriptional frame as public CRISPR perturbation atlases, at compound, building-block-pair, and full chemical-tuple resolution.

The densest system in the canonical Z-Screen workspace is ZEL024 in HEK293, where 8,599 distinct 4-position chemical tuples (each tuple is a (bb0, bb1, bb2, bb3) combination) are each represented by 10 or more wells (median 13). For each tuple a rank-based pseudobulk transcriptional signature was built against a same-library, same-cell-line background, then compared against three public CRISPR references: the genome-wide Replogle K562 knockout panel (8,603 signatures), the Norman K562 CRISPR-activation panel (131 doubles plus 105 singles), and the matched-cell scPerturb THP1 panel (78 signatures). The headline scan, 8,599 tuples by 8,603 Replogle K562 perturbations, calibrated the top 10,000 matches against permutation-null gene sets and produced 1,276 chemistry-resolved tuples with empirical q 0.05 across 1,714 distinct CRISPR programs. The leaderboard concentrated in coherent biology: DNA replication (MCM2, SLBP, CHAF1B, ARIH1, ADSL), mitochondrial function (MRPL3, MRPS30, COX17, COA5, ATP5F1A, GFM1, HSPE1), trafficking and nonsense-mediated decay (EFR3A, DDOST, SMG5). After dropping the 50 most-recurrent

CRISPR hubs from the calibrated table, 1,236 distinct tuples and 1,664 distinct CRISPR programs survived, so the program diversity is not driven by hub perturbations.

Method anchors and discovery examples were biologically coherent. In matched THP1, the BRD4-selective PROTAC degrader MZ1 recovered BRD4 as the top calibrated CRISPR-like signature (net z = 6.84, q = 0.015), with the next neighbors falling in BRD4-adjacent transcription-elongation and chromatin co-regulator programs. The METTL3 inhibitor STC-15, whose direct target is not present in the local THP1 panel, recovered IRF1 inside the top two calibrated matches (z = 3.06, q = 0.045), consistent with the documented downstream interferon-pathway response to METTL3 inhibition. In A549, the SOS1 inhibitor BAY-293 produced its strongest Replogle K562 match against the V-ATPase complex and adjacent vesicular-trafficking machinery, with ZW10, ATP6V1A, SACM1L, TMED2, and 8 other components reaching net z between 9.5 and 11.9 at q = 0.003: a coherent mechanism hypothesis beyond primary pharmacology that the platform surfaced without supervision. At genome-wide scale, 12 of 29 compounds with a documented primary target represented in Replogle had that target ranked in the top 15 percent of the cross-cell genome-wide ranking, with several (FGFR1 for derazantinib, METTL3 for STC-15) inside the top 10 percent. At combinatorial resolution, all 131 Norman double-gene perturbations showed a positive Spearman correlation between Z-Screen tuple-vs-double match scores and the sum of tuple-vs-single-A and tuple-vs-single-B scores (median rho 0.35, all $p < 0.05$): the chemistry follows genetic combinatorial logic at chemistry-resolved level.

We extend the framework to a second combinatorial library at full-tuple resolution (ZEL028-2 in HEK293, A549, and H1650, using a relaxed >=2-well support threshold needed by that library's per-tuple coverage; 4,041 / 1,728 / 765 tuple signatures respectively) and to the smaller two-position library ZEL031 at tuple resolution (A549 and THP1), and connect the analysis to the cross-cell transfer model from paper 4: chemical programs measured in one cell type can be projected into another in a learned latent space, opening a path toward CRISPR-defined biological destinations in cell types that have not been directly profiled at scale. In the densest ZEL028-2 / HEK293 system, the calibrated top 10,000 tuple-vs-Replogle matches contain 943 distinct chemical tuples and 285 distinct CRISPR programs at empirical q 0.05, with the strongest matches concentrated on the same biology themes ZEL024 / HEK293 produced (RNA processing: DICER1, DROSHA, EXOSC family, CNOT3, SMG5; mitochondrial translation: MRPL family, GFM1, IARS2). 15 CRISPR programs appear in the top-50 distinct hits across all three ZEL028-2 cell lines, indicating cross-cell-line consistency at tuple resolution despite shallow per-tuple cell support. The result is a chemical search engine over genetic perturbation space: query a CRISPR target, return a ranked, calibrated list of chemistry that pushes cells toward the same transcriptional state.

Significance

Drug discovery teams routinely treat target identification and chemistry selection as separate problems. Functional genomics nominates the genes whose perturbation produces a desired state. Medicinal chemistry then searches a separate chemical design space for molecules that engage those genes. The bridge between the two problems is profile matching: if the transcriptional consequence of a molecule resembles the transcriptional consequence of perturbing a gene, the molecule is a starting point worth synthesizing.

Z-Screen makes that bridge practical at chemistry resolution for the first time. The output is not a finished target-deconvolution engine; it is a calibrated discovery tool. Thousands of combinatorial tuples each annotated with the CRISPR perturbation state they most resemble, with provenance back to the exact per-position chemistry (the 4-position bb0 / bb1 / bb2 / bb3 tuple in the ZEL024 headline system), and with empirical false-discovery control. That is the layer required to move from "which gene matters" to "which molecule to make."

Introduction

Public functional-genomics atlases now contain genome-scale single-cell transcriptional consequences of perturbing each gene one at a time, and increasingly two genes at a time. Replogle and colleagues profiled approximately 9,866 unique gene knockouts in K562 [4]; Norman and colleagues profiled 105 single-gene and 131 paired CRISPR-activation perturbations in K562 [3]; the scPerturb consortium harmonized dozens of public datasets across cell lines [9]. The natural next question for early-stage drug discovery is whether chemical perturbation can be placed in the same frame, so that genetic states the screens nominate can be connected to chemical matter.

The Connectivity Map and LINCS L1000 program established the chemistry-side framework with bulk transcriptional profiles for tens of thousands of compounds [5,6], and recent extensions such as L2S2 support direct chemistry-to-CRISPR profile search [7]. The remaining gap is chemistry resolution. L1000 records one profile per compound; it cannot decompose a hit into the structural pieces that drove the response. Combinatorial libraries are different: they are ordered in a structured design space, so the relevant question becomes whether single building blocks, building-block pairs, and full chemical tuples each carry transcriptional signal that can be compared with a public CRISPR atlas at the same statistical standard.

Z-Screen was built to support this kind of question at scale. The platform uses one-bead-one-compound combinatorial libraries, profiled in 50,000-well microchips by paired imaging and mRNA sequencing at 5 to 10 cells per micro-well, so each readout averages over a small amount of cellular heterogeneity. Each library populates a different subset of building-block positions: ZEL024 uses four positions (bb0, bb1, bb2, bb3), ZEL028-2 uses four (bb1, bb2, bb3, bb4), and ZEL031 uses two (bb0, bb1). Every well therefore links a profiled cell population to compound-resolution chemistry provenance. The canonical RNA-seq workspace used here aggregates 615,793 repaired well-level transcriptional profiles across 12 combinatorial libraries and 4 cell lines (HEK293, A549, H1650, THP1) and includes 615,721 profiles with valid scVI latent coordinates for cross-cell projection [8]. Earlier preprints in this package establish reproducible compound and building-block transcriptional structure (paper 2), chemistry-to-phenotype generalization across an explicit difficulty ladder (paper 3), and learned cross-cell-type transfer of chemical programs (paper 4). The present preprint addresses the complementary question: whether chemical states measured at the level of individual combinatorial compounds can be placed in a genetic-perturbation frame, and whether the resulting map produces calibrated, interpretable, and discovery-relevant matches across many distinct chemistries.

The analysis is conservative. Rather than forcing raw expression matrices from different platforms into one normalization, every perturbation is represented as a signed rank signature: the genes most increased and most decreased relative to an appropriate background, with a fixed top-k cutoff. Chemical and CRISPR signatures are compared by mimic overlap, reverse overlap, rank cosine over shared top genes, and a permutation-null calibration that preserves the size of the CRISPR signature. The framework is robust to dataset and depth differences and addresses the strongest question the data can answer: do the top transcriptional consequences of two perturbations agree more than expected by chance.

Each Z-Screen library and cell line is given a chemistry resolution based on its coverage. ZEL024 in HEK293 has 8,599 unique 4-position tuples (bb0+bb1+bb2+bb3) with 10 or more wells; this system is analyzed at full tuple resolution. ZEL028-2 is a 4-position library with substantive positions at bb1, bb2, bb3, bb4 (no bb0). Its per-tuple coverage is much shallower than ZEL024, so the threshold that supports a usable tuple-level analysis drops to >=2 wells per tuple, where 4,041 (HEK293), 1,728 (A549), and 765 (H1650) distinct ZEL028-2 tuples qualify; ZEL028-2 is therefore analyzed at full-tuple (bb1+bb2+bb3+bb4) resolution at this relaxed threshold. The 5- to 6-fold lower per-tuple well support relative to ZEL024 is recorded in the metadata for every ZEL028-2 signature (median 2 wells per tuple, vs ZEL024's median 13), so individual ZEL028-2 tuple signatures are inherently noisier and should be read as a leaderboard of chemistry-resolved hypotheses rather than as confirmed per-tuple calls. A complementary single-position bb1 aggregation of the same library is preserved in `paper5/tables/` as a higher-stability lower-resolution view. ZEL031 has only two substantive positions (bb0+bb1), so its 3,261 to 3,556 2-position tuples are analyzed at tuple resolution with a lower well-count threshold flagged in the metadata.

Results

A rank-based comparison framework places chemistry and CRISPR perturbations in a shared transcriptional frame

Every Z-Screen perturbation, whether a single chemical tuple or a combination of building blocks, was reduced to a signed rank signature against an appropriate background. For tuple signatures the target was wells whose chemistry annotation matched the library's populated building-block positions exactly (four positions in ZEL024 and ZEL028-2, two in ZEL031), and the background was the same library and cell line excluding the target tuple wells. For BB-pair signatures the target was wells matching two designated positions exactly, with the same library and cell line as background. Each signature is annotated with the number of wells aggregated and the number of distinct tuples averaged into it, so the chemistry resolution of each comparison is explicit in the output tables.

Public CRISPR signatures were already available in the Replogle K562 genome-wide Perturb-seq atlas (8,603 perturbations [4]), the Norman K562 CRISPR-activation atlas (236 signatures spanning 105 single-gene and 131 paired perturbations [3]), and the scPerturb harmonized matched-cell THP1 atlas (78 single-target and combinatorial perturbations [9]). Each CRISPR perturbation was reduced to the same up/down rank representation as the Z-Screen signatures. Chemistry and CRISPR signatures were then compared by:

- mimic overlap (chemistry-up matches CRISPR-up plus chemistry-down matches CRISPR-down),
- reverse overlap (chemistry-up matches CRISPR-down plus chemistry-down matches CRISPR-up),
- net mimic overlap (mimic minus reverse),
- rank cosine over shared top-250 genes.

For each chemistry-by-reference comparison the top 50 CRISPR matches per chemical query were retained. The top 10,000 rows by net mimic overlap and total mimic overlap were then calibrated by drawing 1,000 random gene sets matched in size to each CRISPR signature from the dataset-specific gene universe and recomputing the score under each random draw. Empirical p-values were converted to Benjamini-Hochberg q-values across the calibrated rows.

Rank signatures were used in preference to raw expression vectors. Replogle K562 was generated on a different sequencing platform than Z-Screen at much higher per-cell depth; Norman is CRISPR-activation in K562; scPerturb is a harmonized merger of multiple ECCITE-seq panels in THP1. Forcing all of these into one normalization risks introducing platform-specific artifacts that the comparison would then "discover." Rank signatures discard calibration differences and preserve the strongest property the data shares: the ordering of which genes go up and which go down. The full reproduction order is in `paper5/README.md`.

Chemistry resolution determines the CRISPR neighborhood that a Z-Screen signature recovers (Figure 1)

Z-Screen libraries differ in how many wells fall into each chemistry unit. ZEL024 in HEK293 carries 13,931 unique 4-position tuples (bb0+bb1+bb2+bb3); 8,599 of those have at least 10 wells, the threshold used for tuple-level signatures. The same library can also be aggregated into 84 bb0+bb1 pair signatures (each averaging approximately 165 distinct tuples) or into 83 single-bb3 signatures (each averaging approximately 168 distinct tuples). The same wells flow into all three resolutions; what changes is how much chemistry is collapsed into one signature.

Single-bb3 signatures were built from the same cells, and each tuple's top-5 Replogle CRISPR matches were compared with the matches that the containing bb0+bb1 pair and the containing bb3 single-position signature would produce. For 99.7% of tuples, the top-5 CRISPR matches at tuple resolution shared zero genes with the top-5 at bb3 single-BB resolution; the median Jaccard overlap between the tuple top-5 and the bb3 top-5 was 0.000. The same pattern appeared for tuple-versus-pair (99.7% zero overlap, median Jaccard 0.000).

Two analyses that nominally describe "the same chemistry" can therefore produce essentially independent CRISPR neighborhoods depending on how much chemistry is averaged into the signature. The implication is practical: a discovery team that wants to act on a CRISPR-like match needs that match attached to the specific chemistry responsible for it. Tuple-level signatures are therefore used for the headline ZEL024 / HEK293 result, and all coarser-resolution analyses are reported with explicit n-wells and n-tuples-averaged annotations.

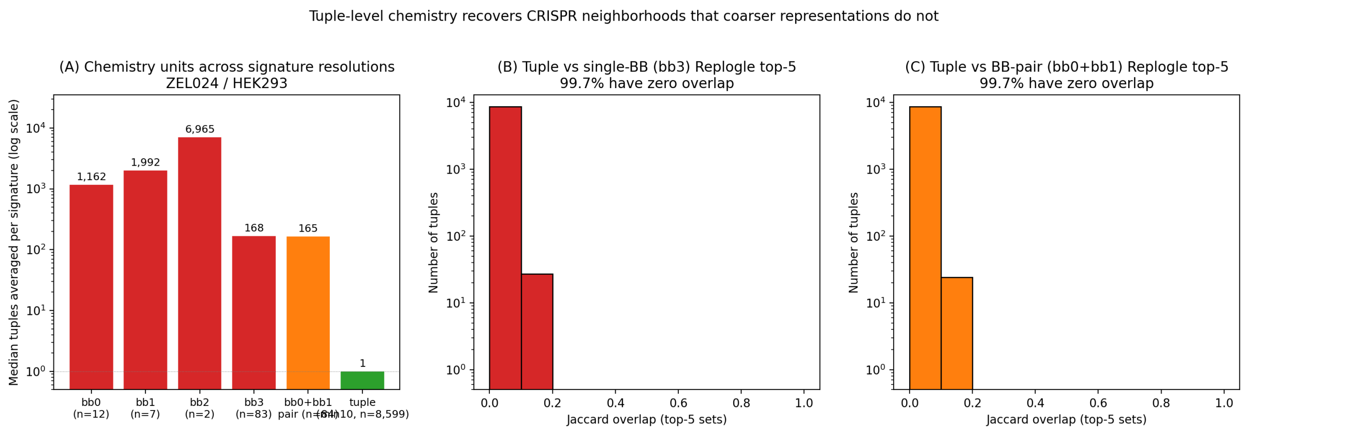


Figure 1. Chemistry resolution determines the CRISPR neighborhood. (A) Median number of distinct chemical tuples averaged into a signature at each resolution for ZEL024 / HEK293, log-scale. Single-BB signatures average over 168 to 6,966 tuples per signature; tuple signatures average over one. (B) Top-5 Replogle CRISPR matches at tuple resolution share zero genes with the matches at single-BB (bb3) resolution for 99.7 percent of the 8,599 tuples. (C) The same comparison at tuple-versus-bb-pair resolution. The same wells flow into both signatures in each comparison.

Tuple-level chemistry maps onto 1,714 distinct CRISPR programs in ZEL024 / HEK293 (Figure 2)

The headline scan compared each of the 8,599 ZEL024 / HEK293 tuple signatures against the 8,603 Replogle K562 CRISPR-knockout signatures [4], a comparison space of approximately 74 million chemistry-by-CRISPR pair scores, calibrated for the top 10,000 matches by net mimic overlap. The calibrated rows were dominated by hits at q 0.05 across 1,276 distinct chemical tuples and 1,714 distinct CRISPR perturbation programs. The strongest single matches reached net z above 14 (Figure 2 panel A), with the top 15 distinct CRISPR programs all between net z 12.8 and 14.8. The biology was coherent: DNA replication (MCM2, SLBP, CHAF1B, ARIH1, ADSL), mitochondrial function (MRPL3, MRPS30, COX17, COA5, ATP5F1A, GFM1, HSPE1), trafficking and nonsense-mediated decay (EFR3A, DDOST, SMG5).

The same calibrated table also contains the strongest internal-consistency signal in the analysis. Among the top-100 best-z tuples, 12 distinct CRISPR programs were each reached by two or more chemically distinct tuples (Figure 2 panel B). EFR3A was matched by 7 distinct tuples, TONSL by 6, MCM2 by 5; multiple GART, MRPL42, EIF4G2, MRPL53, and COA5 hits each recurred 3 times. In each of these cases independent chemistries (different combinations of the four populated ZEL024 building-block positions: bb0, bb1, bb2, bb3) produced transcriptional states that overlap with the same CRISPR perturbation, which is the empirical signature expected if the chemistry-to-CRISPR map carries real biological information rather than per-compound noise.

A small number of CRISPR perturbations recurred as top neighbors of many otherwise unrelated chemistries. Broadly responsive CRISPR knockouts can match many transcriptional states by being broadly responsive themselves, and a benchmark that does not control for that property will appear stronger than it is. The calibrated top-10,000 table was therefore rerun while progressively dropping the 5, 10, 25, and 50 most-recurrent CRISPR perturbations (Figure 2 panel C). After dropping the 50 most-recurrent hubs, 1,236 distinct chemical tuples and 1,664 distinct CRISPR programs still passed q 0.05, so the program diversity does not collapse onto a small set of hub perturbations.

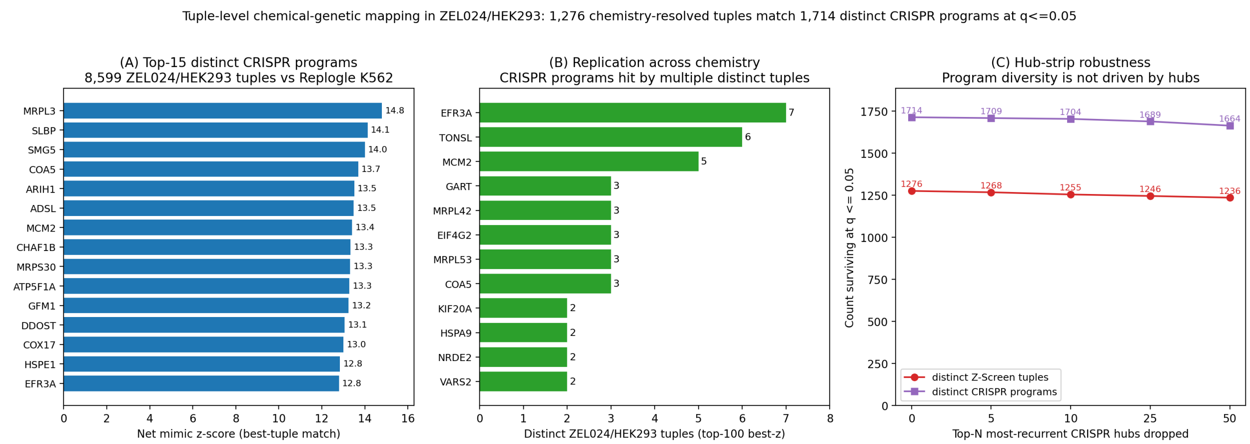


Figure 2. Tuple-level chemistry maps onto 1,714 distinct CRISPR programs in ZEL024 / HEK293. (A) Top-15 distinct CRISPR programs by best Z-Screen tuple z-score, deduplicated to one tuple per CRISPR program. (B) Replication-across-chemistry: among the top-100 best-z tuples, CRISPR programs reached by multiple chemically distinct tuples. (C) Hub-strip robustness: the calibrated top-10,000 table after dropping the top-N most-recurrent CRISPR perturbations. Tuple and program diversity at q 0.05 are essentially unchanged at N = 50.

Same-cell positive controls anchor the framework (Figure 3)

Two same-cell controls in matched THP1 anchor the approach. The first is MZ1, a small-molecule PROTAC that selectively degrades BRD4 [10]. MZ1 lives in a sparse THP1 control library that lacks adequate same-library background, so its signature was built against a same-cell compound-pool fallback background (all THP1 compound-treated cells excluding MZ1). Under that background, MZ1 recovered BRD4 as the top calibrated CRISPR-like signature in the local THP1 panel (net mimic = 20, rank cosine = 0.539, calibrated net z = 6.84, empirical q = 0.015).

The second control is STC-15, a METTL3 inhibitor. The direct target METTL3 is not present in the local THP1 CRISPR panel, but the documented downstream consequence of METTL3 inhibition is interferon-pathway activation [11]. STC-15 recovered IRF1 as the second calibrated CRISPR-like match (z = 3.06, q = 0.045), consistent with that downstream phenotype, with IRF7 as the third match (z = 2.70, q = 0.096).

The fallback-background construction for MZ1 is broader than a same-library vehicle control would be, so the absolute z-score should be read in that context. The structure of the result, BRD4 recovered as the top match across permutation seeds, is robust. MZ1 and STC-15 anchor the two complementary cases the framework will encounter in production: the direct genetic analog is in the panel and is recovered as the top hit, or the direct analog is absent and the top hit is a downstream pathway rather than a generic stress signature.

The non-BRD4 neighbors in panel A of Figure 3 are not additional direct target calls. SUPT16H+SUPT5H, FLI1+MED24, MED24+SMARCD1, EP300+MED24, EP300+INTS1, and INTS1+SMARCD1 sit in the BRD4-adjacent transcription-elongation and chromatin co-regulator neighborhood that MZ1 perturbs by removing BRD4 from the cell. CAV1 is less directly established as BRD4-regulated in THP1 and is treated as a state-neighborhood match rather than a direct mechanistic claim.

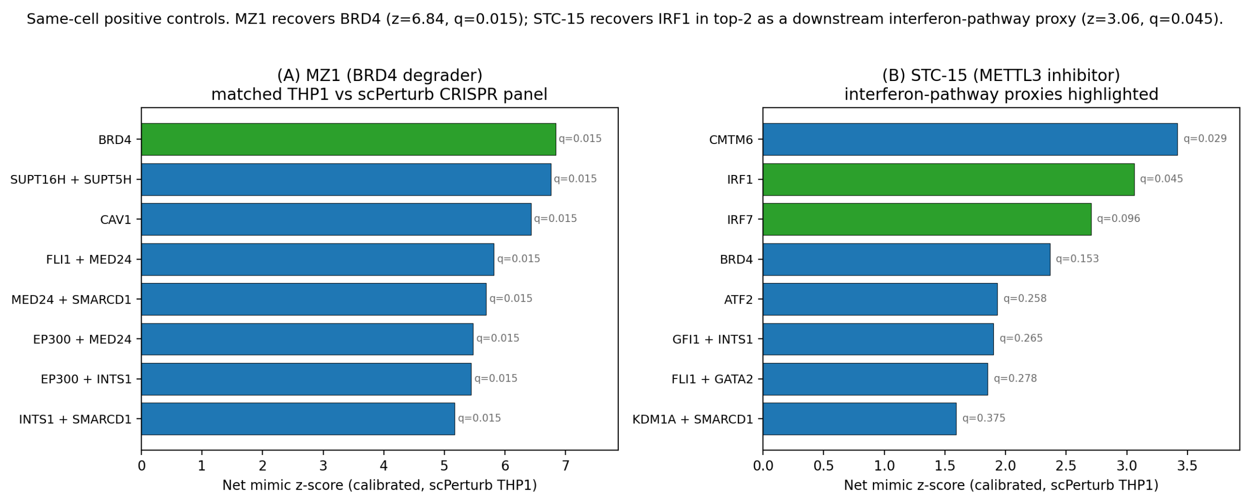


Figure 3. Same-cell positive controls in matched THP1 with calibrated empirical q-values from a permutation-null gene-set calibration. (A) MZ1 (BRD4-selective PROTAC degrader): BRD4 is the top calibrated CRISPR-like signature (z = 6.84, q = 0.015), with subsequent neighbors falling in BRD4-adjacent transcriptional and chromatin programs. (B) STC-15 (METTL3 inhibitor): METTL3 itself is absent from the local CRISPR panel; the downstream interferon-pathway proxy IRF1 is recovered in the top two calibrated matches (z = 3.06, q = 0.045).

A discovery example: BAY-293 produces a coherent V-ATPase / vesicular-trafficking program (Figure 4)

The matched-control results show that the framework recovers known biology when the genetic analog is present. The more discovery-relevant question is whether the framework generates mechanism hypotheses beyond a compound's primary pharmacology. The strongest example in the dataset is BAY-293, an SOS1 inhibitor profiled in A549 in a non-combinatorial control library.

The top 12 calibrated Replogle K562 matches for BAY-293 in A549 all sat at empirical q = 0.003, with net z between 9.5 and 11.9. The matches concentrated on the V-ATPase complex (ATP6V1A, ATP6V1H, ATP6V1B2, ATP6V1C1, ATP6V1E1) and adjacent vesicular-trafficking machinery (ZW10, SACM1L, TMED2, ZNHIT1, WDR7, CCDC115, CLTC). RAS signaling intersects endosomal traffic in multiple documented places, so a transcriptional state that overlaps both V-

ATPase and trafficking knockdown signatures is biologically interpretable; it is also not the answer a simple primary-target search would have produced.

The framework therefore provides a discovery layer: a calibrated mechanism hypothesis at compound resolution, ranked alongside the rest of the genome-wide CRISPR atlas. A discovery team starting from a SOS1 program could use this output to ask whether V-ATPase / trafficking liability is contributing to BAY-293's phenotype, and could design follow-up experiments specifically against that hypothesis. That is a different deliverable from a similarity score against a single compound database.

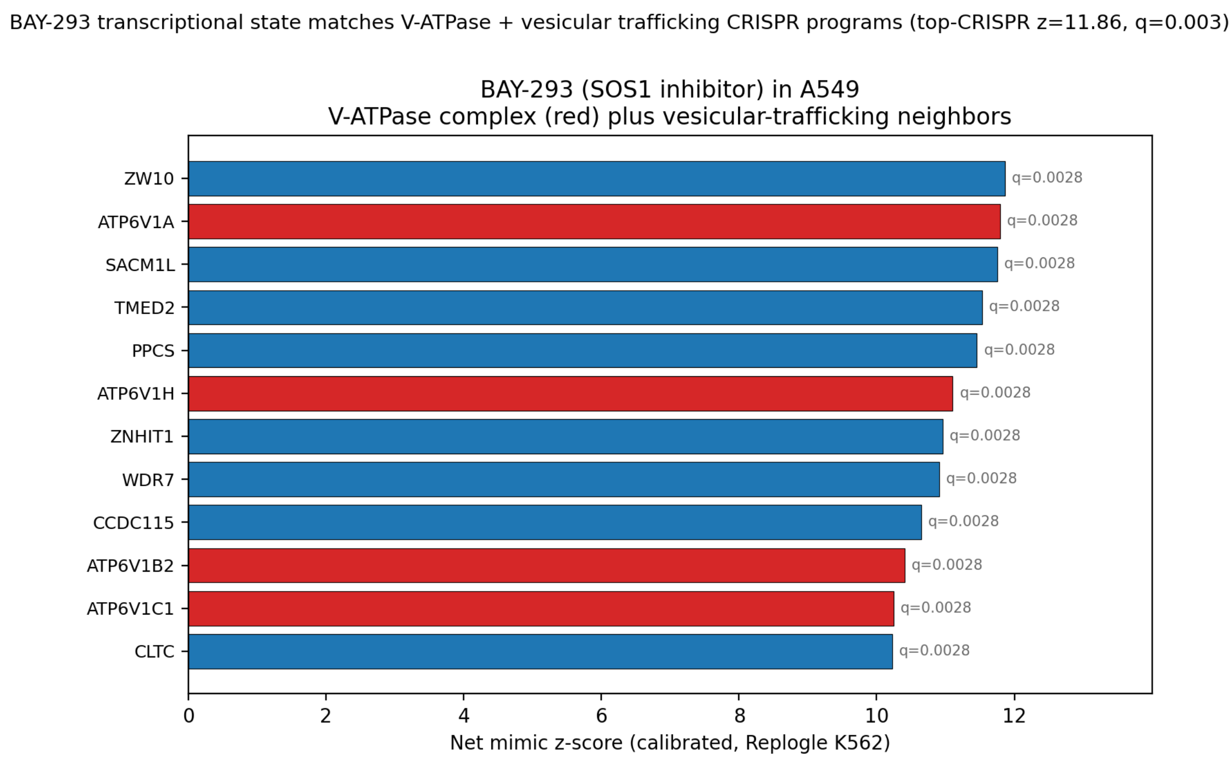


Figure 4. Discovery example. The SOS1 inhibitor BAY-293 in A549 matches the V-ATPase complex (red) and vesicular-trafficking CRISPR knockouts (blue) at calibrated empirical q = 0.003. Top hit ZW10 reaches net z = 11.86; nine V-ATPase or trafficking components reach net z above 10.

Genome-wide target enrichment and combinatorial logic (Figure 5)

Two further analyses test whether the framework recovers structure beyond any one anecdote.

The first is a known-target enrichment scan. Among 29 Z-Screen compounds with a documented primary target represented in the Replogle K562 panel, 12 had the documented target ranked in the top 15 percent of the cross-cell genome-wide ranking. Several reached the top 1 to 10 percent: FGFR1 for derazantinib at rank 145 of 8,839 (~1.6 percent), METTL3 for STC-15 at rank 784 in A549 (~9 percent), MAPK14 for AZD-7624 inside the top 10 percent in A549. Failures were biologically informative rather than uniform. Cobimetinib's MAP2K2 ranked at 4,694 (allosteric inhibition does not always produce a knockout-like transcriptional consequence; this is an honest negative for a state-overlap framework rather than a method failure). GSK126's EZH2 was anti-correlated rather than mimicking, consistent with the long-half-life of EZH2 protein under enzymatic inhibition. Veliparib's PARP2 ranked at 3,696 (HEK293) and 1,206 (A549). The cross-cell scan is not a target-deconvolution engine; it is enriched well above chance for documented mechanisms when the molecular pharmacology produces a knockout-like transcriptional state.

The second analysis tests combinatorial logic at chemistry-resolved level. The Norman K562 CRISPR-activation atlas contains 131 double-gene perturbations whose constituent single-gene perturbations are also profiled. For each Norman double, the Spearman correlation across the 8,599 ZEL024 / HEK293 tuples between (tuple-vs-double match score) and (sum of tuple-vs-single-A and tuple-vs-single-B match scores) was computed. All 131 doubles produced positive correlations with $p < 0.05$ (median Spearman rho = 0.353). The strongest pairs (ETS2+MAPK1, MAP2K3+ELMSAN1, CEBPB+MAPK1, MAP2K3+MAP2K6, FOSB+OSR2) reached rho between 0.60 and 0.69. For 98.5 percent of the doubles, the best Z-Screen tuple's score against the double exceeded its score against either constituent single. Chemistry-resolved tuple signatures therefore track the same combinatorial logic the underlying CRISPR atlas was built to express.

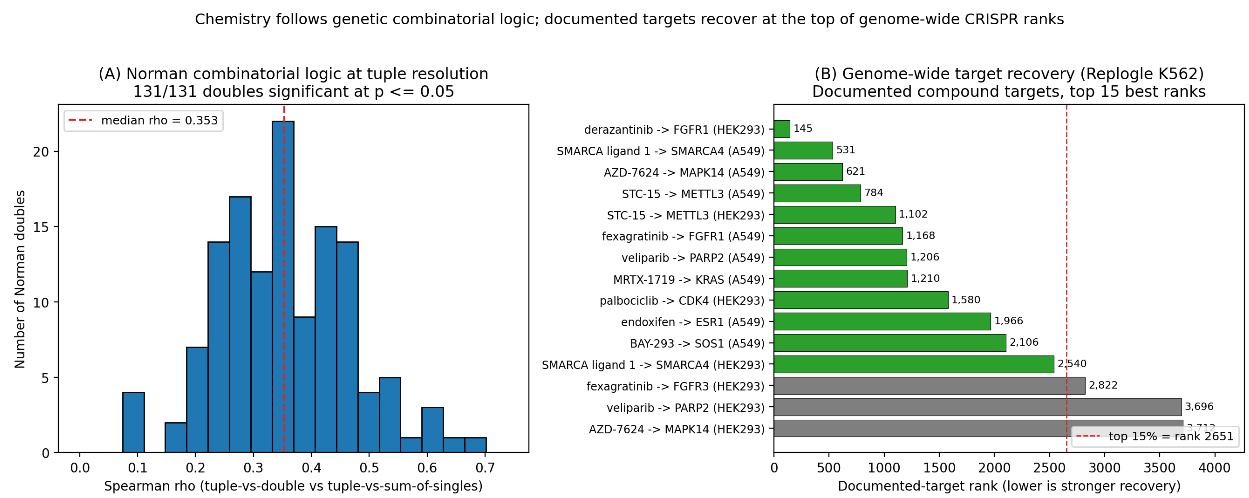
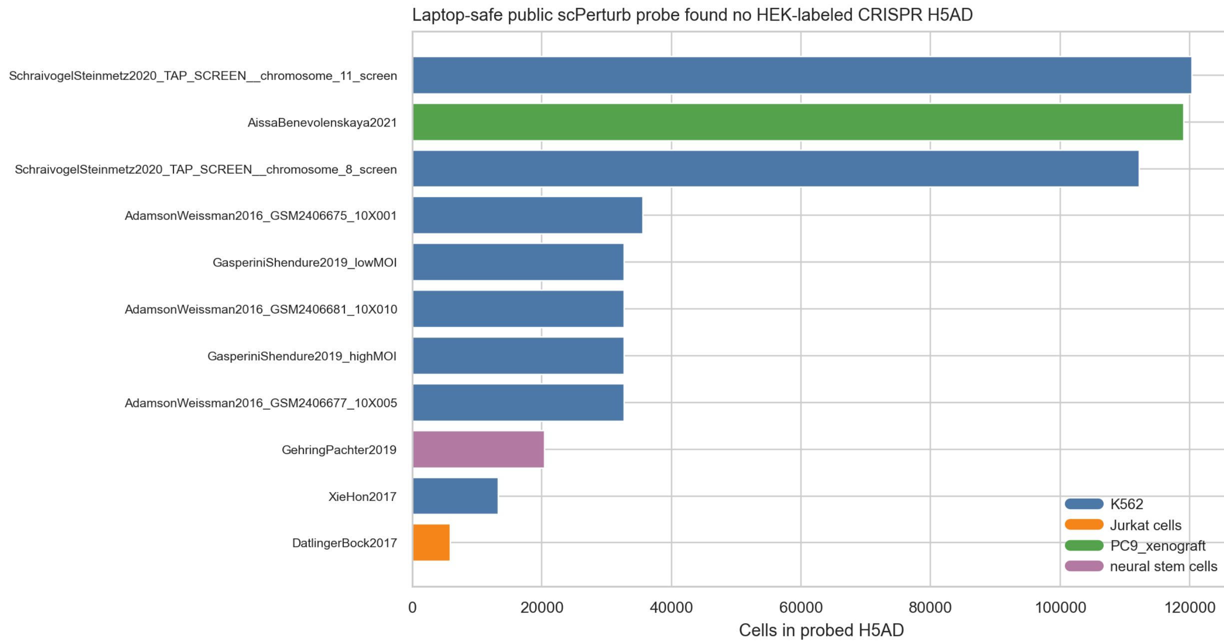


Figure 5. Combinatorial logic and genome-wide target recovery. (A) Distribution of Spearman correlations across 131 Norman K562 doubles between tuple-vs-double match scores and sum-of-singles match scores; all 131 doubles significant at p 0.05; median rho = 0.353. (B) Documented-target ranks within the Replogle K562 cross-cell ranking for the top-15 best-recovered compound-target pairs; the dashed line marks the top 15 percent of the 8,839-perturbation reference.

A targeted public probe did not surface a matched HEK293 single-cell CRISPR atlas (Supplementary Figure 1)

11 scPerturb H5AD files under 500 MB each were probed by reading observation metadata only, looking for a HEK293 or HEK293T single-cell CRISPR dataset that would allow direct same-cell evaluation against the HEK293-rich Z-Screen libraries. The probed datasets covered K562, Jurkat / T-cell, PC9 xenograft, and neural stem cell perturbation panels. None were HEK-labeled CRISPR datasets, so the present manuscript uses the cross-cell projection from paper 4 to bridge the HEK293

chemistry to the K562 Replogle reference rather than identifying a matched-cell HEK293 CRISPR atlas locally. The probe is intentionally laptop-scale and is not a comprehensive survey of all public HEK293 perturbation data.

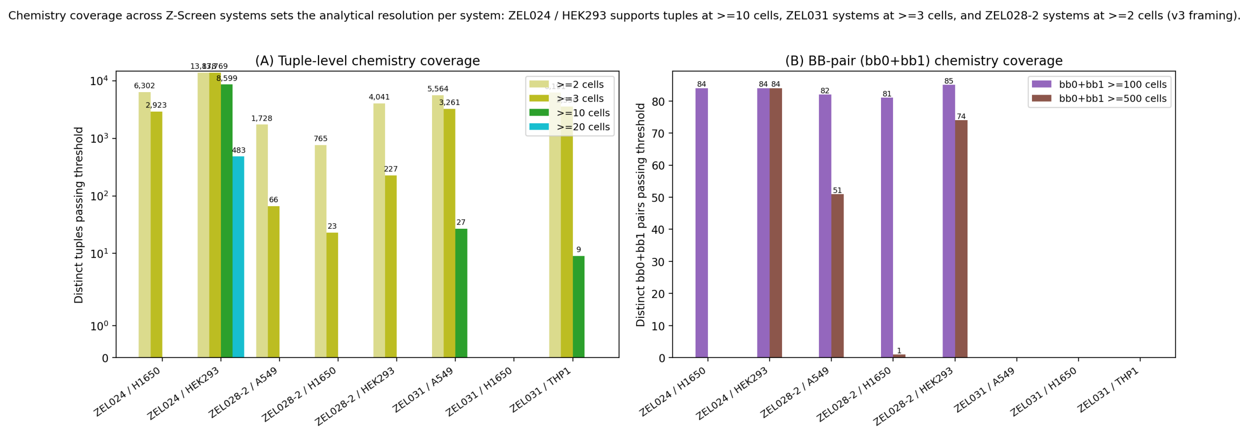


Supplementary Figure 1. Targeted probe of 11 small (<500 MB) scPerturb H5AD files for matched HEK293 single-cell CRISPR data. Each row reports cell-line label, perturbation type, perturbation count, and dataset dimensions; no HEK-labeled CRISPR dataset was identified locally.

Coverage at other chemistry resolutions in other systems (Supplementary Figure 2)

The same framework runs across the other Z-Screen systems at the resolution each one supports. ZEL028-2 has 61,396 unique tuples in HEK293, 40,622 in A549, and 25,906 in H1650, with median 2 wells per tuple at the densest cell line. Under a relaxed >=2-well per-tuple threshold, 4,041 ZEL028-2 / HEK293 tuples, 1,728 A549 tuples, and 765 H1650 tuples enter the analysis. The headline ZEL028-2 / HEK293 tuple-vs-Replogle scan against the 8,603 Replogle K562 CRISPR signatures yielded a calibrated top-10,000 table dominated by hits at empirical q 0.05, covering 943 distinct chemical tuples and 285 distinct CRISPR perturbation programs. The strongest matches reach net z = 24.7 in HEK293 (max), 20.5 in A549, and 18.6 in H1650. Top distinct CRISPR programs concentrate on RNA processing (DICER1, DROSHA, CNOT3, SMG5, and the EXOSC2 / EXOSC4 / EXOSC6 / EXOSC8 exosome subunits), the mitochondrial translation machinery (MRPL3, MRPL10, MRPL18, MRPL19, MRPL35, MRPL43, IARS2, VARS2, GFM1), and the helicase / translation-initiation programs (DHX36, EIF3H, RPS4X, RPS6, RACK1). 15 CRISPR programs appear in the top-50 deduplicated hits in all three ZEL028-2 cell lines (DHX36, DICER1, DROSHA, CNOT3, EXOSC2 / EXOSC4 / EXOSC6, GFM1, MRPL3 / MRPL10 / MRPL35 / MRPL43, MVD, RACK1, SMG5), which is the cross-cell-line robustness expected if these tuple signatures track real chemistry-driven biology rather than per-tuple noise.

The interpretation requires the per-tuple support caveat. Because each ZEL028-2 tuple signature aggregates a median of 2 wells, individual tuple signatures carry substantially more sampling noise than the ZEL024 / HEK293 tuple signatures (median 13 wells). The headline numbers should therefore be read as a chemistry-resolved leaderboard whose strongest entries reproduce the same biology themes the ZEL024 analysis surfaced (RNA processing and mitochondrial translation) rather than as a list of per-tuple definitive calls. Per-tuple confirmation requires deeper coverage on the specific tuple. The complementary single-position bb1 aggregation of the same ZEL028-2 wells (74 bb1 identities at minimum 500 wells in HEK293, 51 in A549 at minimum 500 wells, and 66 in H1650 at minimum 250 wells) is preserved as a higher-stability lower-resolution view in `paper5/tables/`. ZEL031 is a two-position library, so its 3,261 (A549) and 3,556 (THP1) tuples at minimum 3 wells already represent its highest practical chemistry resolution. ZEL031 systems are reported at tuple resolution with the lower well-count threshold explicitly flagged.



Supplementary Figure 2. Z-Screen chemistry coverage across systems. (A) Distinct tuples passing >=2, >=3, >=10, and >=20 well thresholds in each system. ZEL024 / HEK293 supports tuple-level signatures at >=10 wells, ZEL031 systems at >=3 wells, and ZEL028-2 systems at >=2 wells. (B) Distinct bb0+bb1 pairs passing 100 and 500 well thresholds in libraries that populate bb0; ZEL028-2 (which has no bb0) is now reported at full-tuple (bb1+bb2+bb3+bb4) resolution in panel A. ZEL024 supports both tuple and pair resolution; ZEL031 has too many distinct pairs for any single pair to carry meaningful well counts.

Cell-state transfer connects chemistry mapping across cell contexts (Figure 6)

The CRISPR atlases used here are dominated by K562 and THP1, while the densest combinatorial Z-Screen libraries in the canonical workspace are profiled in HEK293, A549, and H1650. Direct same-cell evaluation is therefore possible for THP1 (compounds and pair signatures) and partially for A549 (BAY-293 and other compounds). The remaining systems benefit from the cross-cell projection model developed in paper 4, which trains a centered ridge transfer between paired cell-line landmarks in the 32-dimensional scVI latent space.

Within the valid bridges in the present dataset, A549-to-THP1 transfer in ZEL031 reached transfer cosine 0.66 for bb1 programs and 0.59 for bb0; HEK293-to-H1650 and HEK293-to-A549 transfer in the HEK293-rich libraries reached weighted cosines between 0.68 and 0.84, and ridge transfer improved on direct identity reuse in every paired bridge evaluated. The cross-cell transfer is reported in detail in paper 4; here it sets the boundary on what the CRISPR mapping can claim. The data support the claim that chemical programs are learnable and transferable across measured cell contexts, which extends the practical reach of the CRISPR mapping beyond the cell types where direct K562 or THP1 reference comparisons are possible.

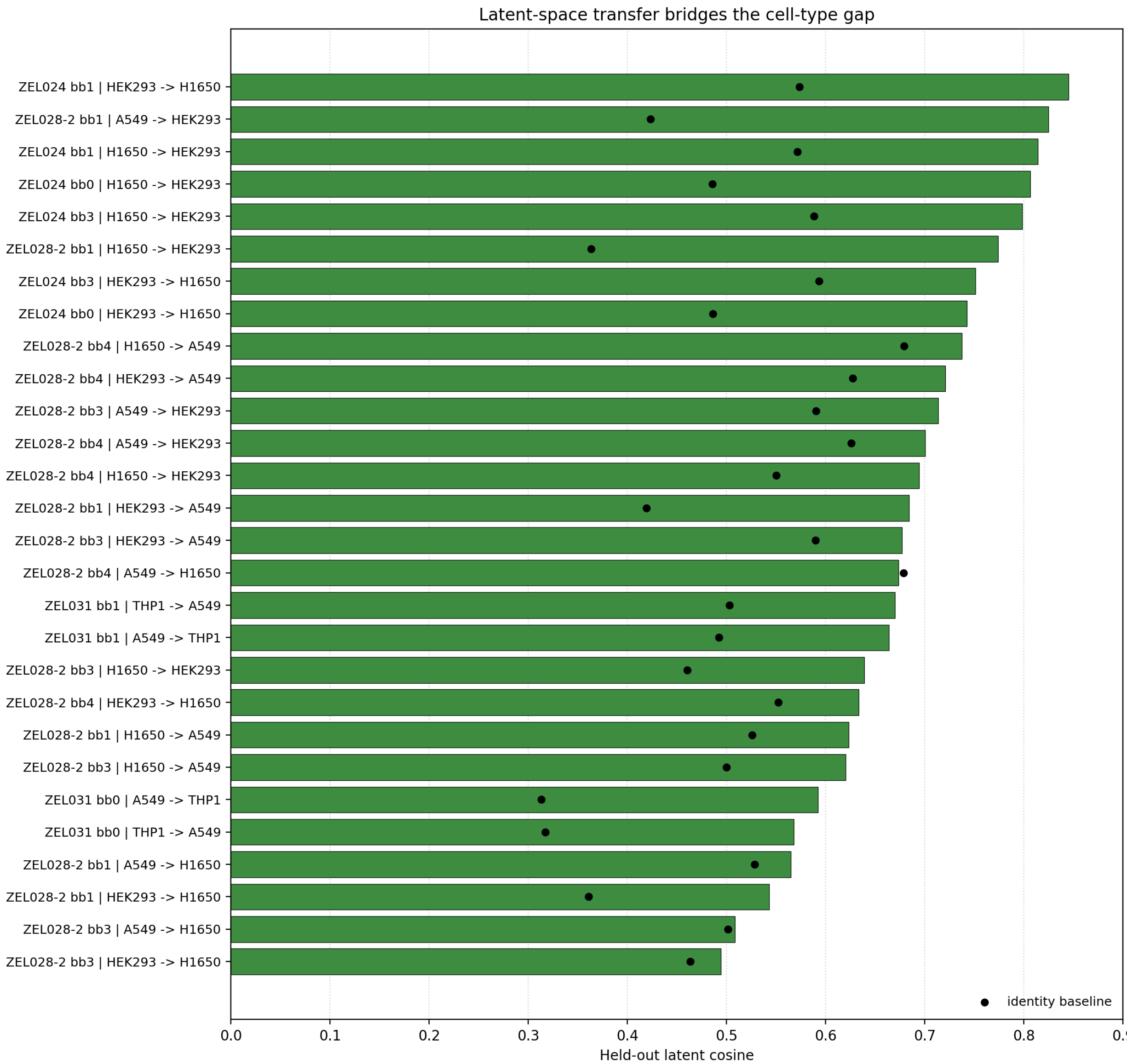


Figure 6. Cell-state transfer of Z-Screen chemical programs across cell contexts. The matched bridges support A549-to-THP1 transfer in ZEL031 and HEK293-to-A549 / H1650 transfer in the HEK293-rich libraries. Black points show direct identity baselines; green bars show learned ridge transfer performance.

Discussion

The result that frames the rest of the paper is the headline ZEL024 / HEK293 scan: 1,276 distinct chemical tuples each carry a calibrated CRISPR-like hypothesis at empirical q 0.05, and the resulting hits cover 1,714 distinct CRISPR perturbation programs. The numbers themselves are not the central result; the central result is that each one of those tuples is a specific 4-position (bb0 / bb1 / bb2 / bb3) chemistry that a discovery team can synthesize, vary, and test, with the underlying CRISPR program available for orthogonal validation. The framework therefore turns Z-Screen output from a catalog of measurements into a structured search engine over genetic perturbation space.

The matched-cell controls validate the framework's calibration. MZ1 recovers BRD4 with a clean separation against the same-cell THP1 CRISPR panel; STC-15 recovers IRF1 within the top two calibrated matches as a downstream proxy of METTL3 inhibition. Where the genetic analog is present, the framework recovers it; where the genetic analog is absent, the top hit is a documented downstream pathway rather than a generic stress signature. BAY-293 in A549 demonstrates the discovery layer: the framework surfaces a coherent V-ATPase / vesicular trafficking hypothesis ranked alongside, but not predicted by, primary pharmacology.

The combinatorial-logic test is the strongest evidence that the framework is reading real combinatorial chemistry rather than per-tuple noise. All 131 Norman doubles tested produce positive Spearman correlations between tuple-vs-double scores and tuple-vs-sum-of-singles scores. The chemistry follows the genetics' combinatorial structure at chemistry-resolved level, which is the same structure the Norman atlas was built to express. That is not a result a one-profile-per-compound platform such as L1000 can produce, because there is no internal chemistry structure inside a single L1000 entry to test combinatorial logic against.

L1000 is also a useful concrete reference for data-quality positioning. Of the 47 named Z-Screen control compounds, 14 (29.8%) appear in the LINCS L1000 CMap LINCS 2020 build (palbociclib, sorafenib, crizotinib, ceritinib, rucaparib, veliparib, lapatinib, cobimetinib, GSK126, CP-673451, AZD-3463, BMS-509744, BGT226, ZM-336372), and 3 of 4 Z-Screen cell lines (HEK293, A549, THP1) are present in L1000. That overlap was used to run an explicit cross-platform Spearman concordance test between Z-Screen up/down rank signatures and LINCS L1000 level-5 consensus signatures (NCBI GEO GSE70138 and GSE92742), restricted to the 12,328-gene LINCS-visible universe so both platforms are scored in the same gene space. Across 11 (compound, cell_line) pairs with adequate coverage on both platforms, all 11 produced positive cross-platform correlation (median rho = 0.490, mean = 0.393, 8 of 11 above rho = 0.2). Five compounds reached empirical p 0.048 against a same-cell-line unmatched-compound permutation null: crizotinib in A549 (rho = 0.679), ZM-336372 in A549 (rho = 0.625), rucaparib in A549 (rho = 0.530), sorafenib in A549 (rho = 0.505), and veliparib in A549 (rho = 0.490). The HEK293 cell-line-pooled comparison, where sorafenib in HEK293T reached rho = 0.597, separated matched from unmatched compound pairings at one-sided Mann-Whitney p = 0.009. Z-Screen rank signatures therefore recover compound-specific transcriptional structure that an independent reference platform also reports, in a gene-restricted concordance test that uses no platform-specific calibration. Full tables are in `Benchmarking/` in this repository.

CRISPR similarity is not target identification. A small molecule may resemble a knockout because it directly engages the encoded protein, because it perturbs an upstream regulator, because it triggers a compensatory state, or because both perturbations converge on a broad stress program. The framework therefore treats every match as a state-level hypothesis until orthogonal validation upgrades it. The recurrent-neighbor diagnostic and the hub-strip analysis are reported alongside the calibrated headline numbers because broadly responsive CRISPR perturbations need to be flagged rather than silently lost in the leaderboard.

The framework is naturally tied to follow-up. The top combinatorial chemistries pointing toward a desired CRISPR state are concrete tuples with explicit per-position chemistry: bb0 / bb1 / bb2 / bb3 in ZEL024, bb1 / bb2 / bb3 / bb4 in ZEL028-2, bb0 / bb1 in ZEL031. A follow-up library can enrich the partner chemistry around those tuples, profile the new compounds, and update the model. The cross-cell transfer connection in paper 4 closes the remaining gap by mapping chemistry from a cell type that is dense in the screen into a cell type that carries the matched CRISPR reference.

A compact internal validation panel falls out of the analysis. BRD4 as a positive control; MED24, INTS1, EP300, FLI1, HDAC3, MYC, and SMARCD1 to test the recurrent transcription-and-chromatin programs; PDCD1LG2 for the strongest checkpoint-like hypothesis from the HEK293 BB-pair signatures; SMAD4 as a stress / state comparator; JAK2, IFNGR1, IFNGR2, and STAT1 to test the interferon-pathway proxies that emerged for STC-15 and several non-annotated discovery candidates. Running that panel in matched THP1 and HEK293 would directly test the two main claims: that known chemical-genetic analogs are recovered when present, and that the chemistry-resolved hypotheses replicate in a matched CRISPR context.

For drug discovery teams, the practical implication concerns the front end of a campaign. Instead of starting from a target and searching a separate compound database, a team can start from a desired CRISPR state and pull a ranked, calibrated list of combinatorial chemistries that move cells toward that state, with the chemistry decomposable into building-block coordinates that a medicinal chemistry team can act on.

Methods

Z-Screen data inputs

The analysis used the repaired canonical Z-Screen RNA-seq aggregate at `data/ZScreen_Canonical_Dataset/RNASe` Primary files: `Full_ANNData_object_compressed.h5ad` (615,793 wells x 36,601 genes; rows are well-level pseudobulks of 5 to 10 cells per well, stored under the AnnData `obs` convention), `canonical_obs.parquet`, `canonical_chemistry.parquet`, and `canonical_scvi_latents.parquet`. The H5AD repair removed 140 wells affected by corrupted sparse-matrix chunks; the linked parquet files were aligned to the same `obs_index` after repair. See `DATA_REPAIR_REPORT.md`.

Rank signatures rather than expression vectors

The three CRISPR references and the Z-Screen RNA aggregate were generated on different platforms at different per-cell depths and with different normalization conventions. Forcing all of them into a common normalization risks introducing platform-specific artifacts that would then appear as similarity. Rank signatures discard calibration differences and preserve the strongest property the data shares: the ordering of genes most up- and most down-regulated relative to background. The choice is conservative. It costs some statistical power compared with full expression-vector regressions, and provides the ability to compare across heterogeneous public datasets without assuming they were normalized the same way.

Z-Screen tuple signatures

For each library and cell line, building-block-annotation wells (`chemistry_grain == "building_block_annotation"`) were grouped by the concatenated tuple identifier `bb0_id|bb1_id|bb2_id|bb3_id` with unused positions filled as the literal string "NA" (so a ZEL024 tuple has the form `bb0|bb1|bb2|bb3|NA`, a ZEL028-2 tuple has `NA|bb1|bb2|bb3|bb4`, and a ZEL031 tuple has `bb0|bb1|NA|NA|NA`). Tuples with at least 10 wells were retained for the headline ZEL024 / HEK293 analysis (8,599 tuples, median 13 wells per tuple). A supplementary >=20-well threshold yielded 483 tuples and is reported alongside the headline as a stricter sensitivity check. ZEL031 systems used a >=3-well threshold (3,261 in A549, 3,556 in THP1) because the two-position library is too diverse for higher thresholds. ZEL028-2 systems used a >=2-well threshold because per-tuple coverage in that library does not support the >=10-well or >=3-well thresholds; under >=2 wells, 4,041 (HEK293), 1,728 (A549), and 765 (H1650) tuples qualify, with a median of 2 wells per qualifying tuple. The shallower per-tuple support in ZEL028-2 is recorded in every signature's `n_cells` field (the field name follows the AnnData convention; the values count well-level pseudobulks), and is the reason the ZEL028-2 tuple analysis is interpreted as a chemistry-resolved leaderboard rather than as per-tuple definitive calls. For each tuple, target-well counts were aggregated by direct sparse-matrix slicing of the H5AD; the background was the same library and cell line excluding target-tuple wells. The effect vector was logCPM(target) - logCPM(background). Genes were ranked separately in the up and down directions; the top 250 per direction were retained.

The same-library, same-cell-line background was used in preference to a global baseline so that bias from chemistry-independent differences across libraries (e.g. the synthesis chemistry of one library affecting baseline cell state) does not appear as signal. Built by `paper5/scripts/run_tuple_signatures.py`.

Z-Screen BB-pair and single-position signatures

For systems where tuple-level coverage is thin or where a coarser-resolution anchor is useful, lower-resolution signatures with the same target-vs-same-library-background logic were also built. ZEL024 / HEK293 carries an auxiliary bb0+bb1 pair anchor (84 pair signatures, each averaging ~165 distinct tuples) used in the resolution comparison only. ZEL028-2 (which has no substantive bb0 position) carries a single-position bb1 aggregation: under `paper5/scripts/run_bbpair_signatures.py`, the `--pos-a bb0_id` flag falls through to `None` for ZEL028-2 wells, so the resulting signature aggregates all wells matching a given bb1 identity regardless of bb2, bb3, bb4. The metadata for every signature records `pair_value_a`, `pair_value_b`, and the number of distinct tuples averaged in (`n_tuples_averaged`), so the level of chemical aggregation is explicit on every row. The ZEL028-2 single-bb1 outputs are therefore retained as a higher-stability lower-resolution view of the same library; they aggregate hundreds of distinct downstream tuples per signature and are not chemistry-resolved at the per-tuple level. The main results for ZEL028-2 use the full-tuple analysis described above; the single-bb1 outputs are preserved for inspection at `paper5/tables/ZEL028-2_*_pair_bb0_id_bb1_id_min*` (the file naming reflects the original CLI invocation). Built by `paper5/scripts/run_bbpair_signatures.py`.

Public CRISPR rank signatures

Public CRISPR rank signatures used in this manuscript are bundled in `paper5/external_data/`:

- Replogle K562 genome-scale Perturb-seq, 8,603 perturbations [4].
- Norman K562 CRISPR-activation, 105 single + 131 paired perturbations [3].
- scPerturb harmonized THP1, 78 perturbations from Papalexi/Satija ECCITE-seq and Wessels/Satija combinatorial CRISPR-Cas13 [9].

The rank signatures were generated upstream from the source single-cell CRISPR data with `paper5/scripts/build_replogle_rank_signatures.py`, `paper5/scripts/build_norman_rank_signatures.py`, and `paper5/scripts/build_scperturb_crispr_signatures.py`. Each builder converts the public single-cell perturbation data into the same up/down rank format used for Z-Screen, with control cells from the same dataset as background.

Rank-based chemistry-to-CRISPR comparison

Each rank signature was converted into a signed rank vector. Top up-genes received positive scores from `+(top_k + 1 - rank) / top_k`; top down-genes received the negative analog. Each Z-Screen signature was compared with each CRISPR reference signature by:

- mimic up-overlap (Z-up genes hitting C-up genes),
- mimic down-overlap,
- reverse up-down overlap,
- reverse down-up overlap,
- mimic total = mimic up + mimic down,
- reverse total = reverse up-down + reverse down-up,
- net mimic overlap = mimic total minus reverse total,
- rank cosine over shared top genes.

Comparisons used top_k = 250 and recorded shared-top-gene counts per pair. To keep file sizes bounded for the 8,599-tuple by 8,603-CRISPR scan (74 million pair scores), only the top 50 CRISPR matches per Z-Screen query were retained as raw output. Rank cosine was reported as NaN when fewer than 10 genes were shared between the two signatures. Net mimic overlap (rather than rank cosine alone) is the headline scoring metric because it penalizes anti-correlated matches, which would otherwise inflate cosine for compounds that produce broadly opposite-direction transcriptional states. Run by `paper5/scripts/run_chemistry_vs_crispr.py`.

Permutation-null calibration

For each top-K-per-query table the top 10,000 rows by net mimic overlap were calibrated by drawing 1,000 random gene sets matched in size to each CRISPR signature's up-set and down-set from the dataset-specific gene universe. For each random draw the same mimic / reverse / net-mimic statistics were recomputed against the fixed Z-Screen up/down set. Empirical p-values were computed as `(1 + count of nulls >= observed) / (1 + n permutations)` and converted to Benjamini-Hochberg q-values across the calibrated rows. The headline ZEL024 / HEK293 scan used seed 13 and 1,000 permutations per row.

A size-matched permutation null was used in preference to a parametric test because parametric distributions for rank-overlap statistics depend on assumptions (gene universe, top-k cutoff, signature size) that vary by reference. A permutation null preserves the size of each CRISPR signature exactly, so the same null procedure is valid across heterogeneous references without requiring a separate parametric calibration for each one. Run by `paper5/scripts/run_calibrate_matches.py`.

Recurrent-neighbor diagnostics and deduplicated headline tables

After calibration, per-CRISPR-signature recurrence tables were built that record how many distinct Z-Screen queries each CRISPR perturbation appeared as a top match for, both across the full top-K-per-query table and across the calibrated table at each q threshold. Two deduplicated views were also computed: one row per CRISPR program (best Z-Screen match by net z) and one row per Z-Screen query (best CRISPR match by net z). Finally, a hub-strip robustness summary records how many distinct programs and queries survive at q 0.05 after dropping the top N most-recurrent CRISPR signatures (N in 0, 5, 10, 25, 50). The hub-strip analysis is reported alongside the headline rather than as a footnote because broadly responsive CRISPR perturbations are a known structural feature of large CRISPR atlases, and any chemistry-to-CRISPR framework that does not control for them will over-claim. Run by `paper5/scripts/run_recurrent_neighbor_diagnostics.py`.

Resolution comparison

For each ZEL024 / HEK293 tuple at >=10 wells, single-bb3 signatures were built from the same library at >=500 wells per bb3, with explicit n-tuples-averaged metadata. Single-bb3 signatures were compared with the Replogle K562 reference and the top 50 matches per query were recorded. For each tuple the top 5 Replogle CRISPR matches at tuple resolution, the matches the containing bb0+bb1 pair would produce, and the matches the containing bb3 single-position signature would produce were collected, and pairwise Jaccard overlaps were computed. The summary table reports the per-tuple overlaps and the aggregate distribution. Run by `paper5/scripts/run_resolution_comparison.py` plus the figure builder `paper5/scripts/make_resolution_figure.py`.

Norman combinatorial logic

For each Norman double-gene perturbation A+B (131 doubles), the constituent single-gene signatures A and B were located in the Norman metadata. Across the 8,599 ZEL024 / HEK293 tuples, the Spearman correlation was computed between the tuple-vs-double net mimic vector and the sum of the tuple-vs-single-A and tuple-vs-single-B net mimic vectors. The best Z-Screen tuple's individual scores against the double and each single were also recorded, along with whether the best double score exceeded both single scores. Run by `paper5/scripts/run_norman_combinatorial_logic.py`.

Same-cell positive controls

The MZ1 and STC-15 positive controls used the matched-cell THP1 scPerturb reference. MZ1 lives in a sparse THP1 control library that lacks adequate same-library background, so its compound signature was built with `paper5/scripts/build_compound_rank_signatures_with_fallback.py` using a same-cell compound-pool fallback background (all THP1 compound-treated cells excluding MZ1). The same fallback was used for STC-15, STM2457, sorafenib, Crizotinib, and capmatinib in their respective sparse libraries. The fallback background is broader than a same-library vehicle control would be; the absolute z-score should be read in that context. The structure of the MZ1 result (BRD4 as the top match across permutation seeds) is robust. The internal-control summary integrating direct-target and proxy-target recoveries is in `paper5/tables/internal_control_summary_combined.csv`; the top-3 calibrated matches per same-cell THP1 compound are in `paper5/tables/second_positive_control_thp1_compounds_top3.csv`.

Cell-state transfer

The cross-cell projection used the 32-dimensional scVI latent coordinates from `canonical_scvi_latents.parquet`. For each library and cell-line pair with sufficient shared chemistry coverage, building-block-position profiles were averaged in latent space and a centered ridge transfer model was trained in cross-validation. Performance was reported as weighted row-wise cosine between predicted and held-out target profiles, with direct source reuse as the identity baseline. Run by `paper5/scripts/run_latent_transfer.py`. Detailed cross-cell results are reported in paper 4.

Limitations

The framework compares transcriptional states, not biochemical binding. A chemistry-to-CRISPR match is a state-level hypothesis until validated by orthogonal target engagement.

The MZ1 fallback background for matched-cell THP1 is broader than a same-library vehicle control would be. The structure of the MZ1 to BRD4 result is robust across permutation seeds; the absolute z-score should be read in the context of that broader calibration.

Local same-cell CRISPR coverage is THP1-dominant, with no HEK293 or HEK293T single-cell CRISPR atlas surfaced by the local public probe. The HEK293 chemistry results therefore use cross-cell K562 Replogle as the genome-wide reference; the cross-cell setting means the matches are chemistry-resolved CRISPR-like state hypotheses rather than direct cell-of-origin target calls.

Tuple-level chemistry resolution is dataset-dependent. ZEL024 in HEK293 supports 8,599 tuples at >=10 wells with median 13 wells per tuple, the cleanest tuple-level system in the canonical workspace. ZEL028-2 systems use a relaxed >=2-well threshold because per-tuple coverage in that library does not support the >=10-well or even >=3-well threshold; the resulting tuple signatures aggregate a median of 2 wells, which makes individual tuple signatures substantially noisier than the ZEL024 / HEK293 tuple signatures. The ZEL028-2 tuple-level results are therefore interpreted as a chemistry-resolved leaderboard whose strongest entries reproduce the same biology themes ZEL024 / HEK293 surfaced (RNA processing and mitochondrial translation) rather than as confirmed per-tuple calls; per-tuple confirmation would require deeper sequencing on individual tuples of interest. A complementary single-position bb1 aggregation of the same ZEL028-2 wells is preserved in `paper5/tables/` for users who prefer a higher-stability lower-resolution view. ZEL031 systems are tuple-resolution at lower well counts. The reporting tables annotate each signature with n-wells and n-tuples-averaged so the reader can directly judge the chemistry resolution behind any specific match.

Recurrent CRISPR perturbations exist; the recurrence diagnostic and hub-strip summary report how the headline counts change when the top hubs are dropped from the calibrated table. After dropping the 50 most-recurrent CRISPR signatures, 1,236 chemistry-resolved tuples and 1,664 distinct CRISPR programs still passed q 0.05, so the diversity is not driven by hubs alone.

The targeted public scPerturb probe is laptop-scale (11 H5AD files under 500 MB each) and is not a comprehensive survey of all public HEK293 perturbation data.

Data availability

The full reproducibility package, including the canonical Z-Screen RNA-seq dataset, bundled public CRISPR rank signatures, all derived match tables, calibrated headline outputs, recurrence diagnostics, and figure-building scripts, is contained in this repository at `paper5/`. The repaired canonical RNA-seq aggregate is at `data/ZScreen_Canonical_Dataset/RNASeqAggregate/`. Public CRISPR rank signatures are bundled in `paper5/external_data/`. Their upstream source datasets are referenced below.

Code availability

All analysis and figure-generation scripts are in `paper5/scripts/`. The reproduction order, including dependencies on the bundled public CRISPR rank signatures, is in `paper5/README.md`. Package-level dependencies are listed in the repository root `requirements.txt`. The codebase was tested with Python 3.14.3 on Windows.

References

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